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Nasim et al.

(54) **SECURE VIRTUALIZED TELEMETRY WITH  
SAVE AND RESTORE OVER GLOBAL  
COUNTERS**

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**Related U.S. Application Data**

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11, 2024.

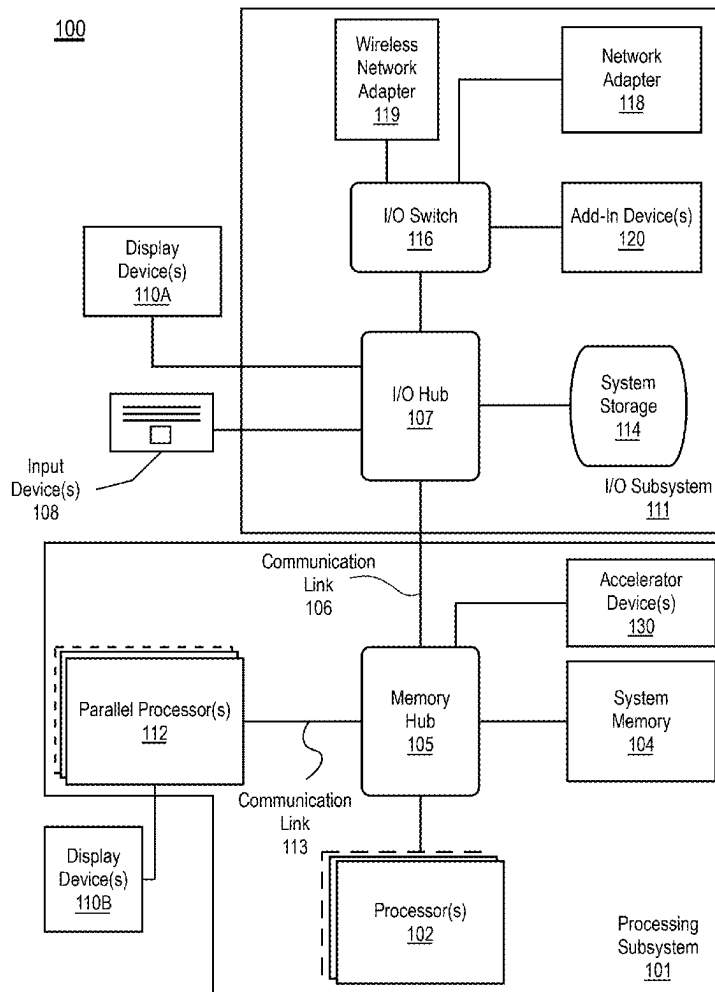
**Publication Classification**

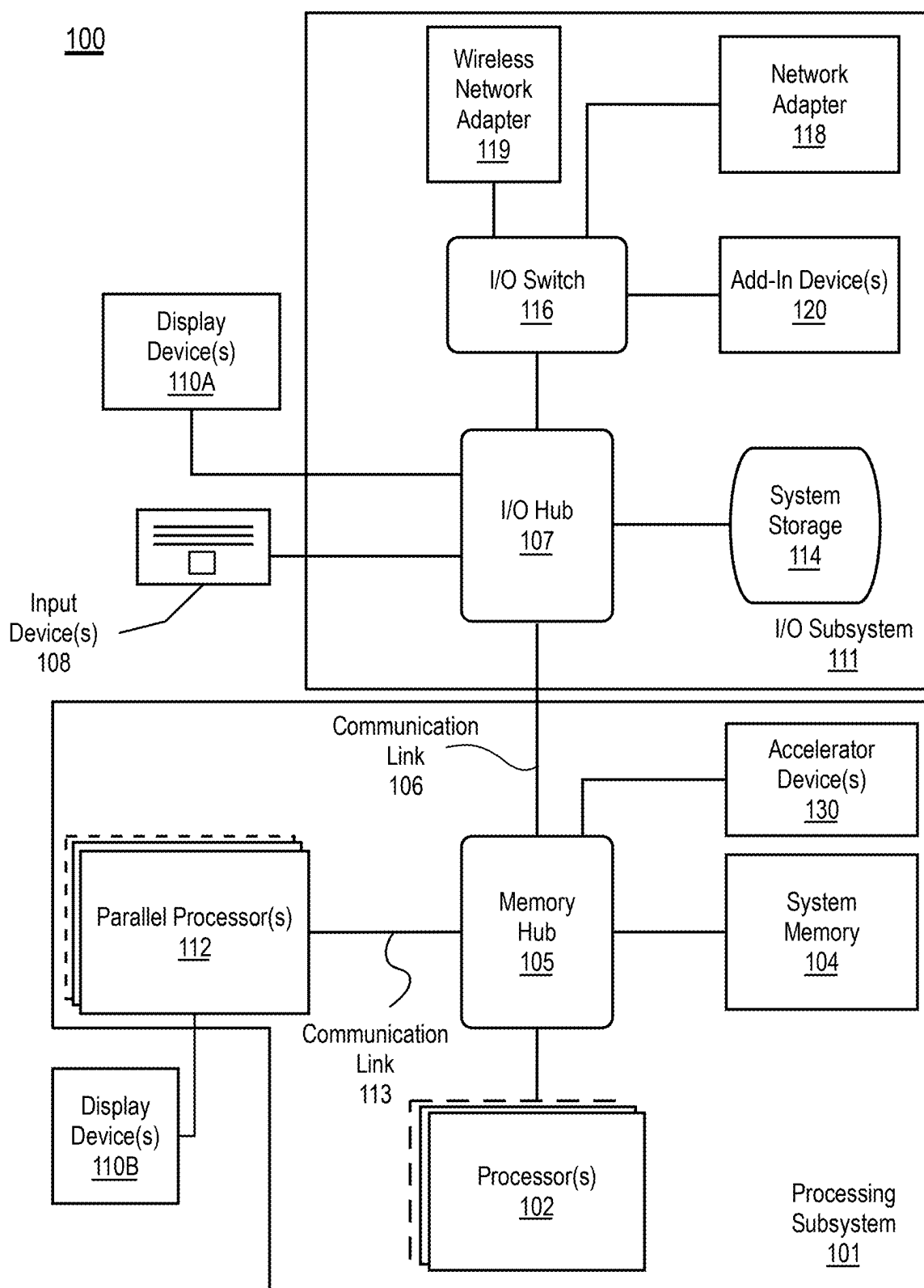
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(52) **U.S. Cl.**  
**CPC** ..... **G06F 9/45558** (2013.01); **G06F**  
**2009/45587** (2013.01)

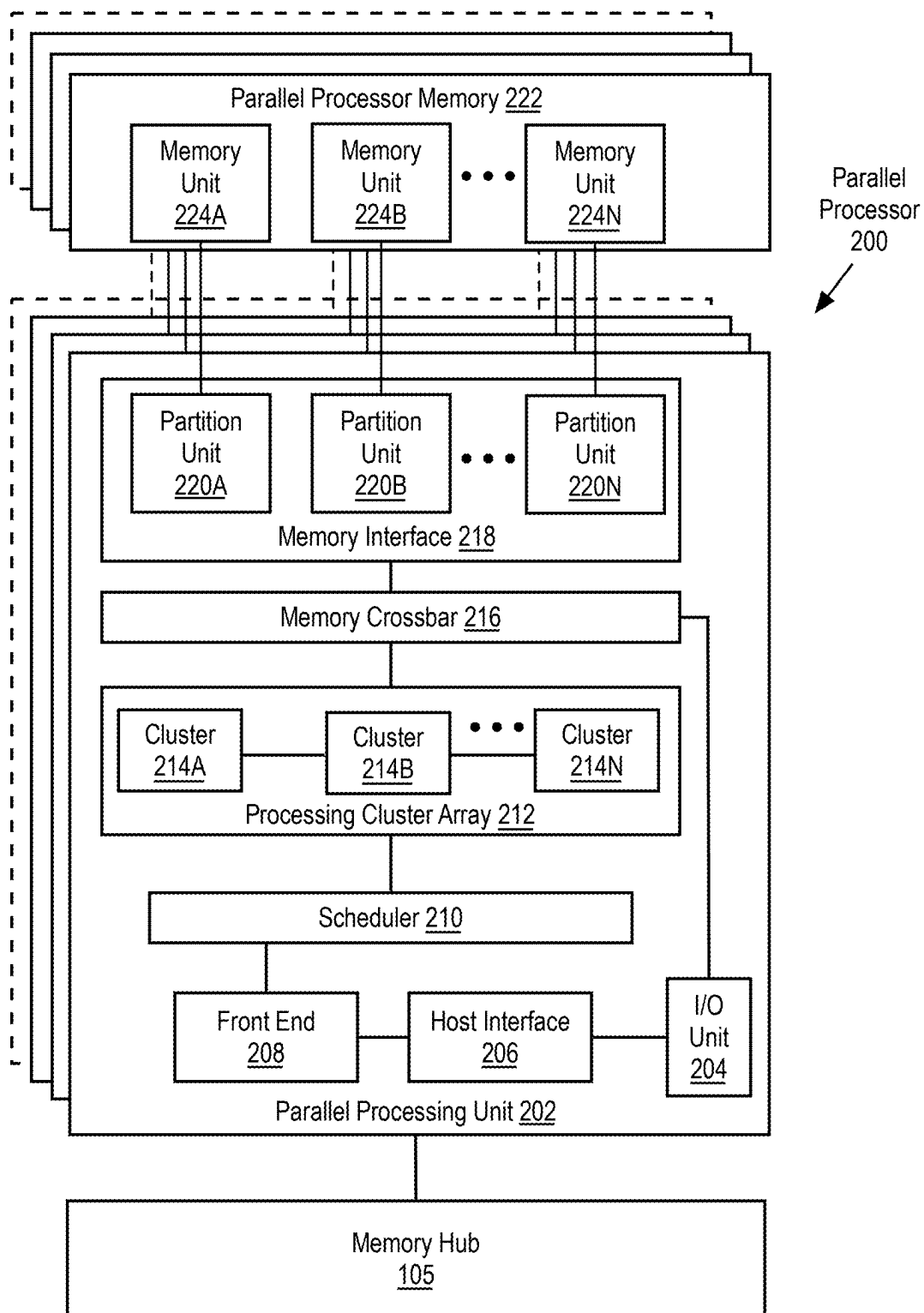
(57) **ABSTRACT**

An apparatus having secure virtualized telemetry with save and restore over global counters is disclosed. The apparatus includes processor circuitry comprising: execution circuitry to provide a trusted execution environment (TEE) to host a virtual machine (VM), the TEE to: responsive to attesting secure telemetry endpoint circuitry as part of TCB of the TEE, map, within VF BAR space of the VM, the secure telemetry endpoint circuitry as a logical endpoint for the VM hosted by the TEE; and establish a trusted session between the VM and the secure telemetry endpoint circuitry; and the secure telemetry endpoint circuitry to: procure telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space; and provide a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.





**FIG. 1**



**FIG. 2A**

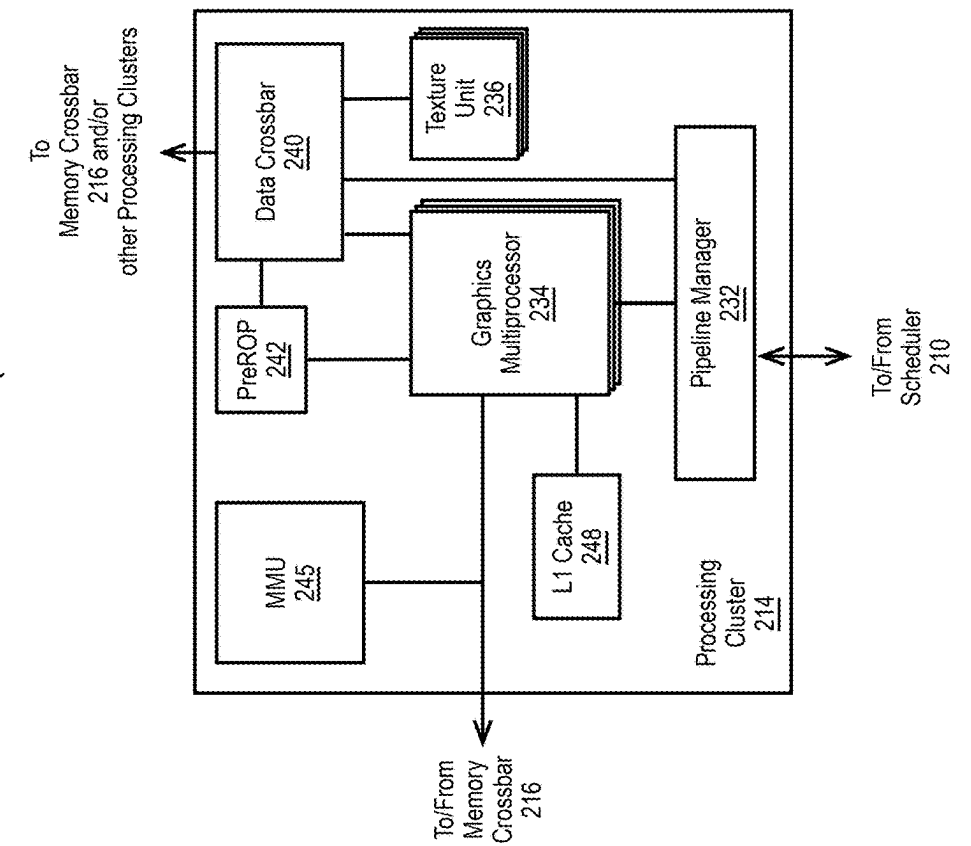


FIG. 2C

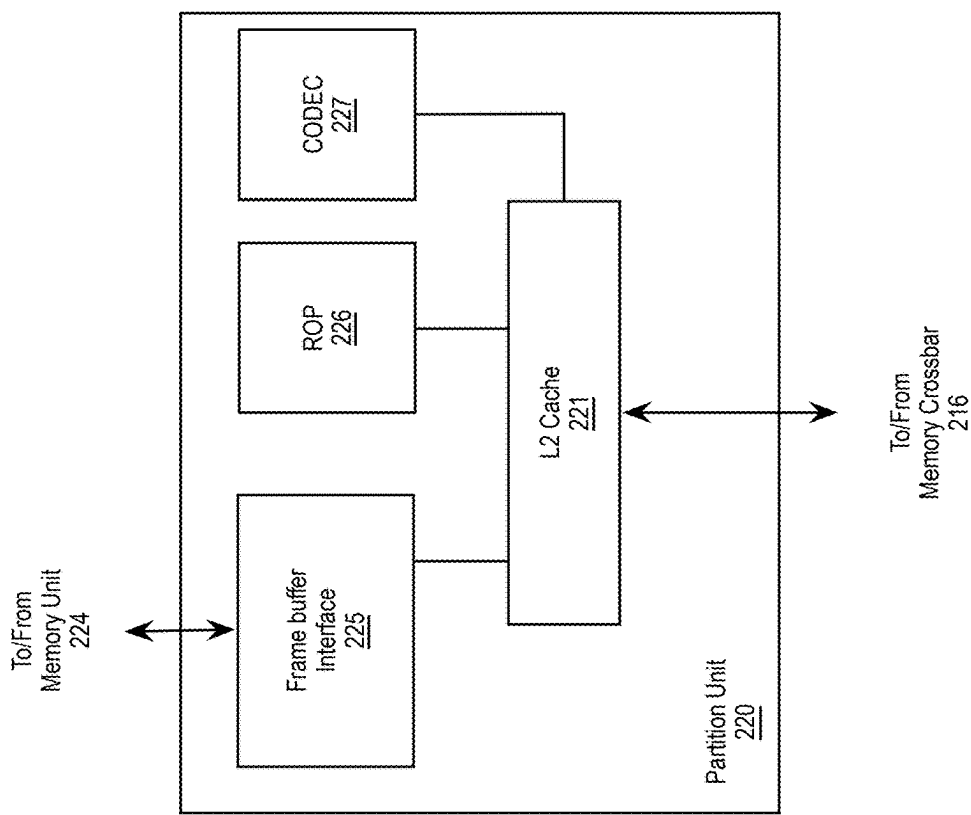
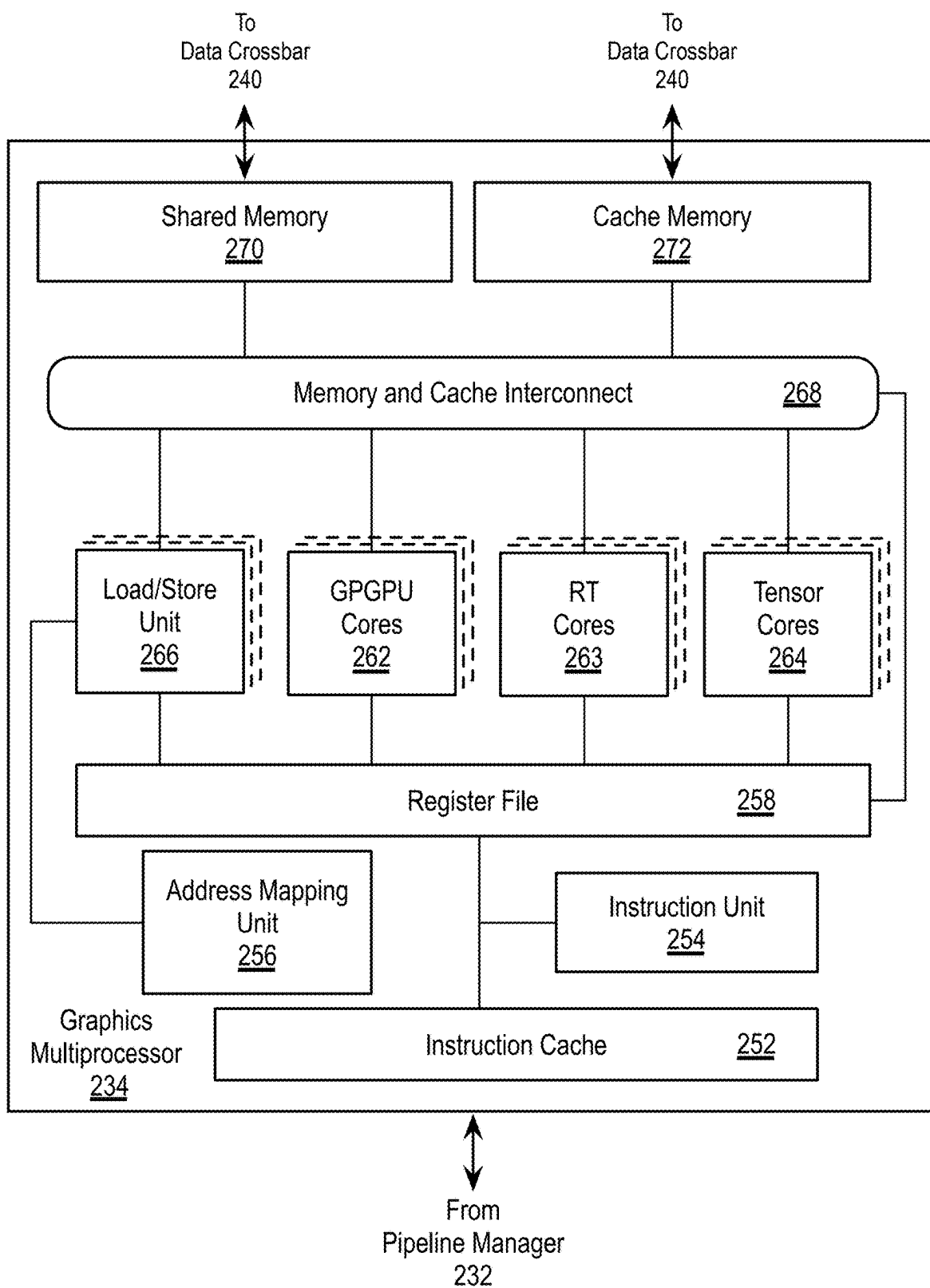
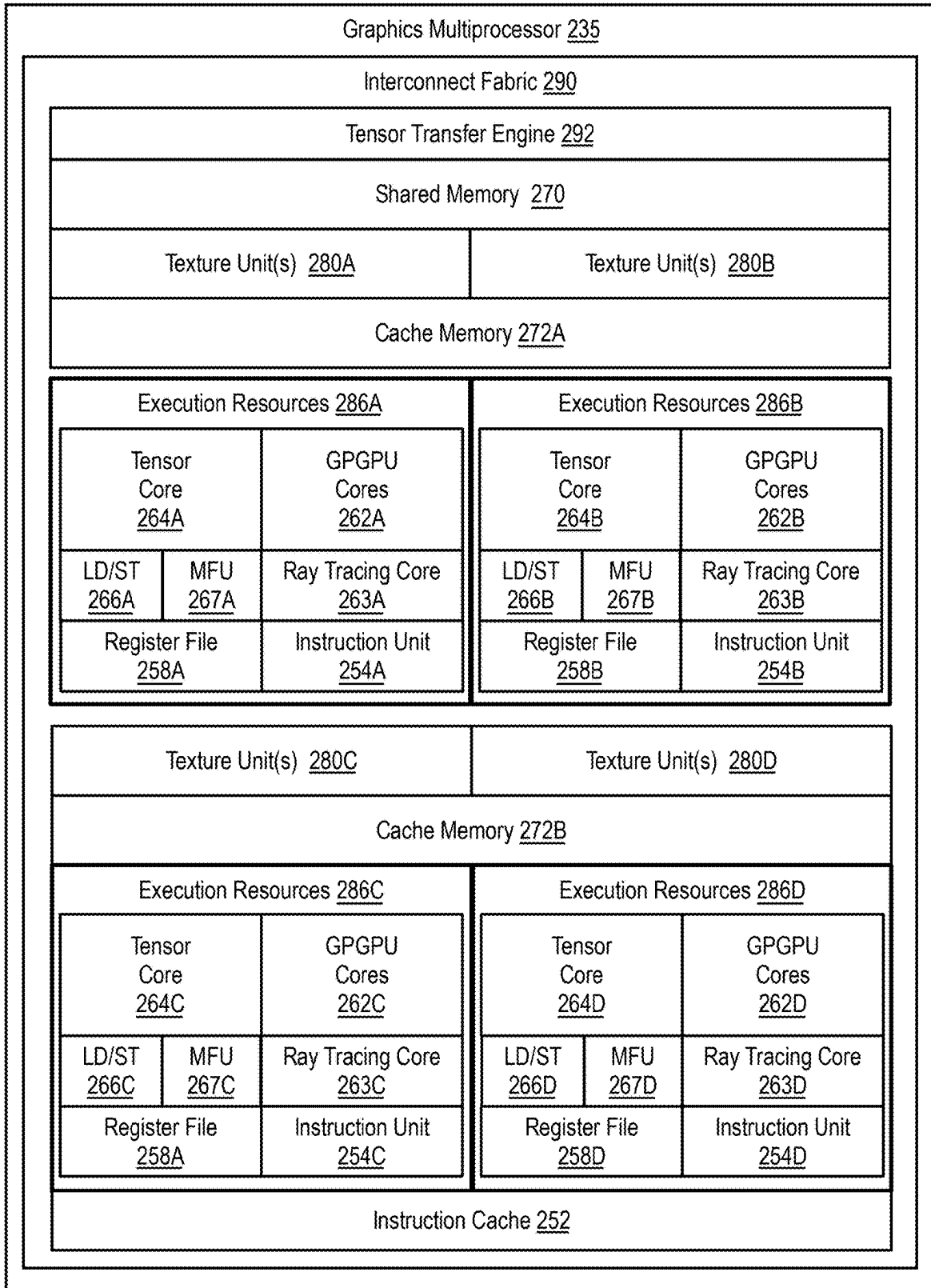


FIG. 2B

**FIG. 2D**



**FIG. 2E**

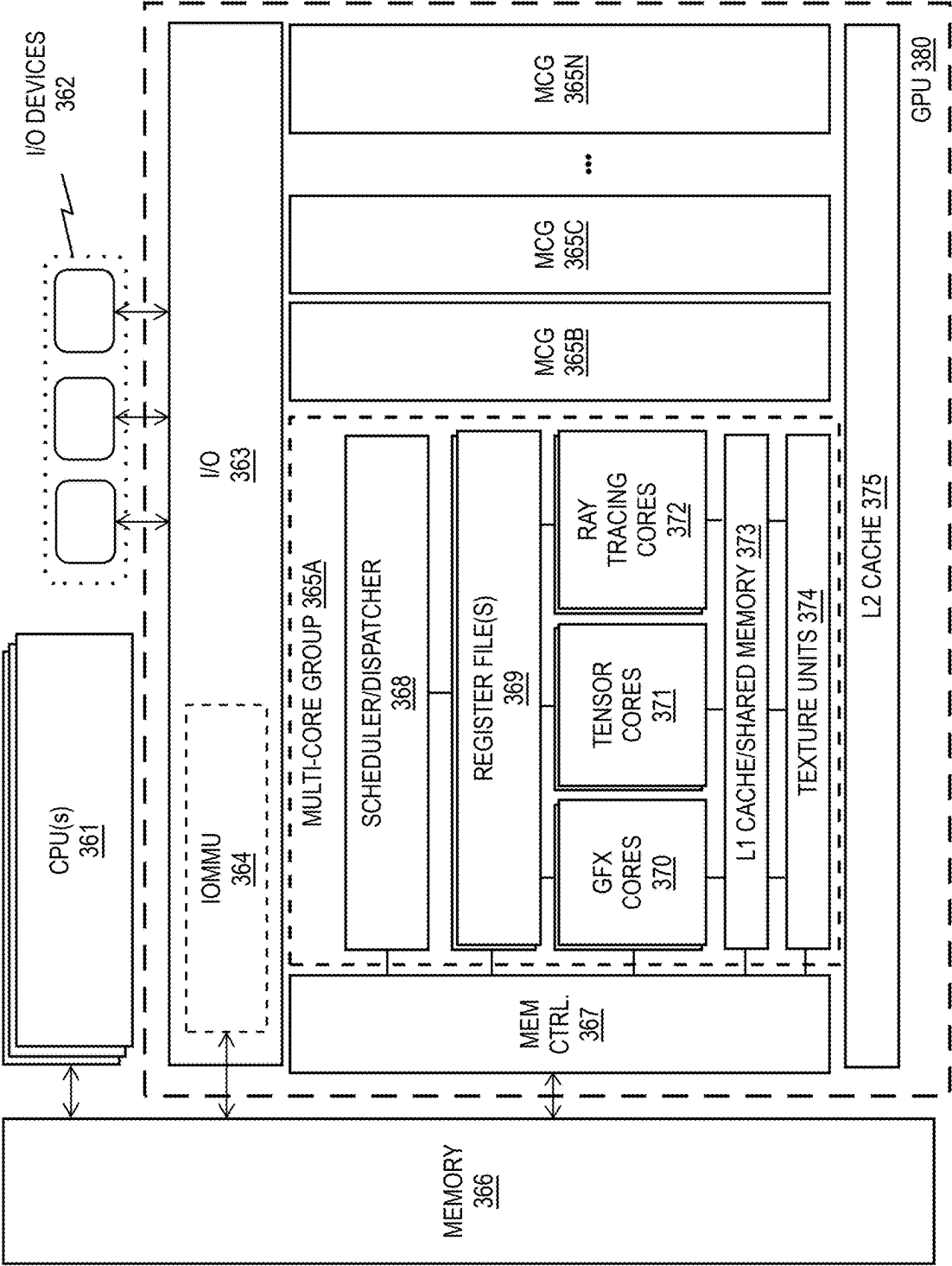


FIG. 3

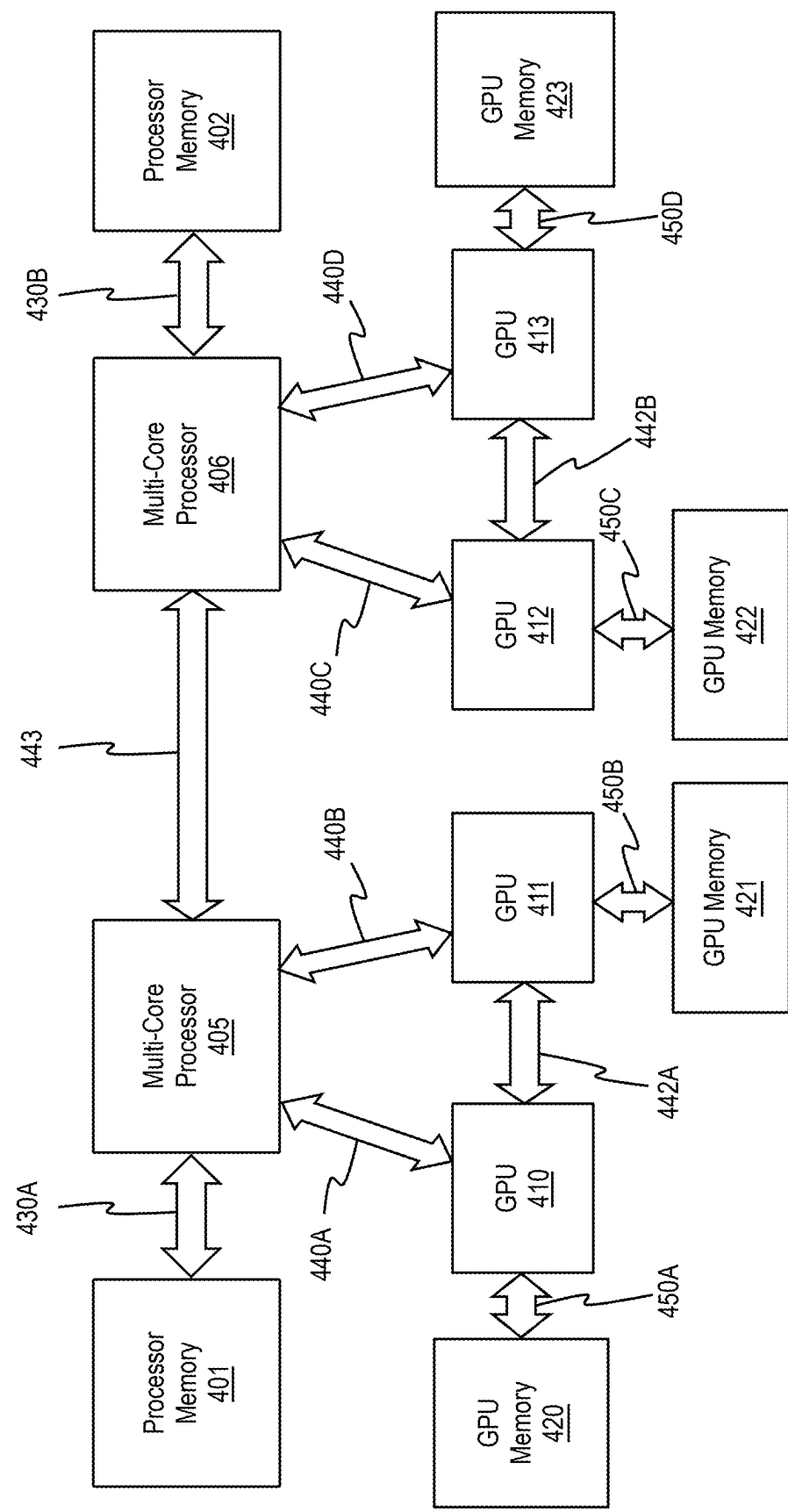


FIG. 4A

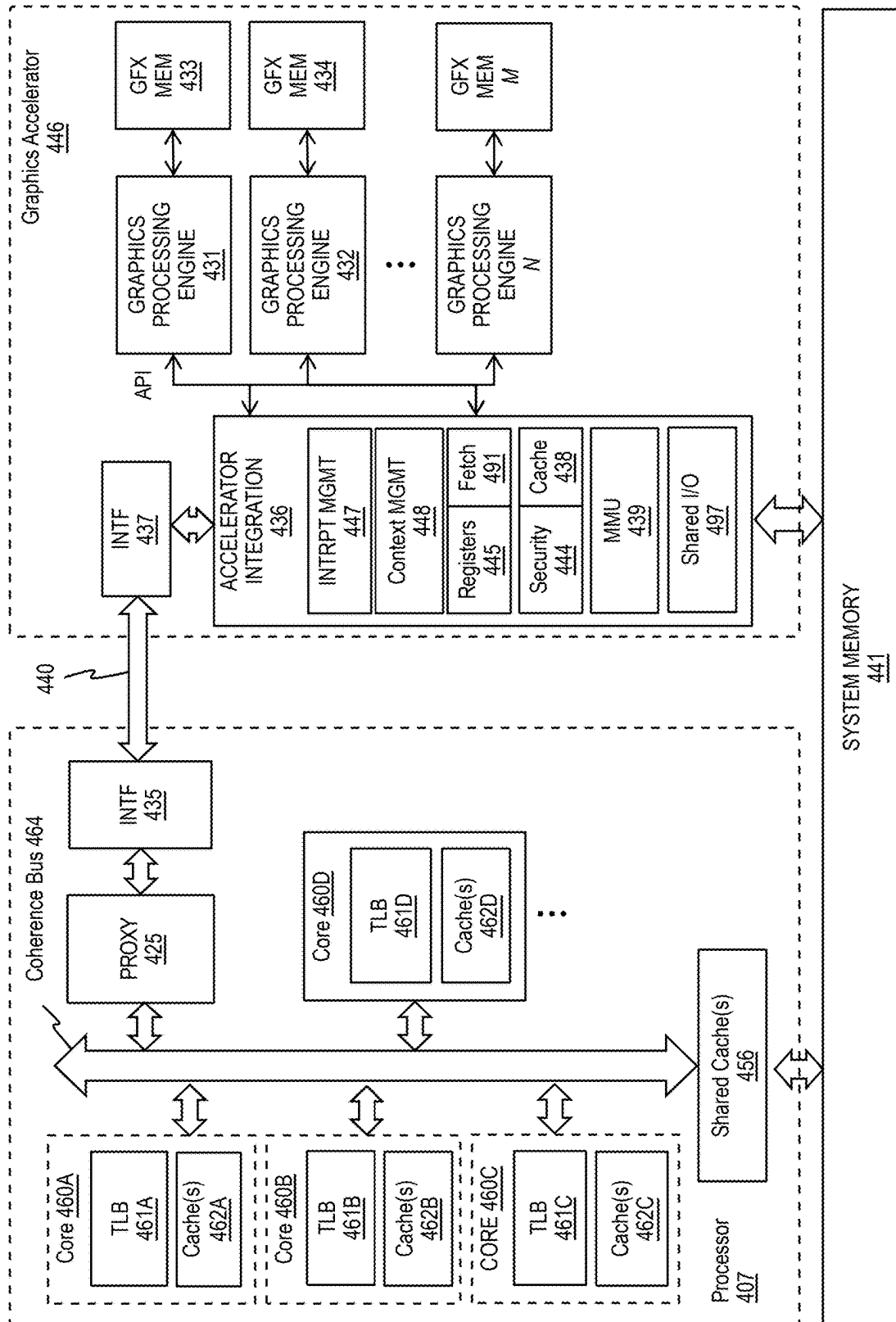


FIG. 4B

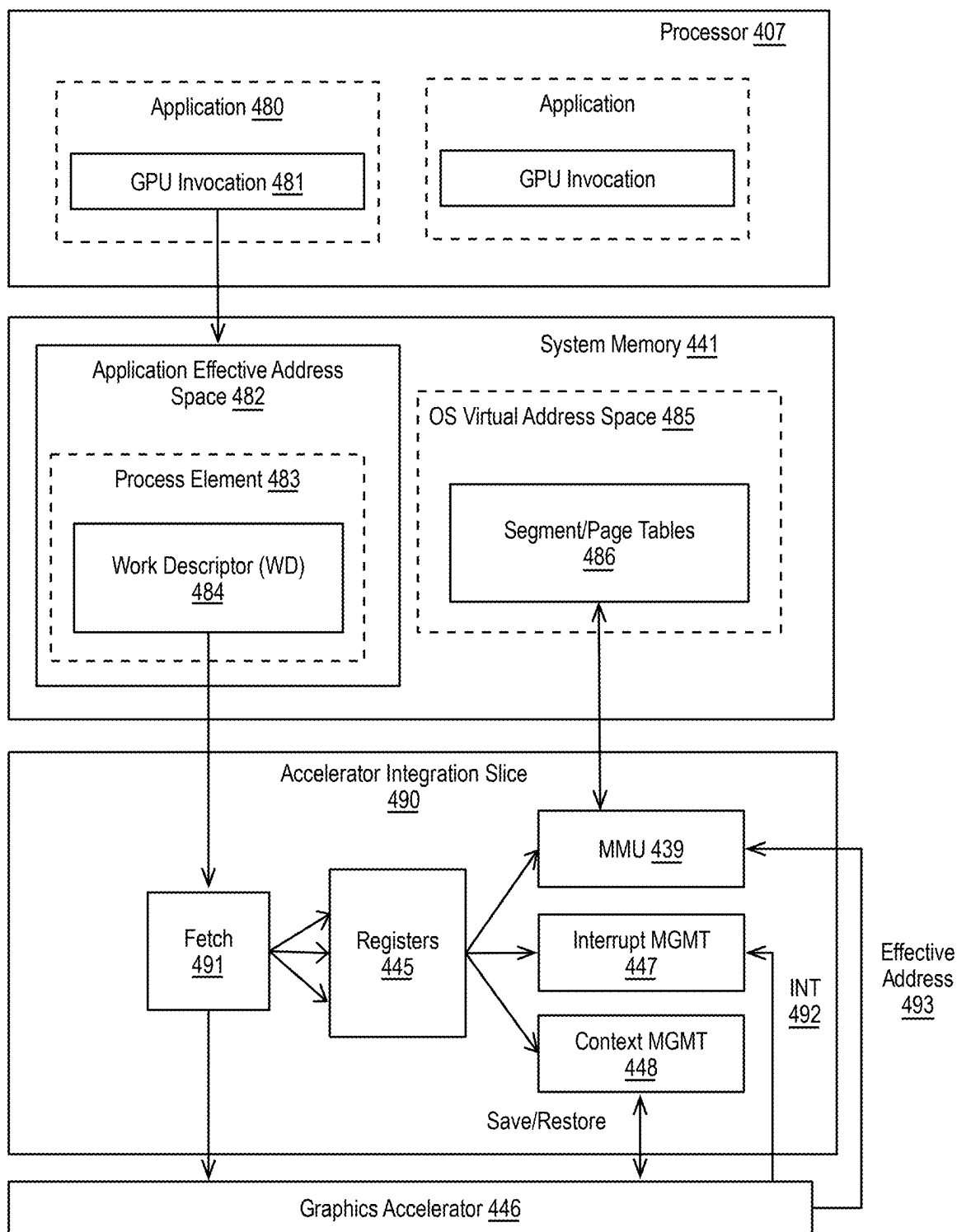


FIG. 4C

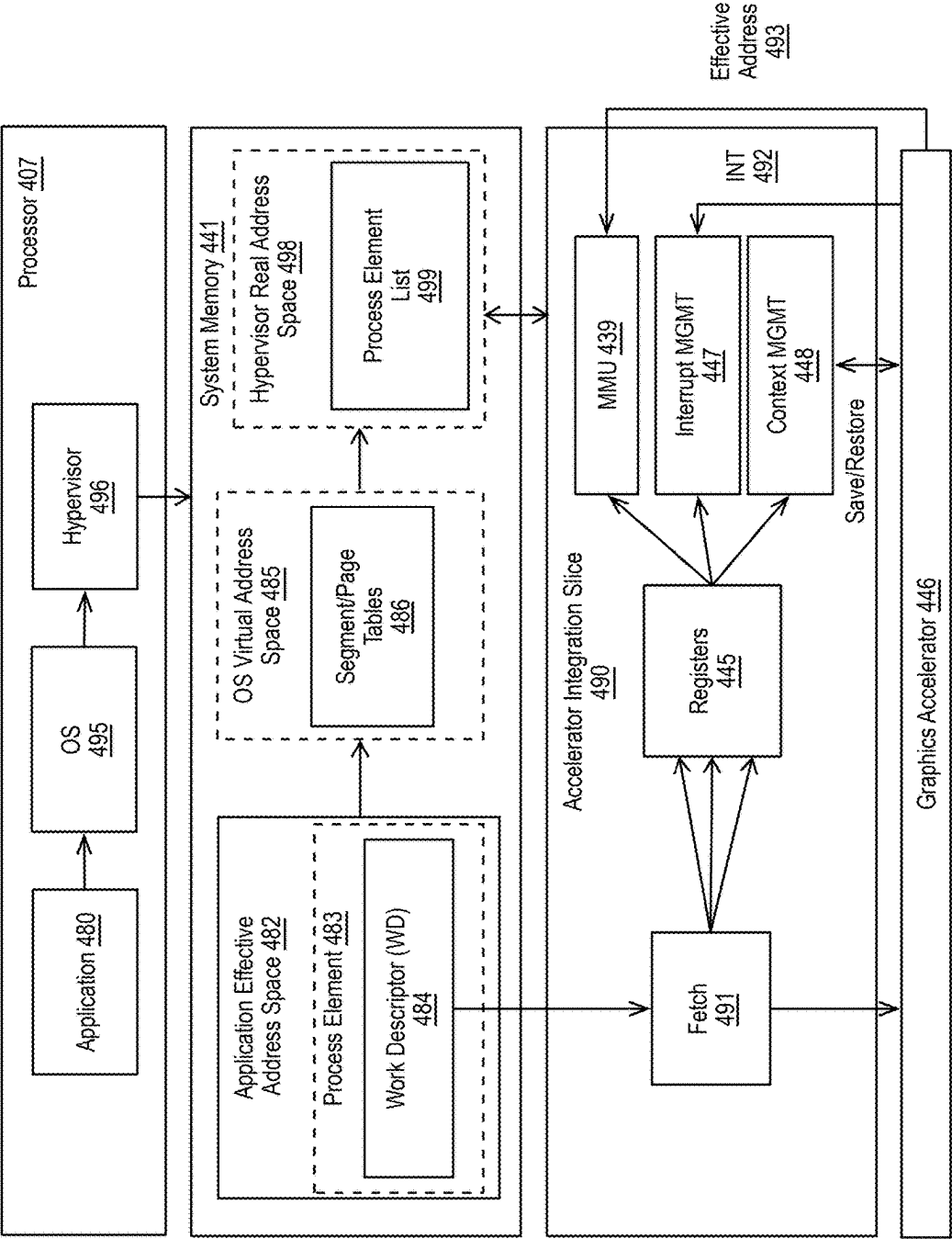


FIG. 4D

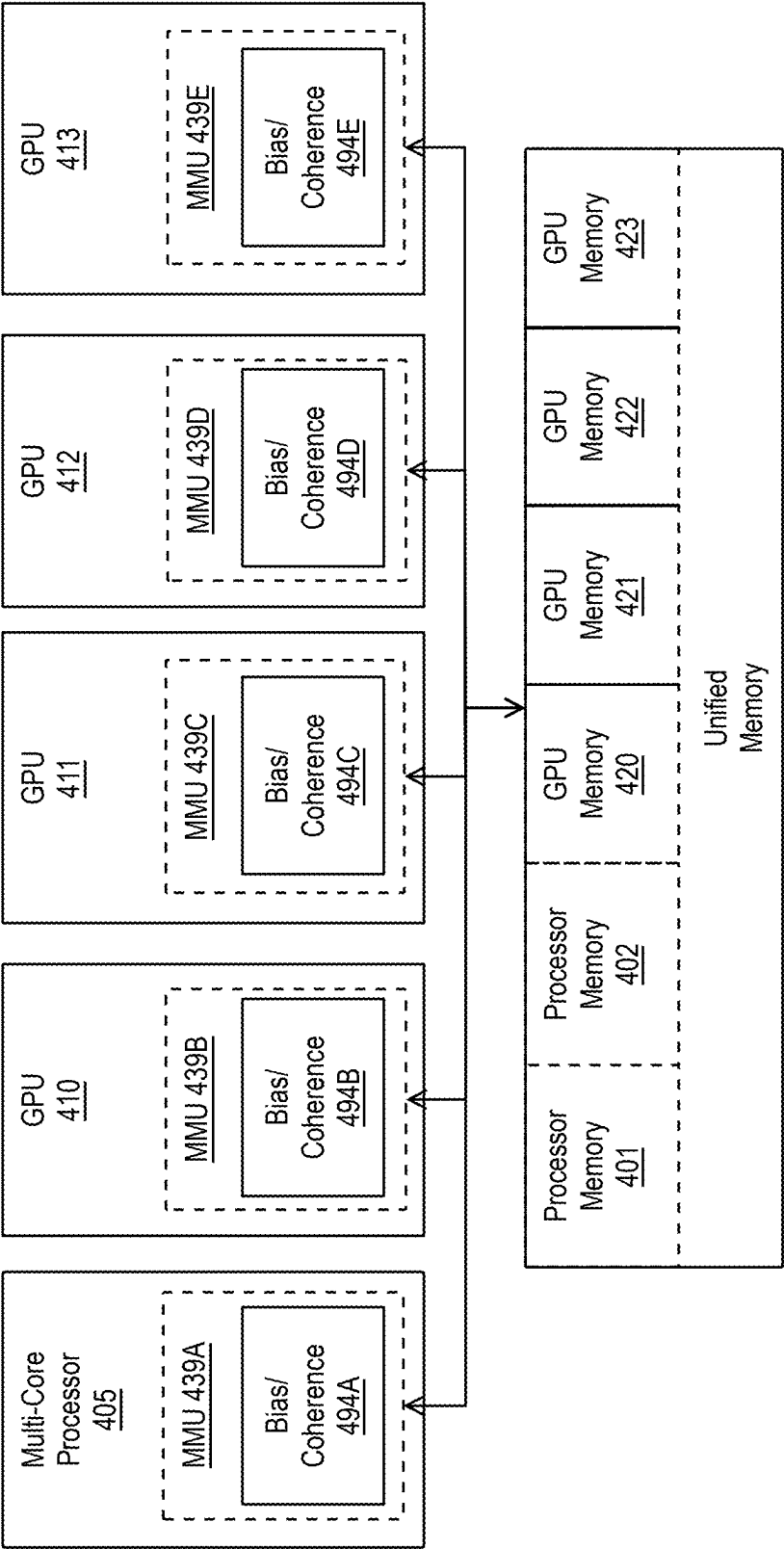


FIG. 4E

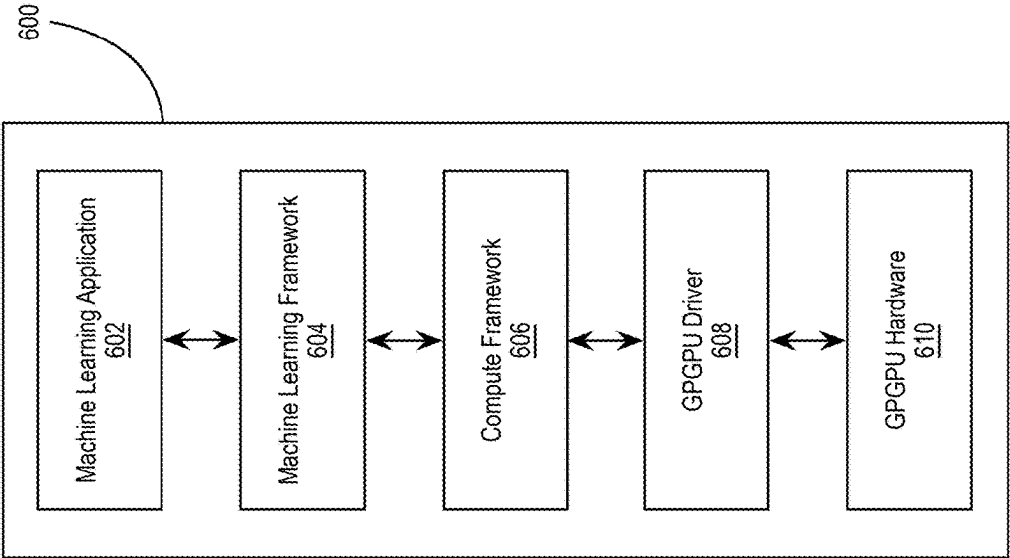


FIG. 6

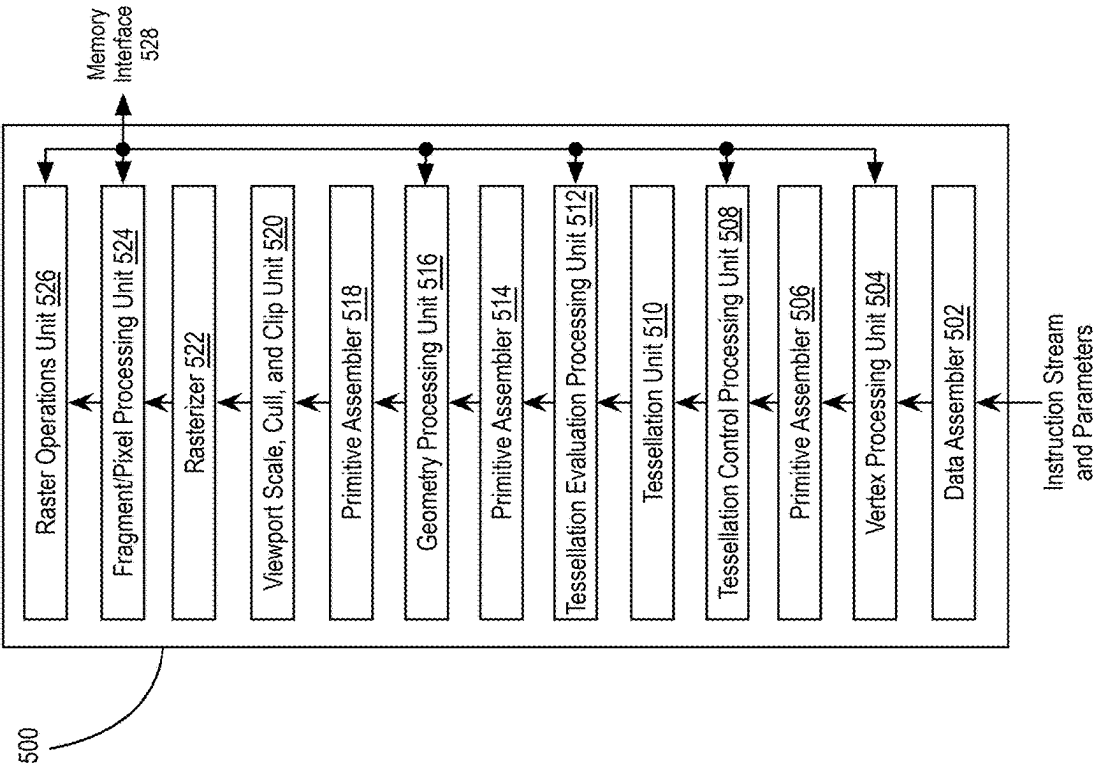


FIG. 5

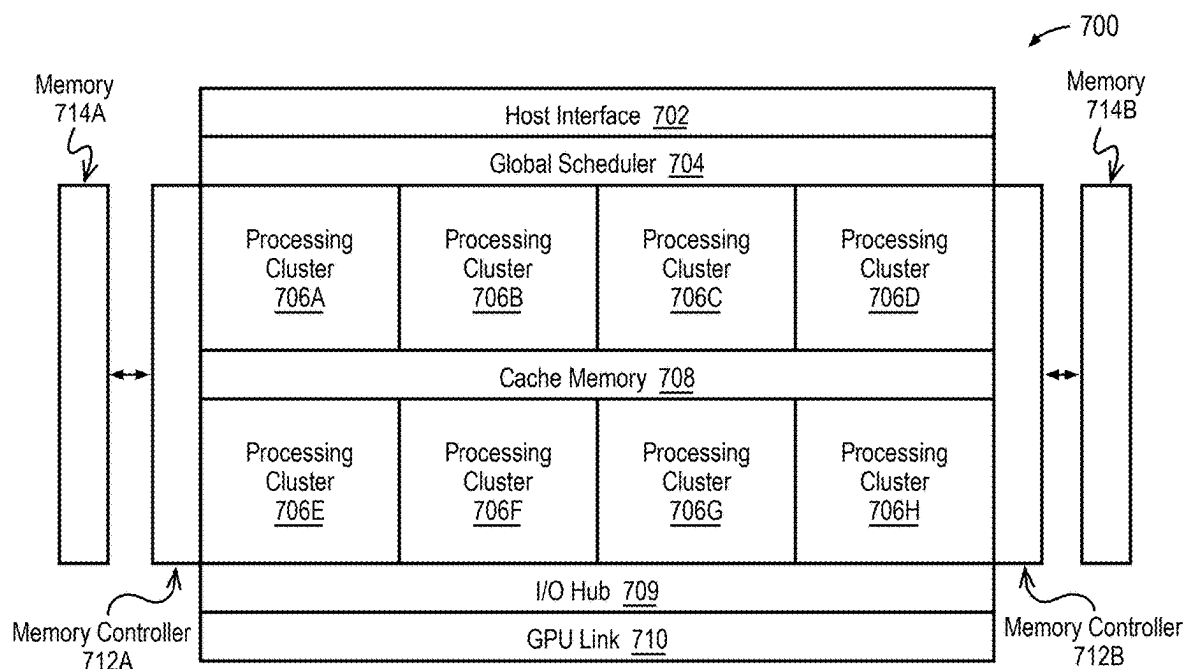


FIG. 7

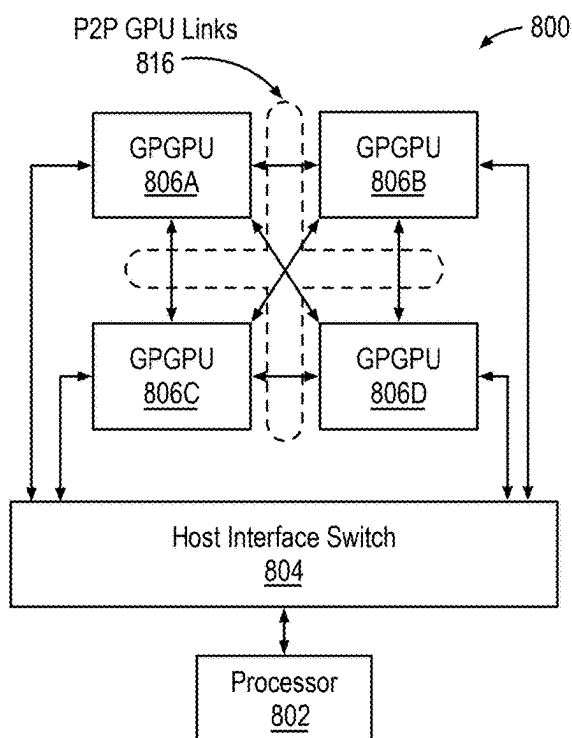
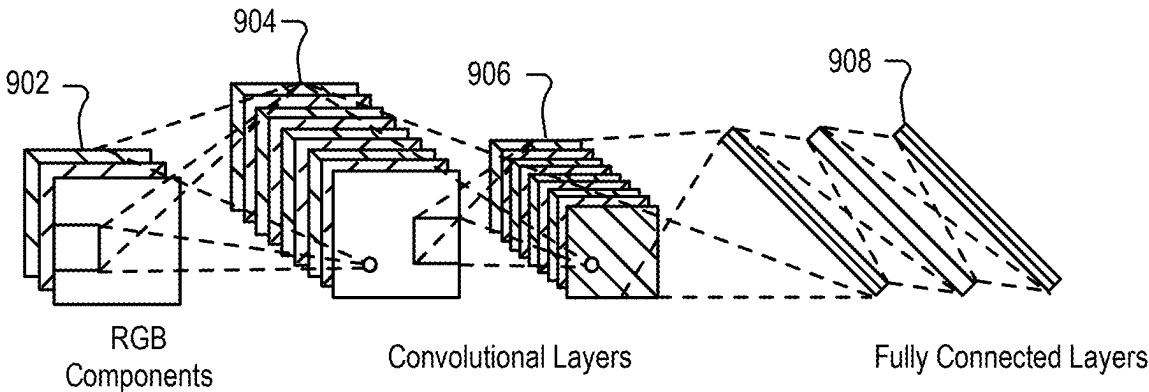
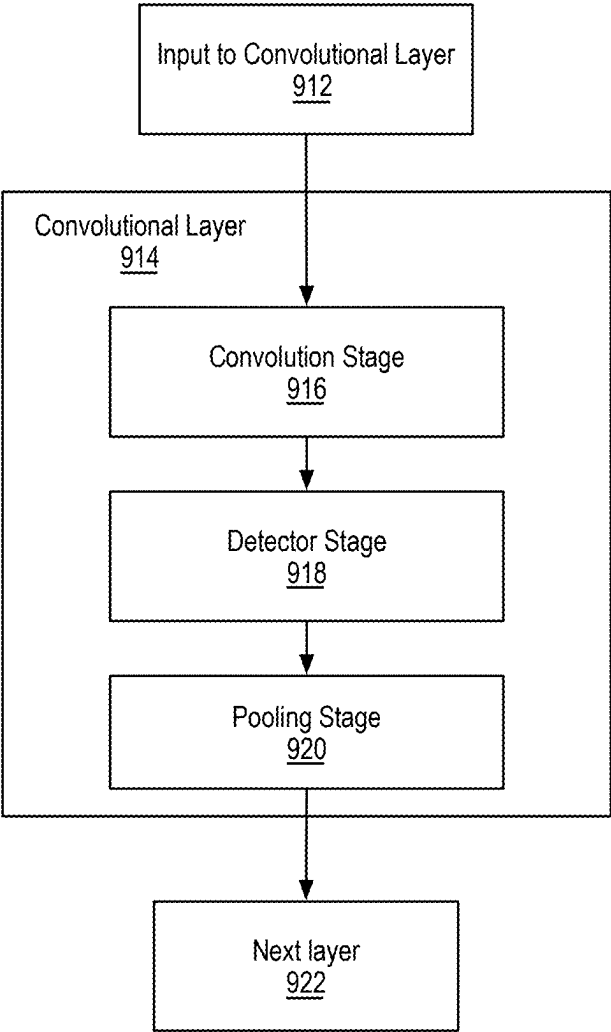


FIG. 8



**FIG. 9A**



**FIG. 9B**

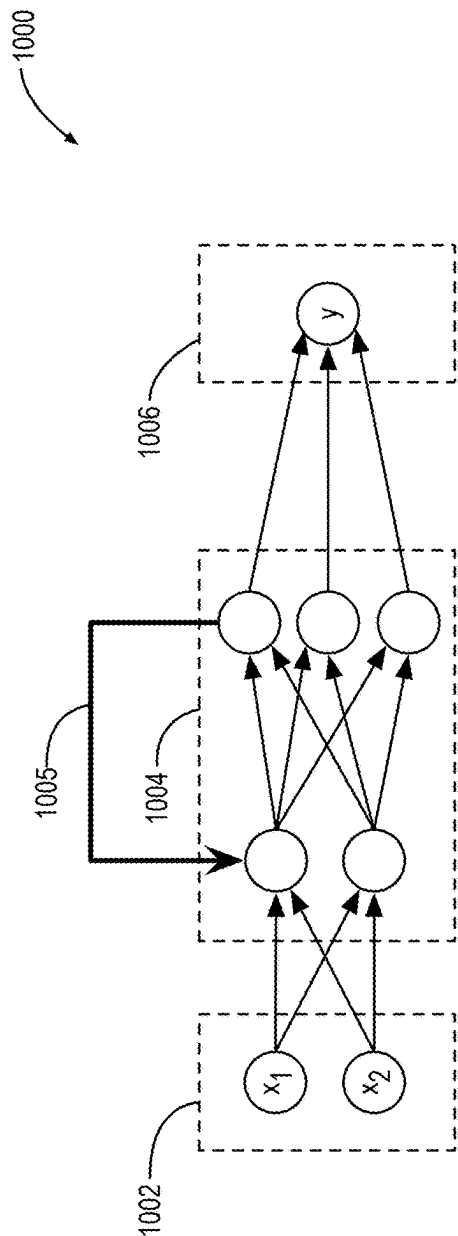


FIG. 10A

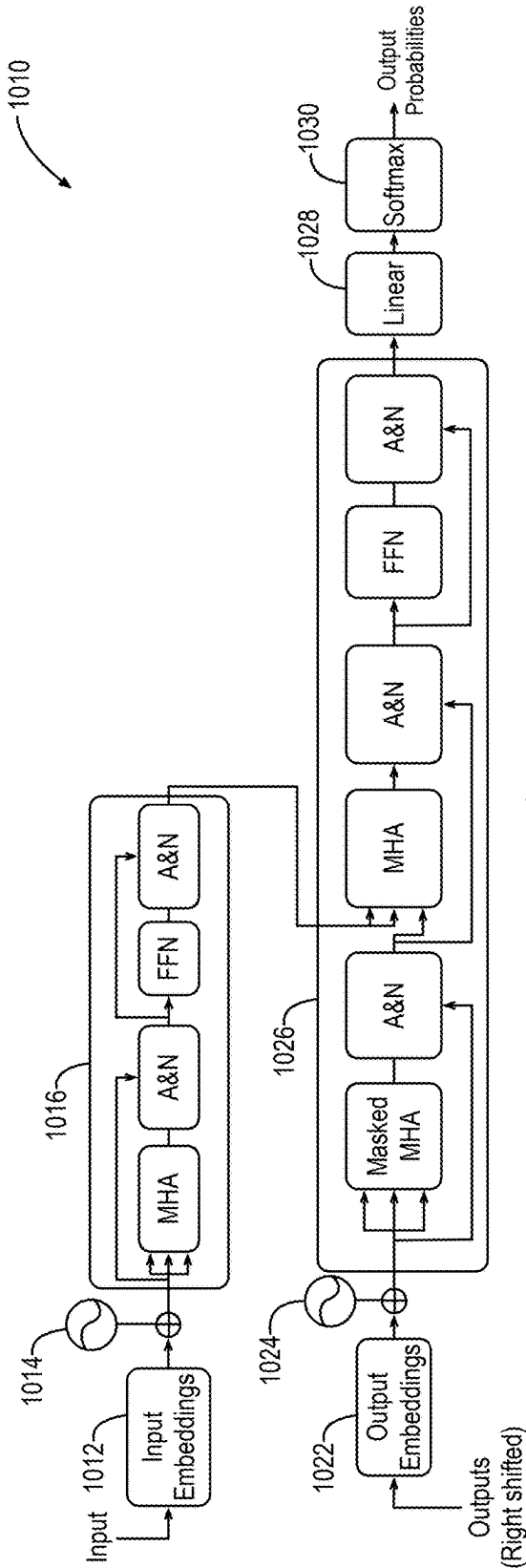


FIG. 10B

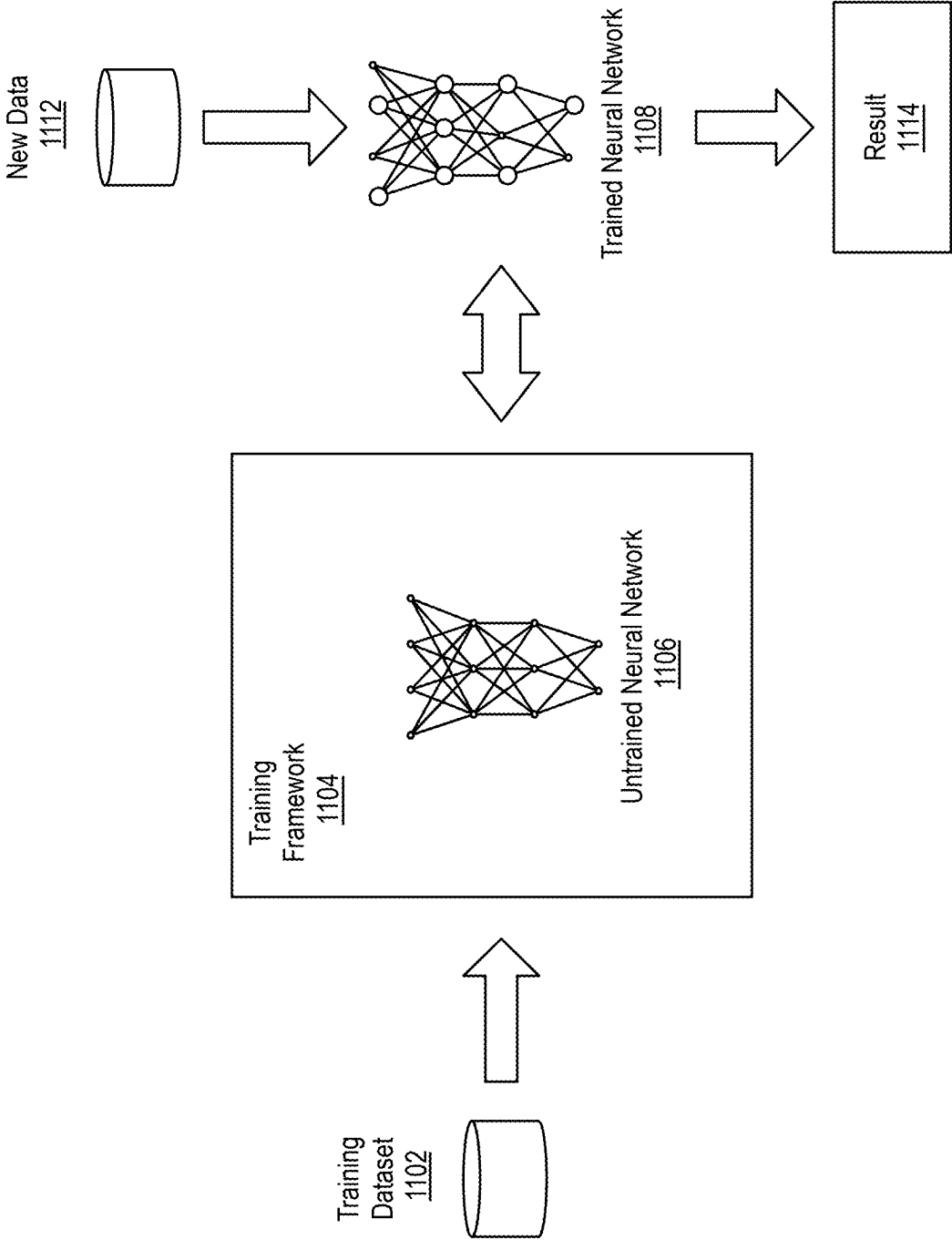


FIG. 11

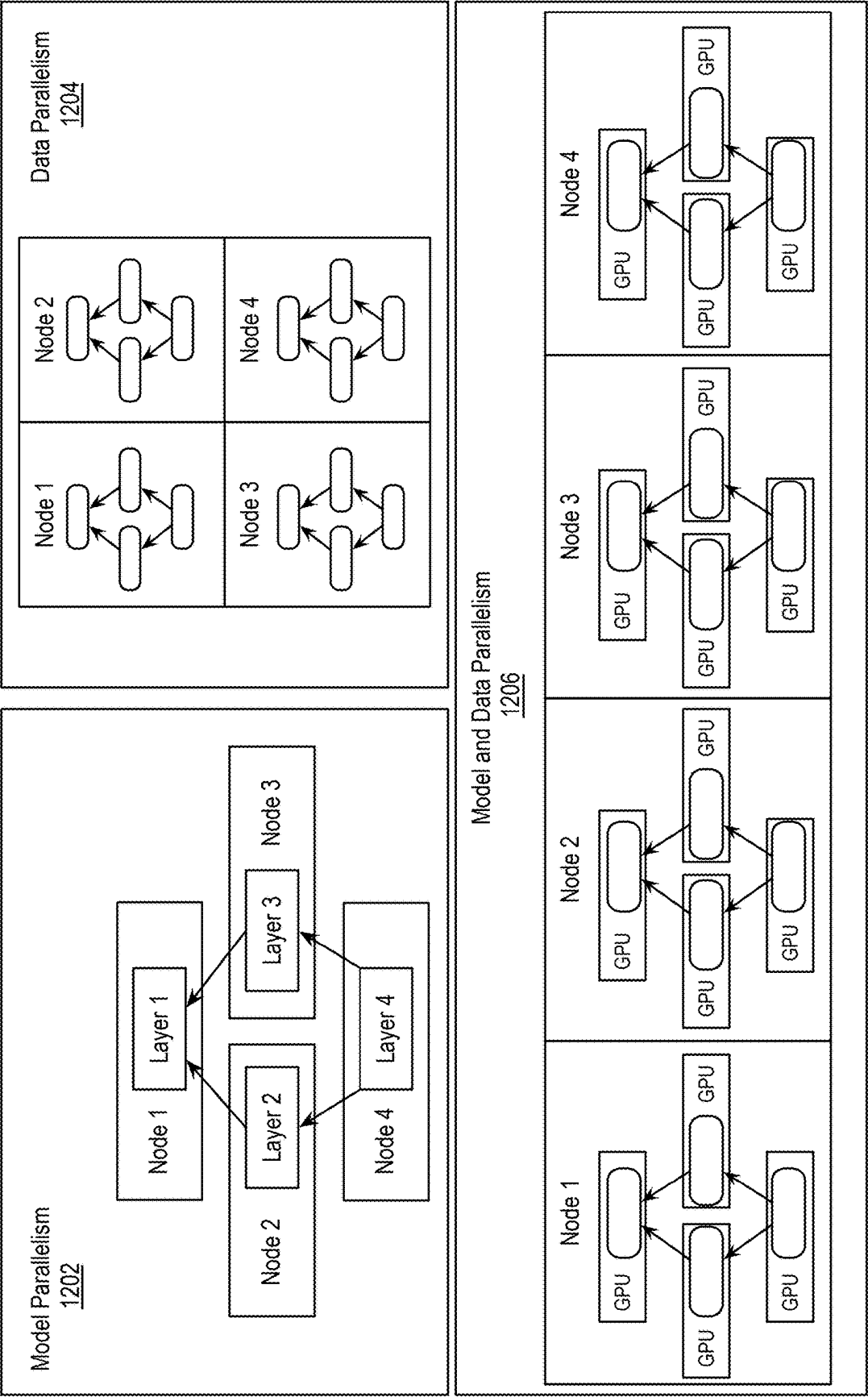


FIG. 12

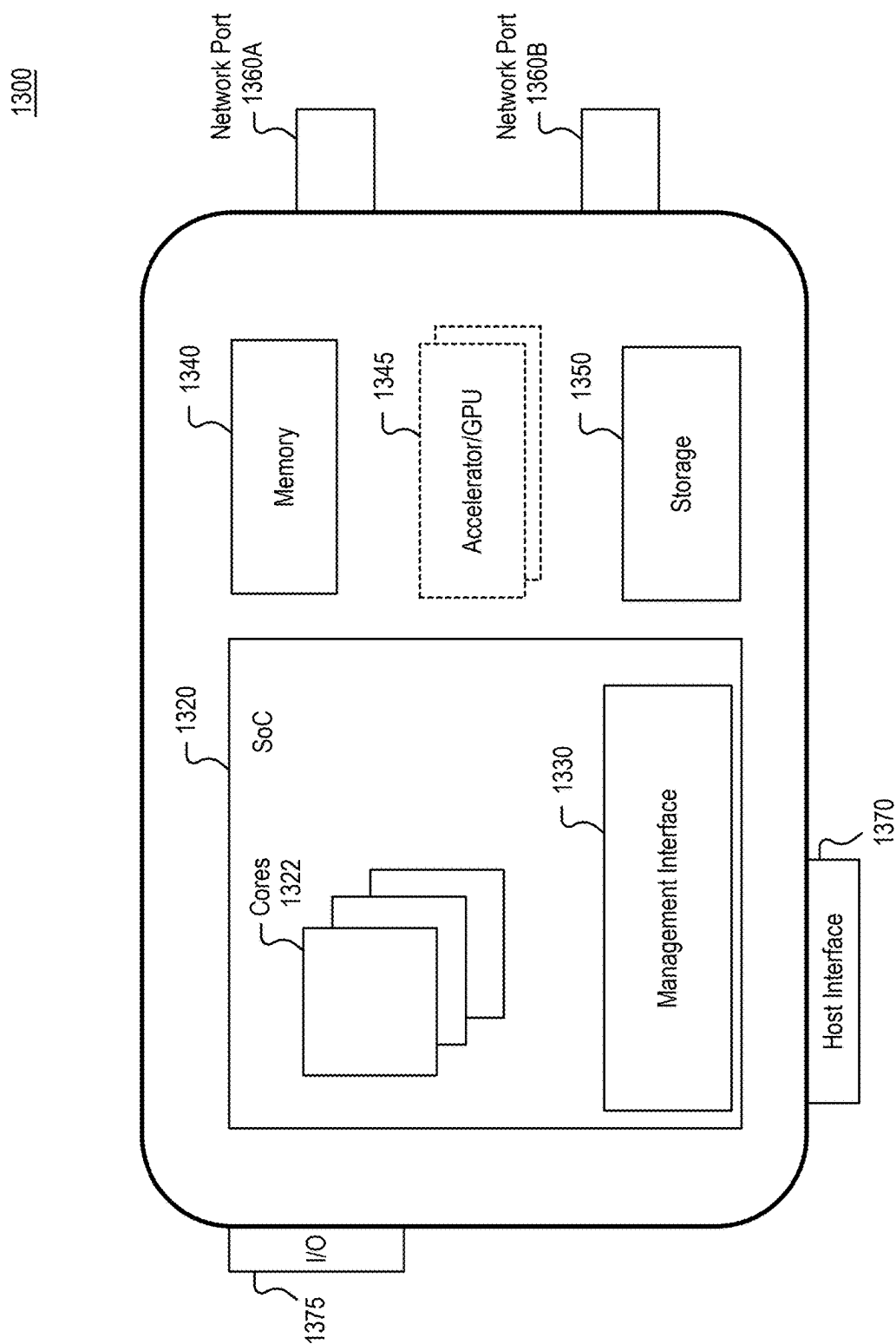
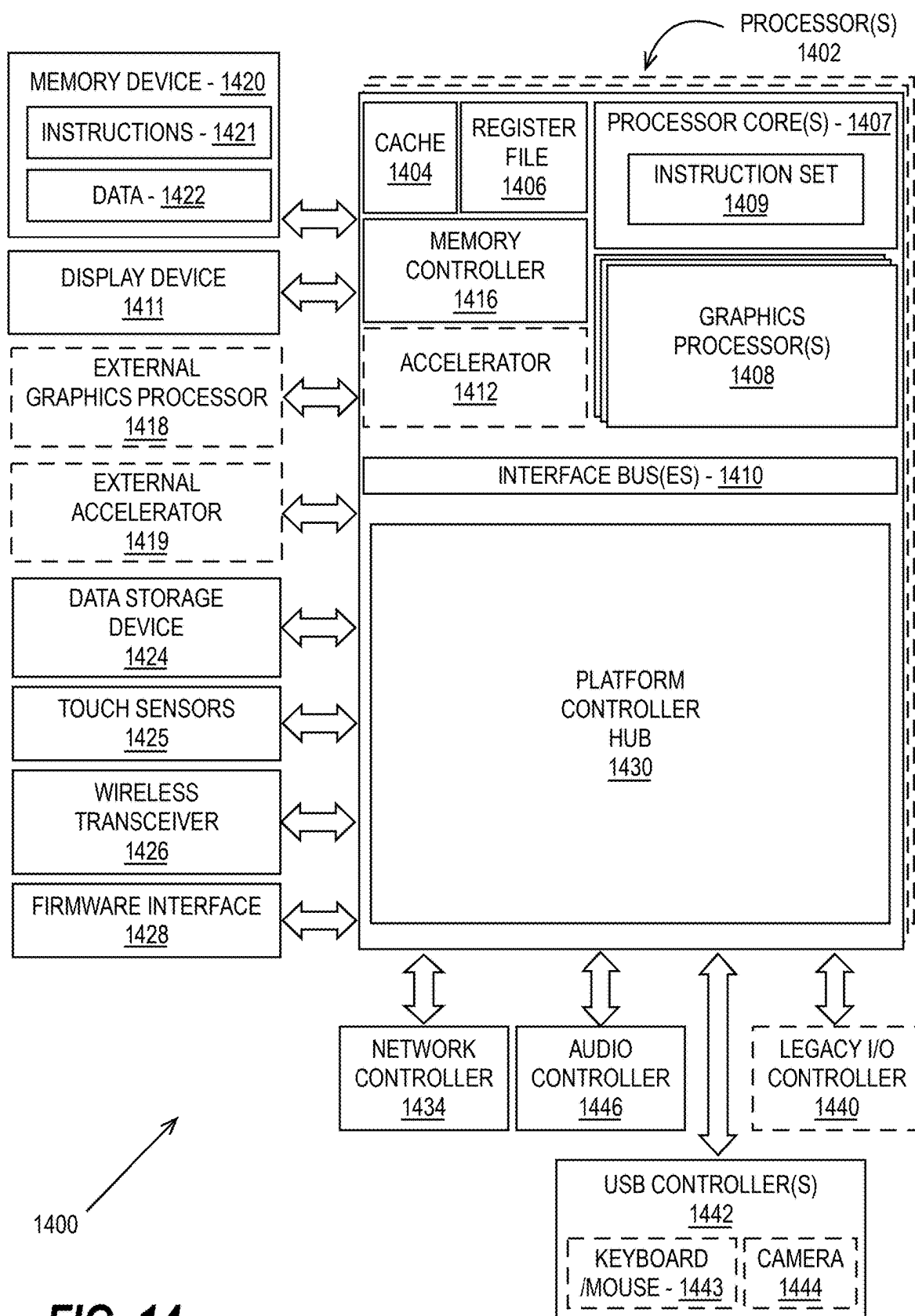


FIG. 13



**FIG. 14**

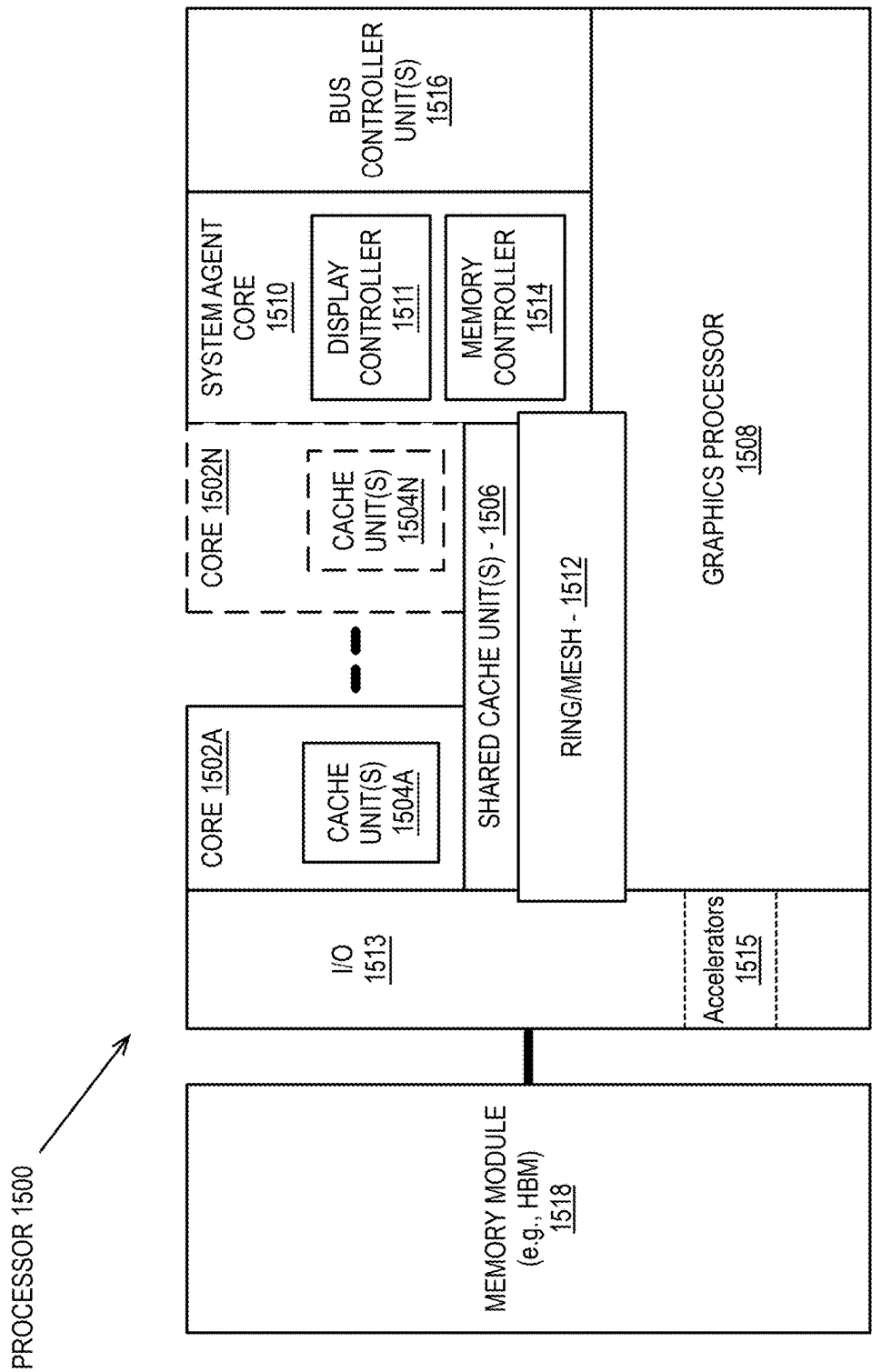


FIG. 15A

1519

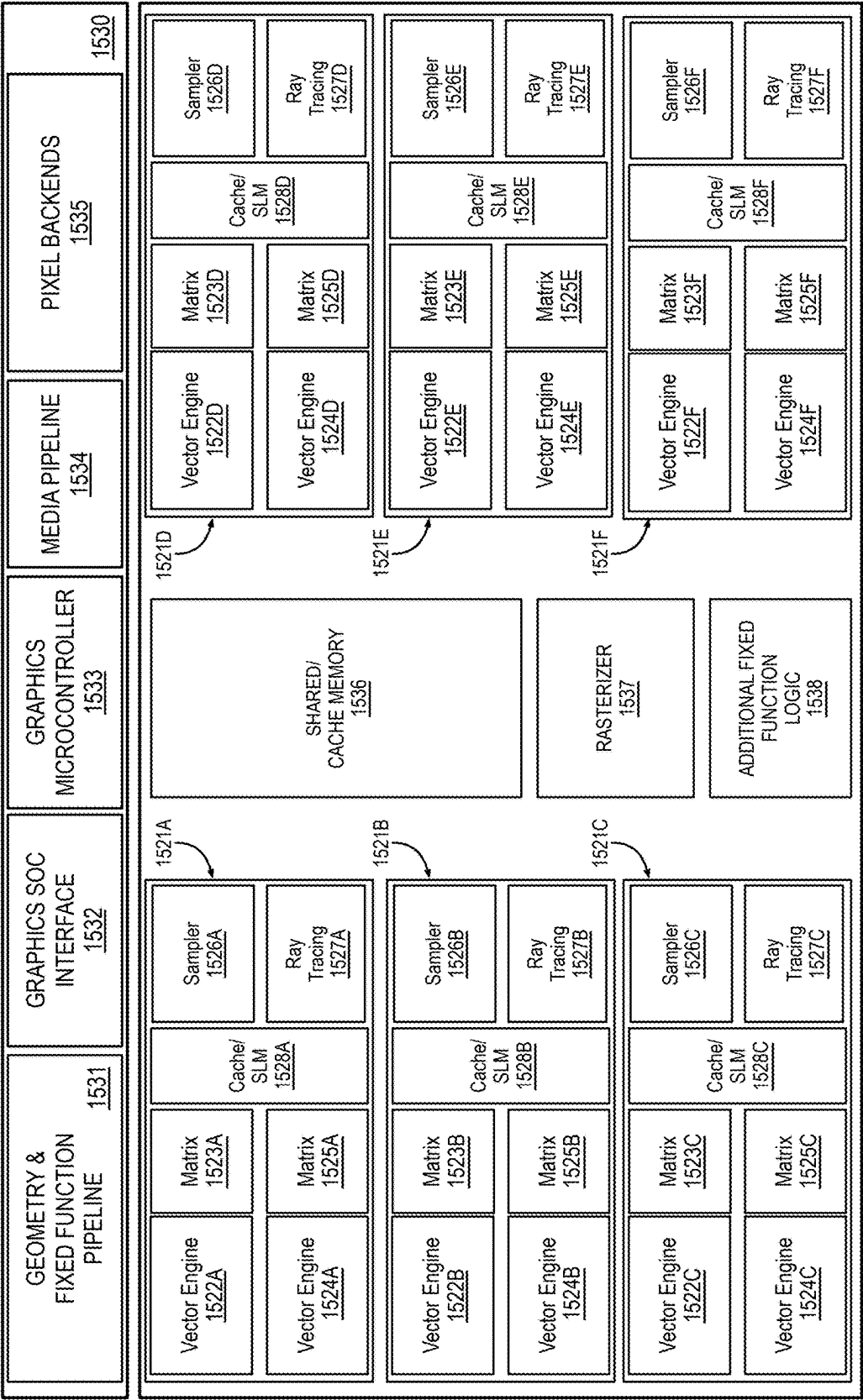
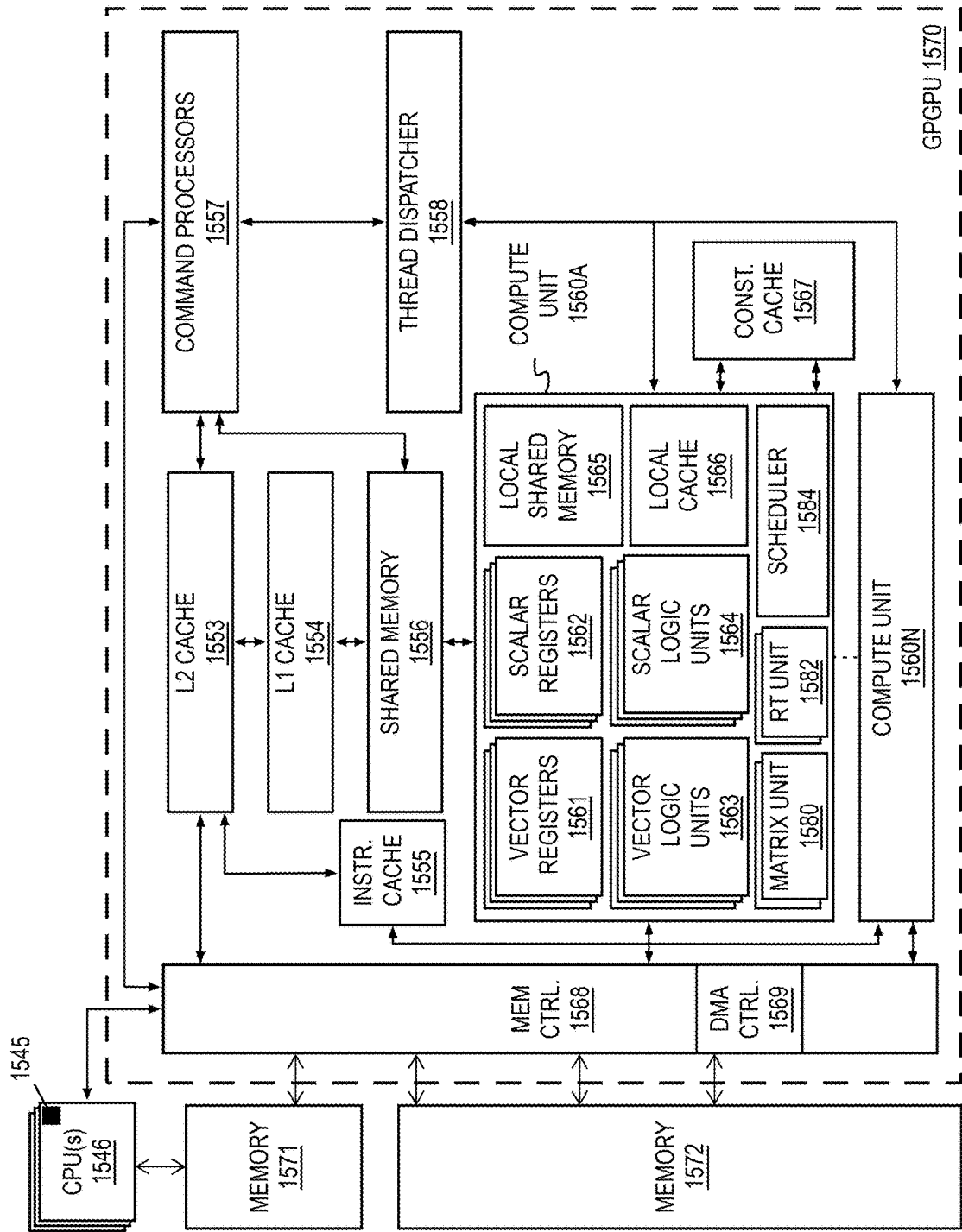
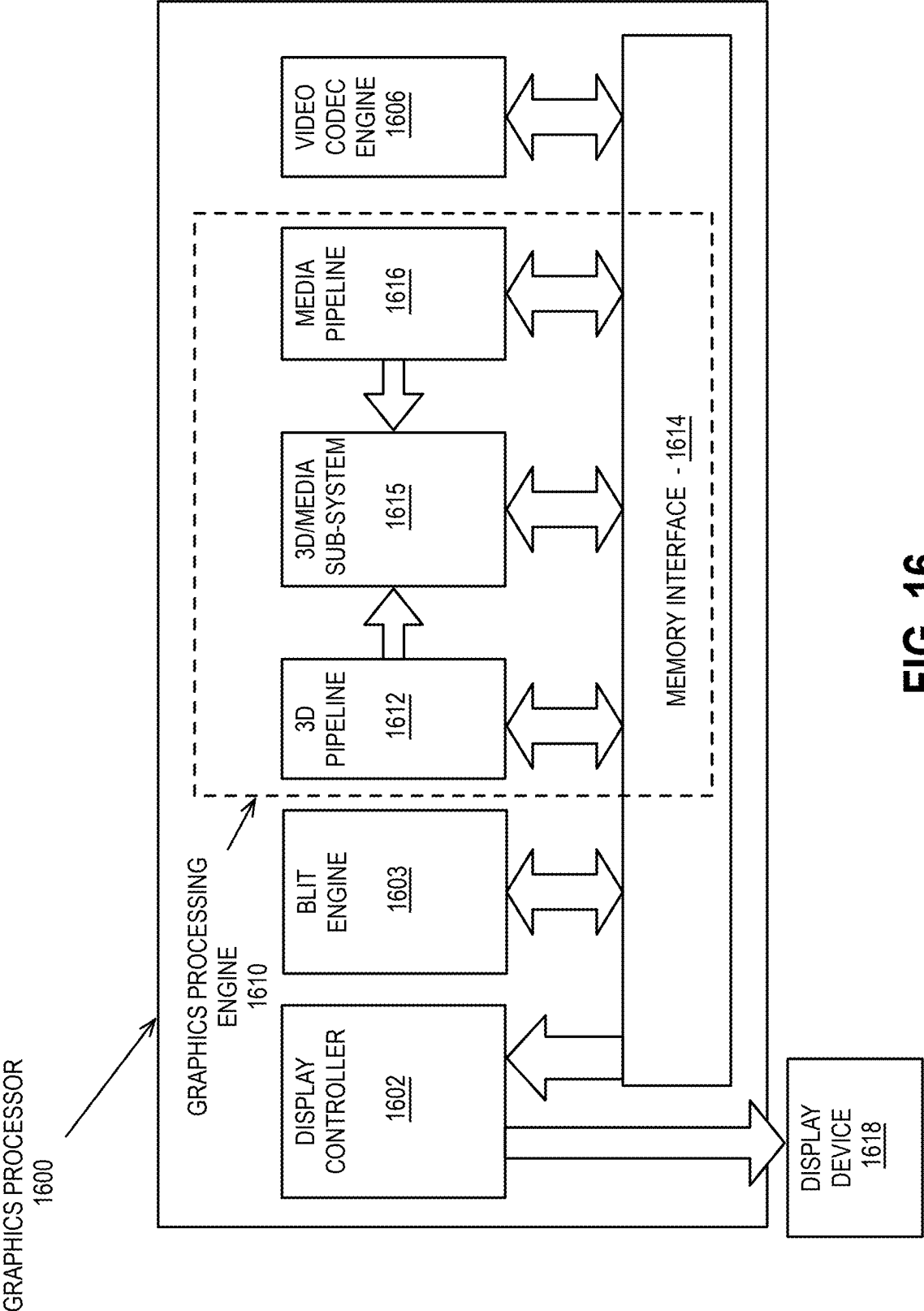


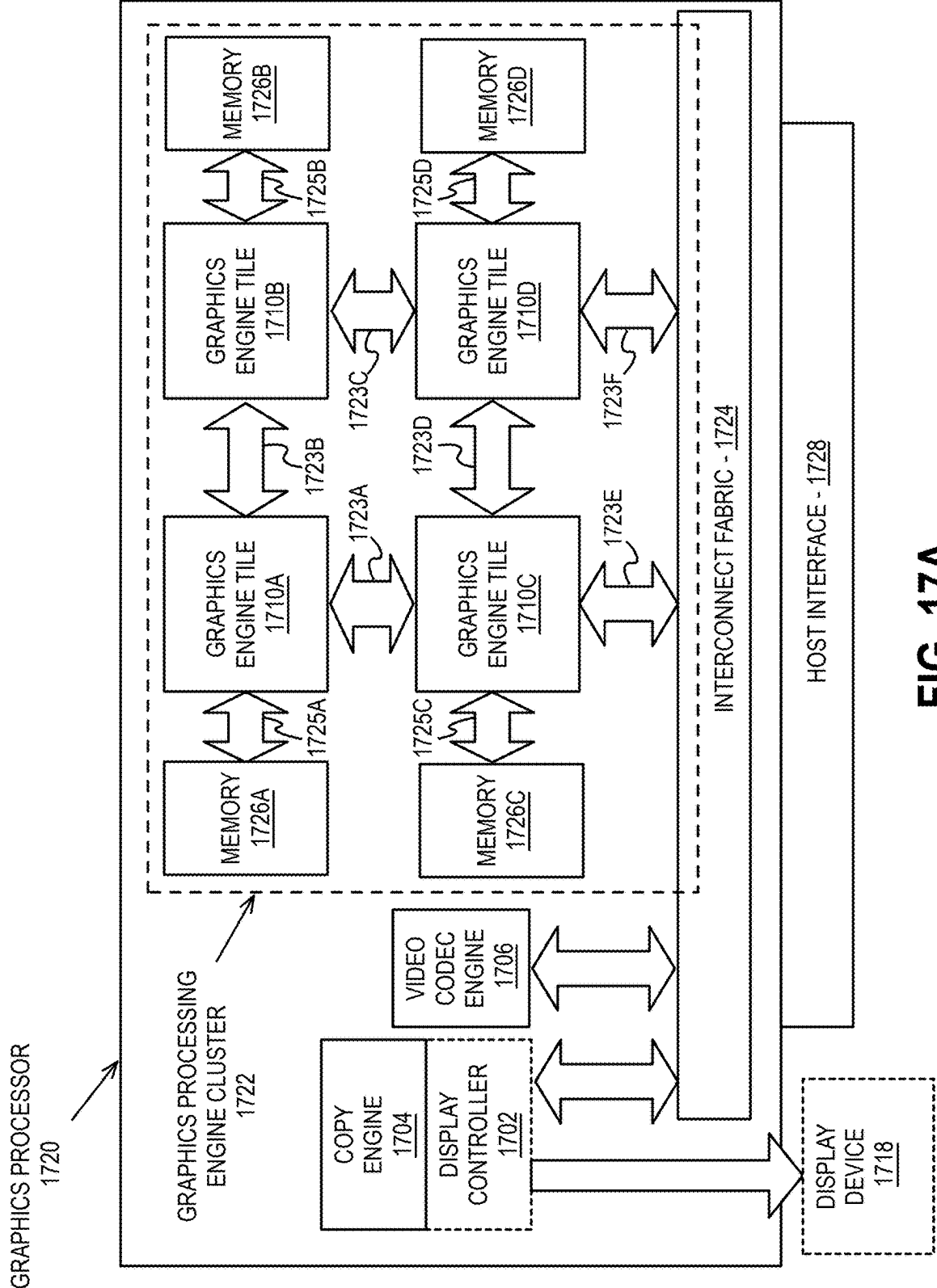
FIG. 15B



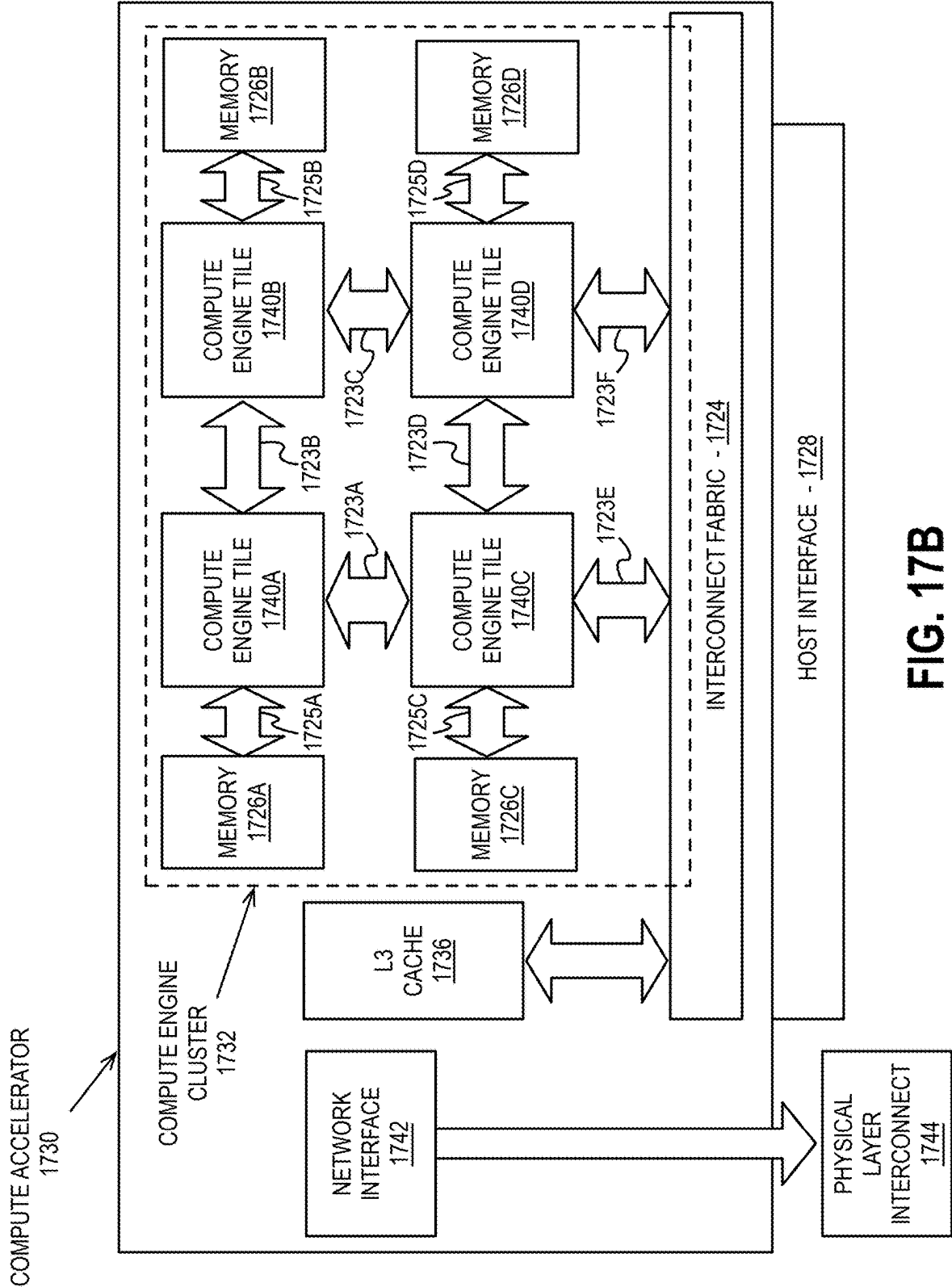
**FIG. 15C**



**FIG. 16**



**FIG. 17A**



**FIG. 17B**

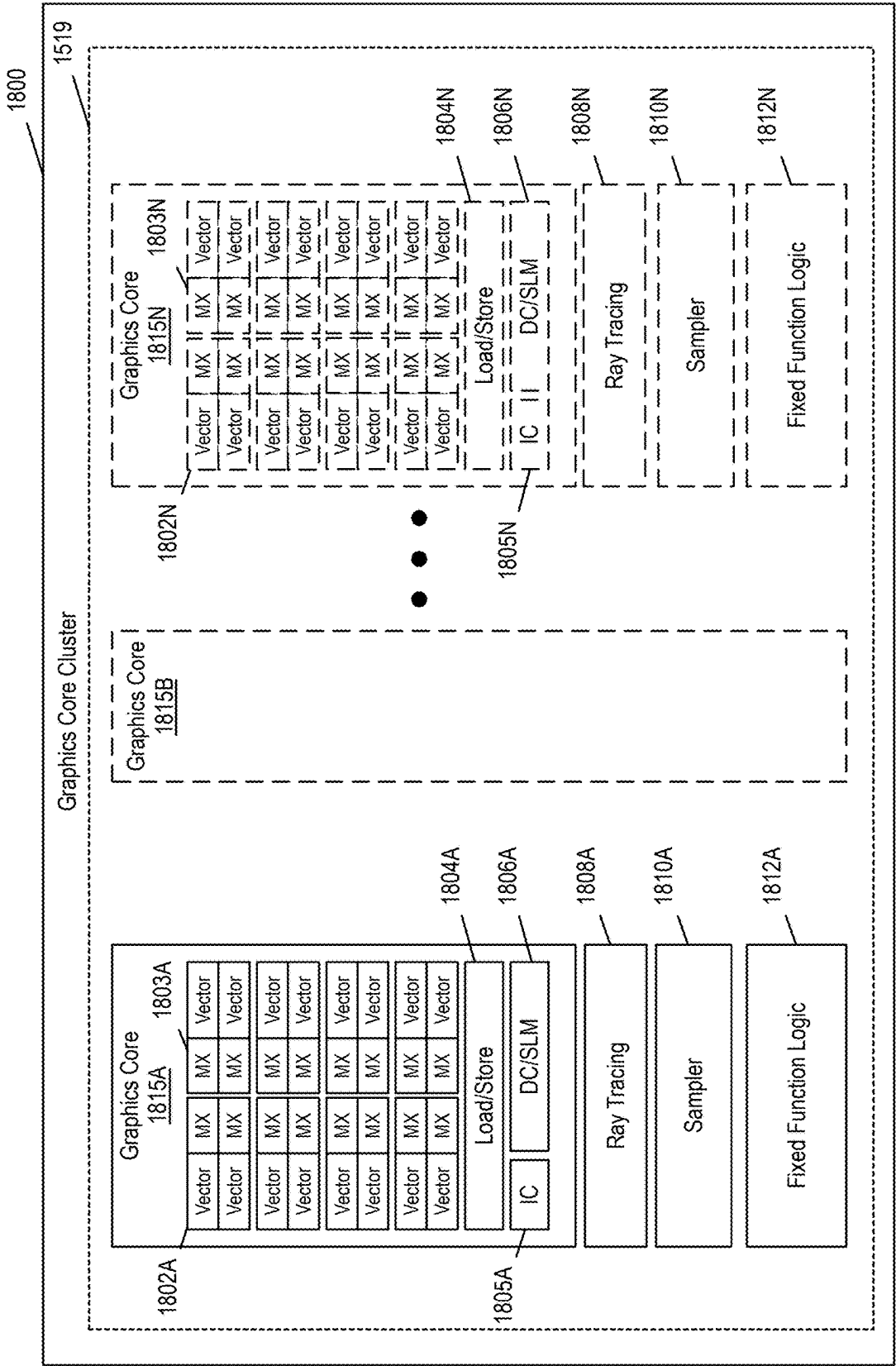
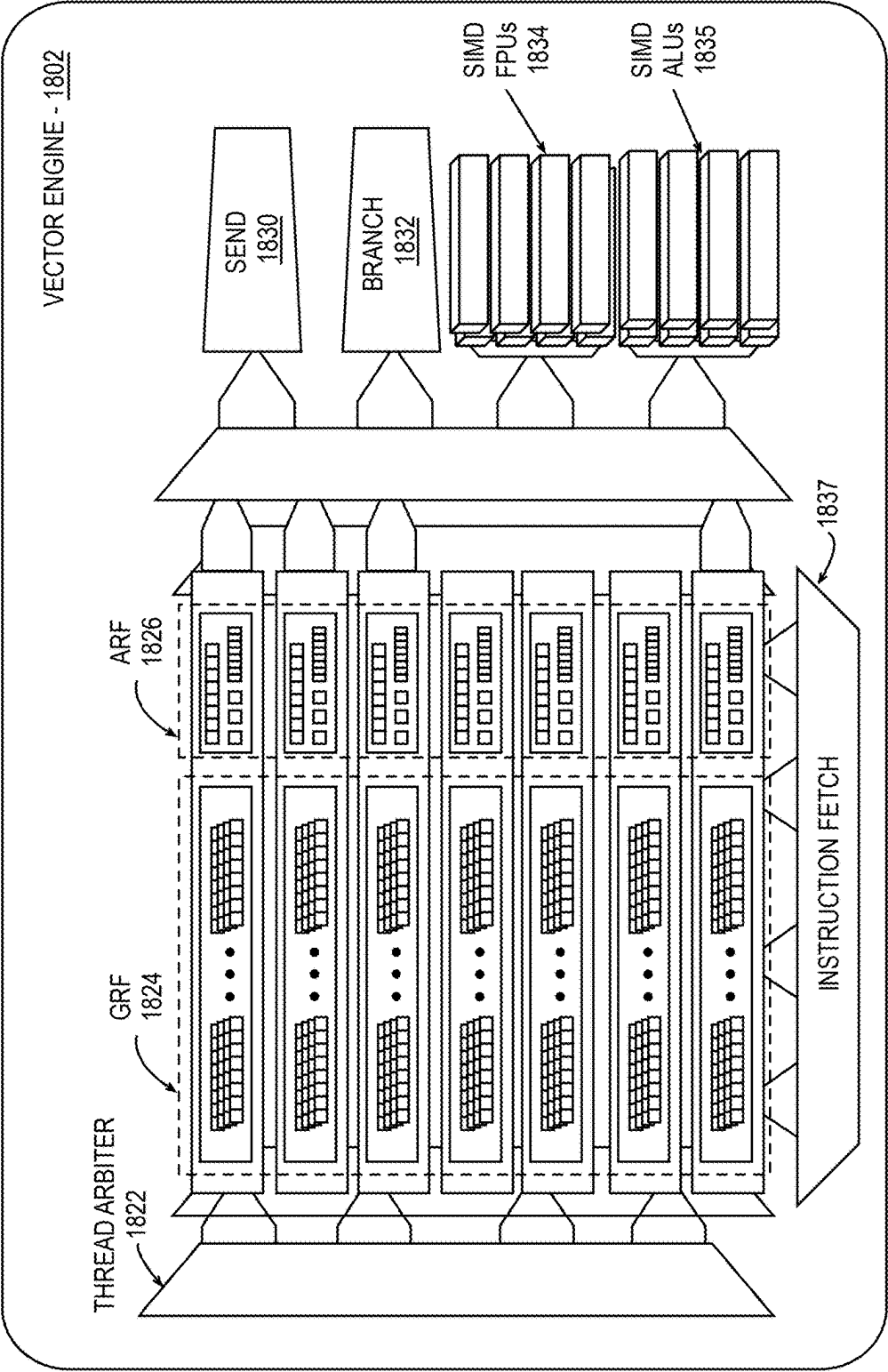


FIG. 18A



**FIG. 18B**

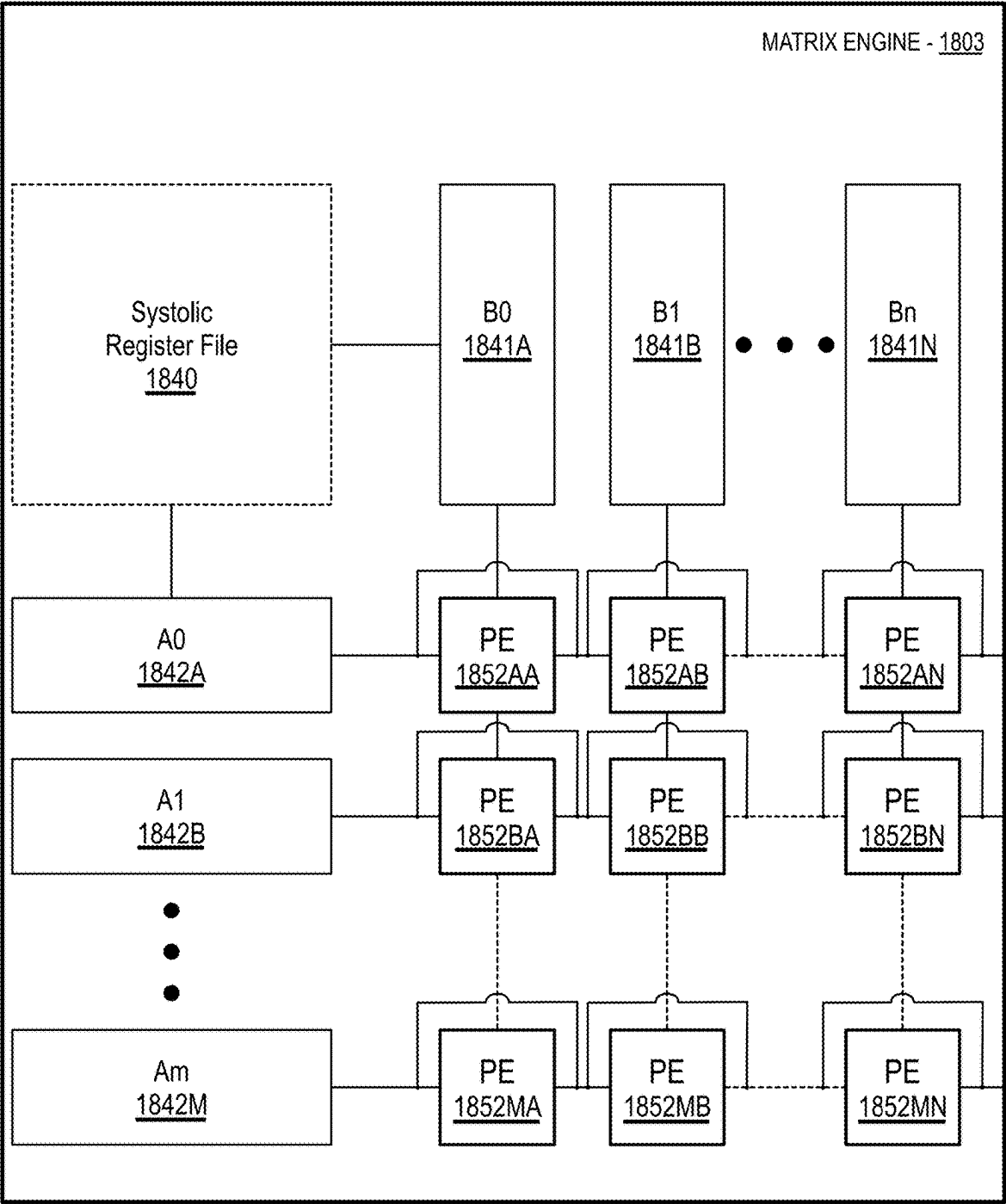


FIG. 18C

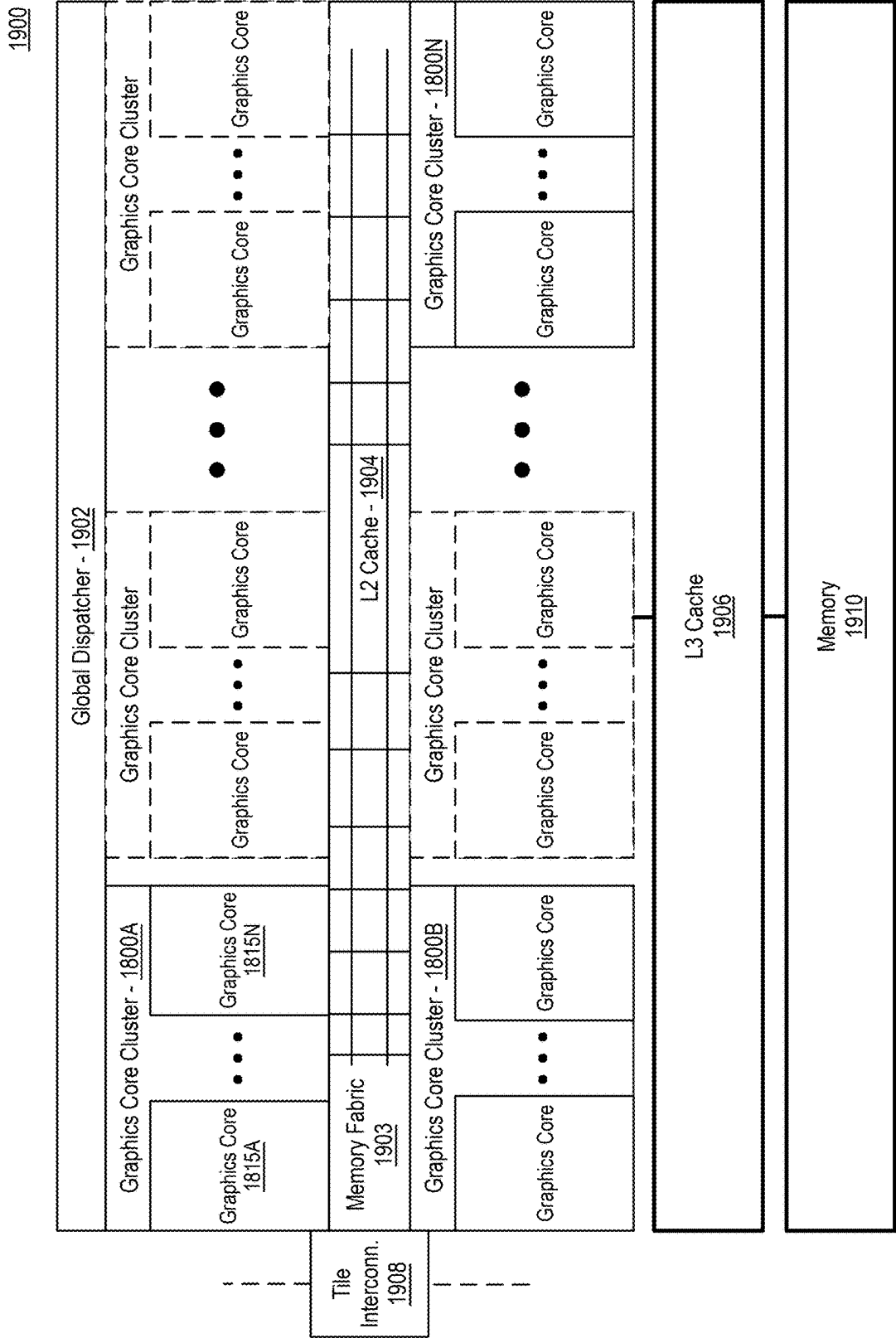


FIG. 19

GRAPHICS PROCESSOR INSTRUCTION FORMATS

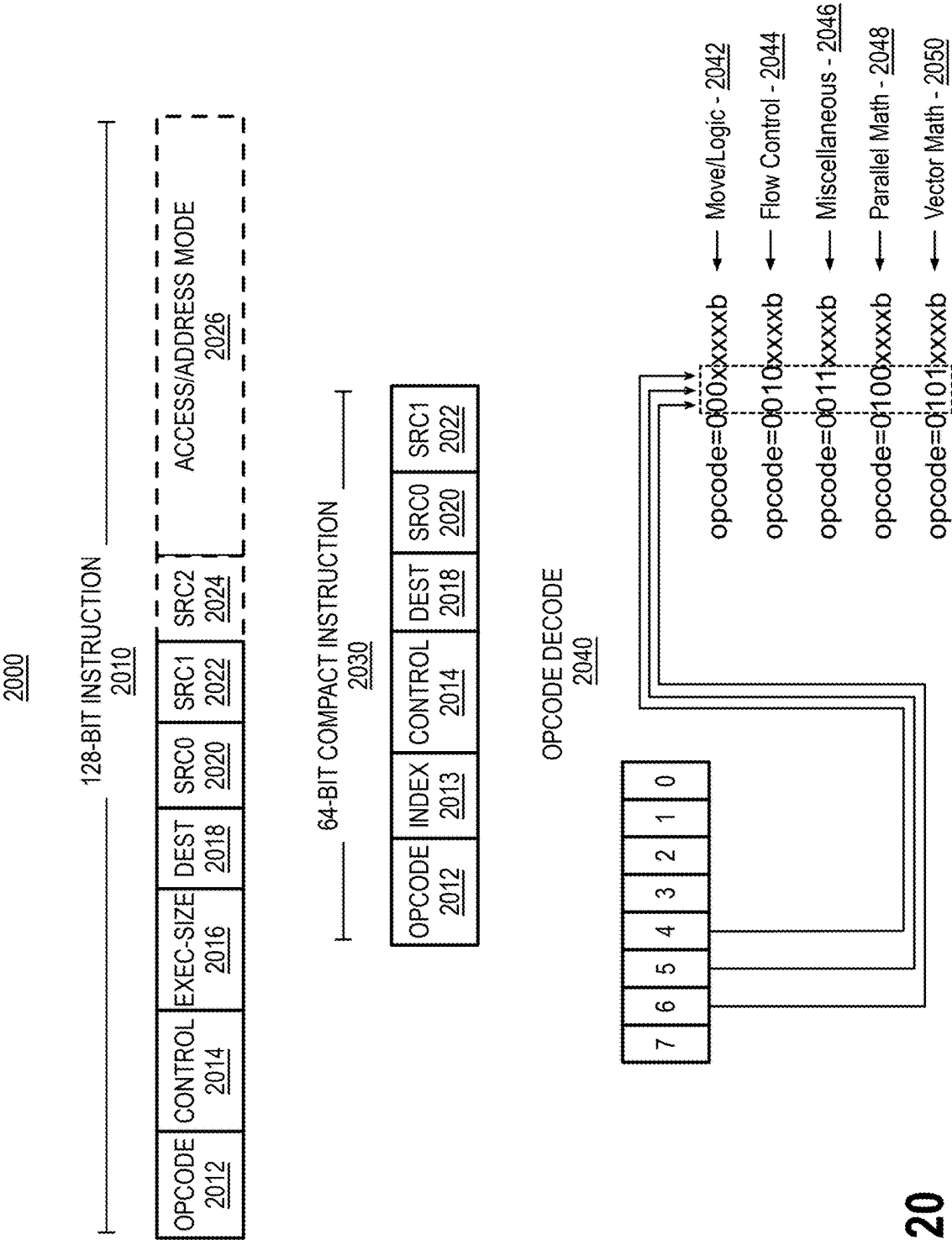


FIG. 20

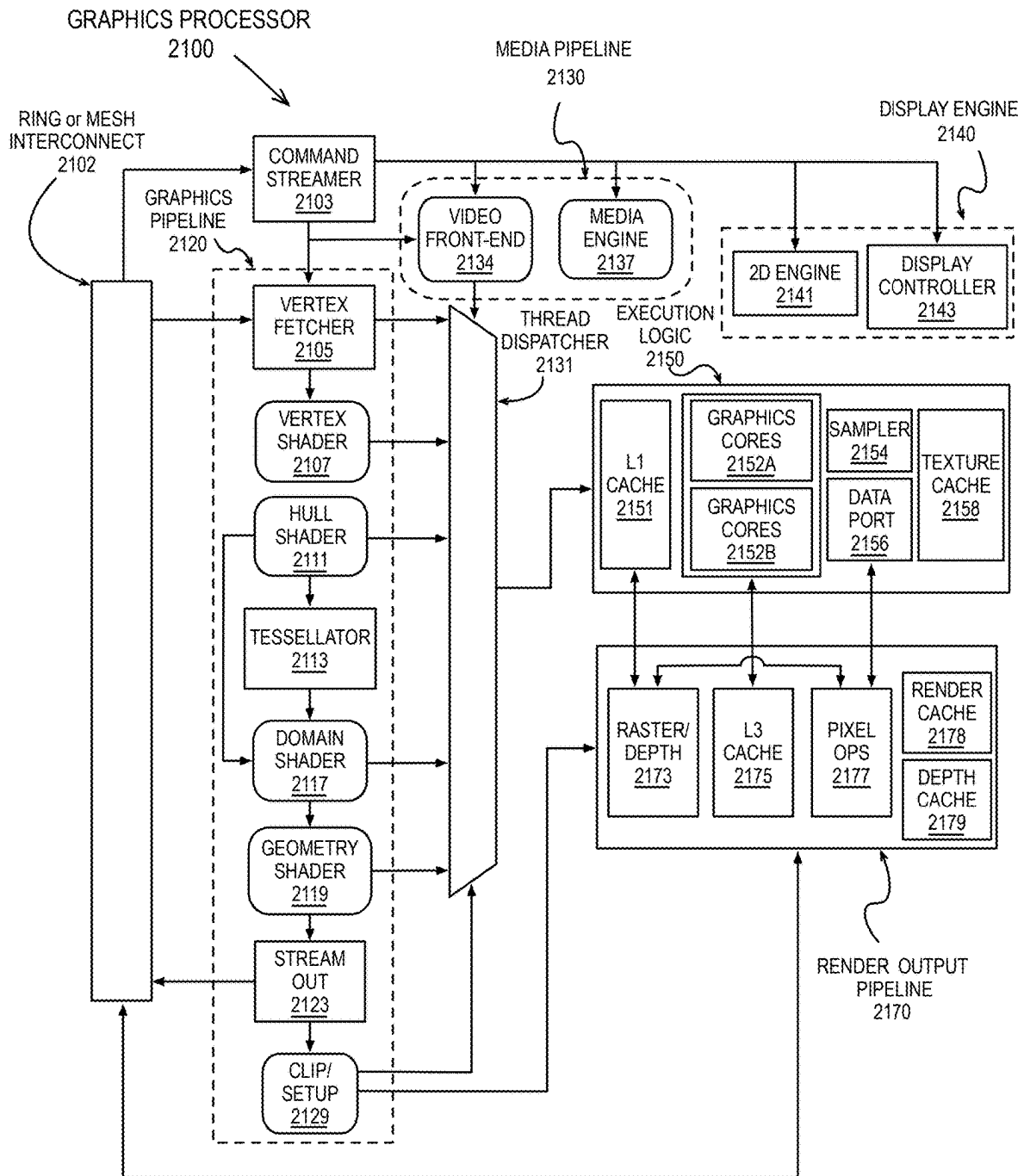
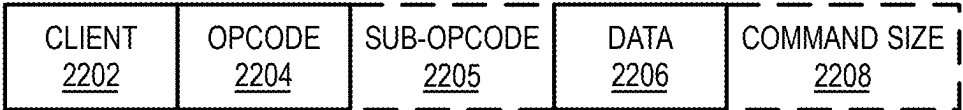
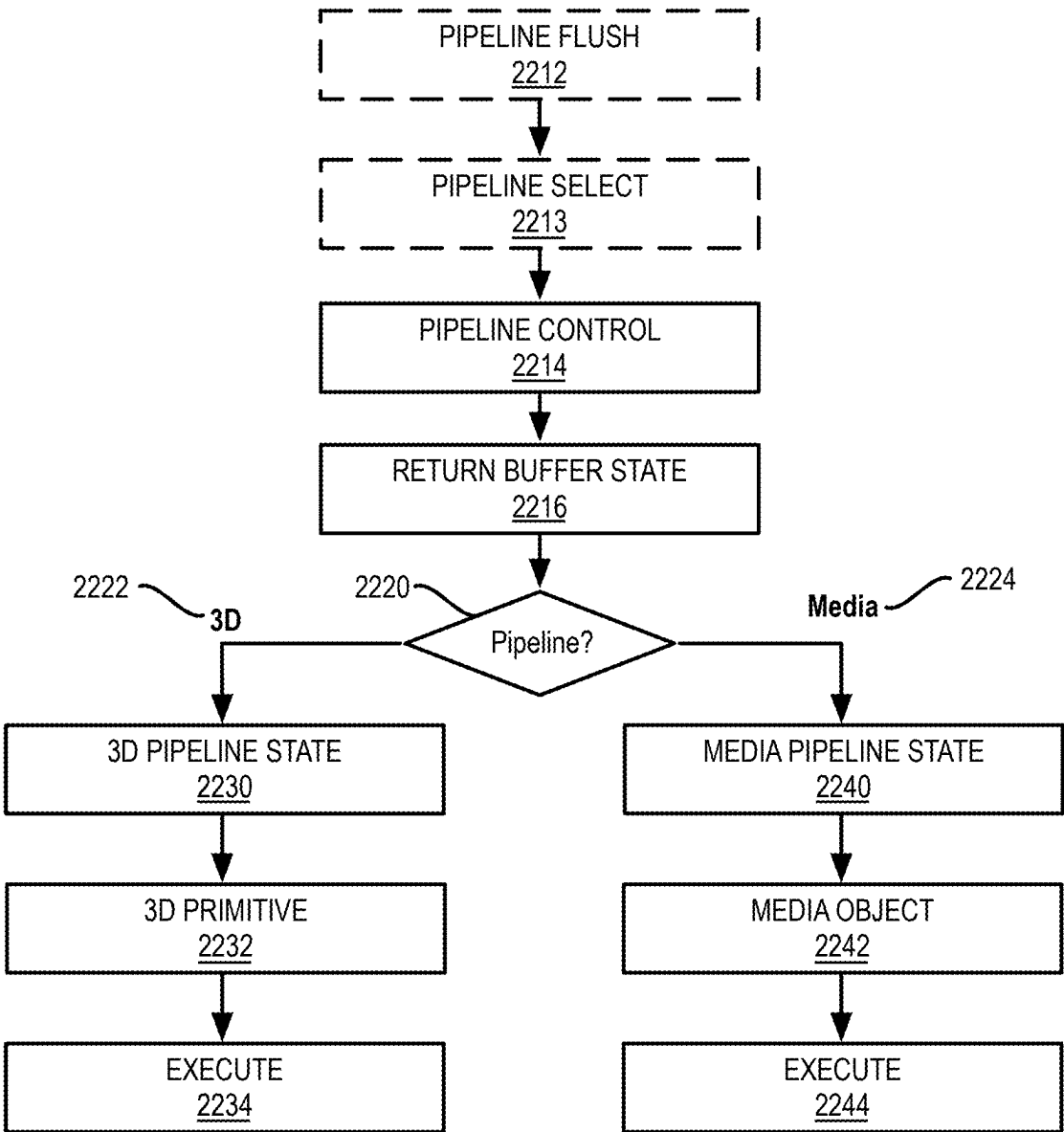


FIG. 21

**FIG. 22A**      GRAPHICS PROCESSOR COMMAND FORMAT  
2200



**FIG. 22B**      GRAPHICS PROCESSOR COMMAND SEQUENCE  
2210



DATA PROCESSING SYSTEM - 2300

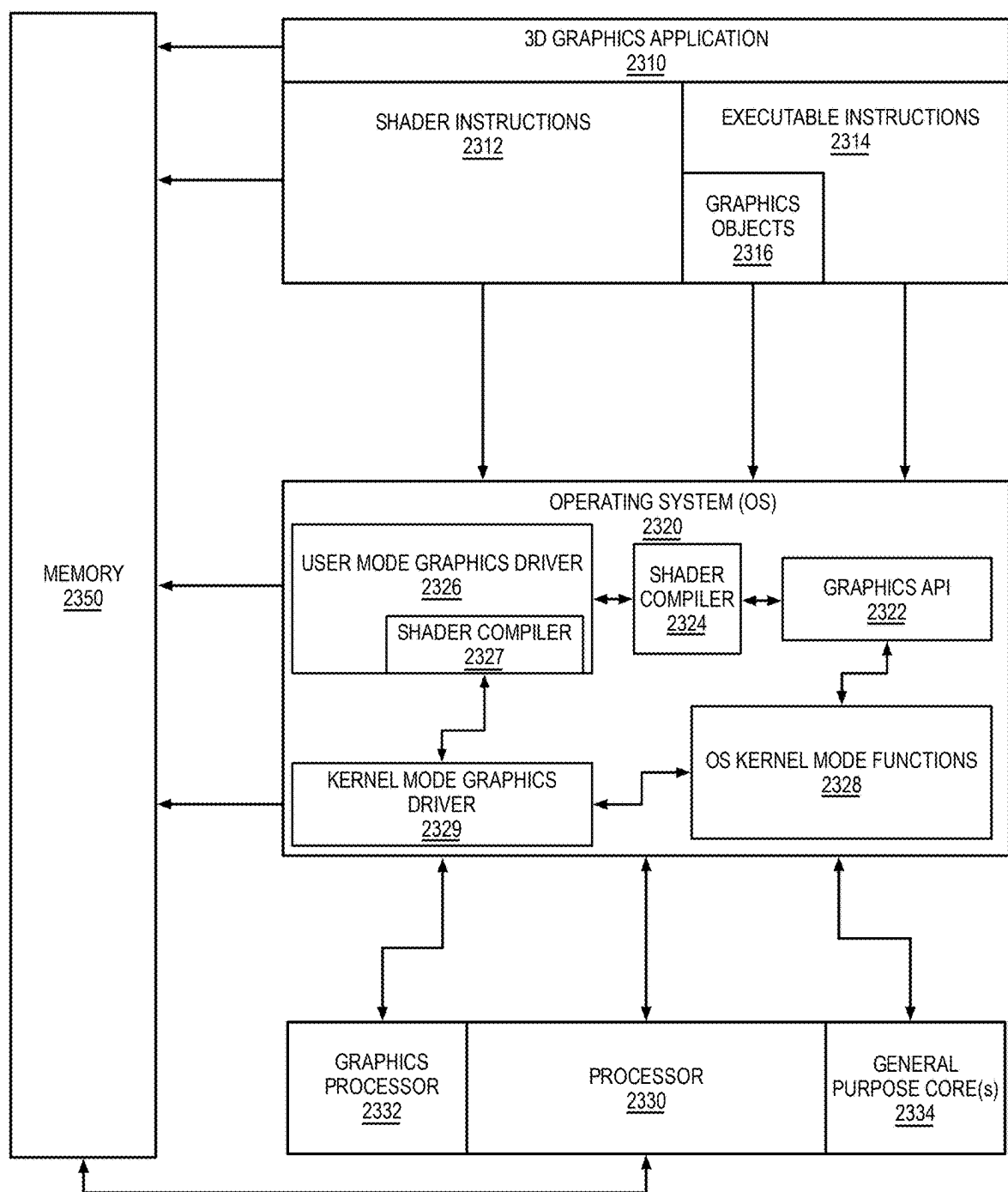


FIG. 23

IP CORE DEVELOPMENT - 2400

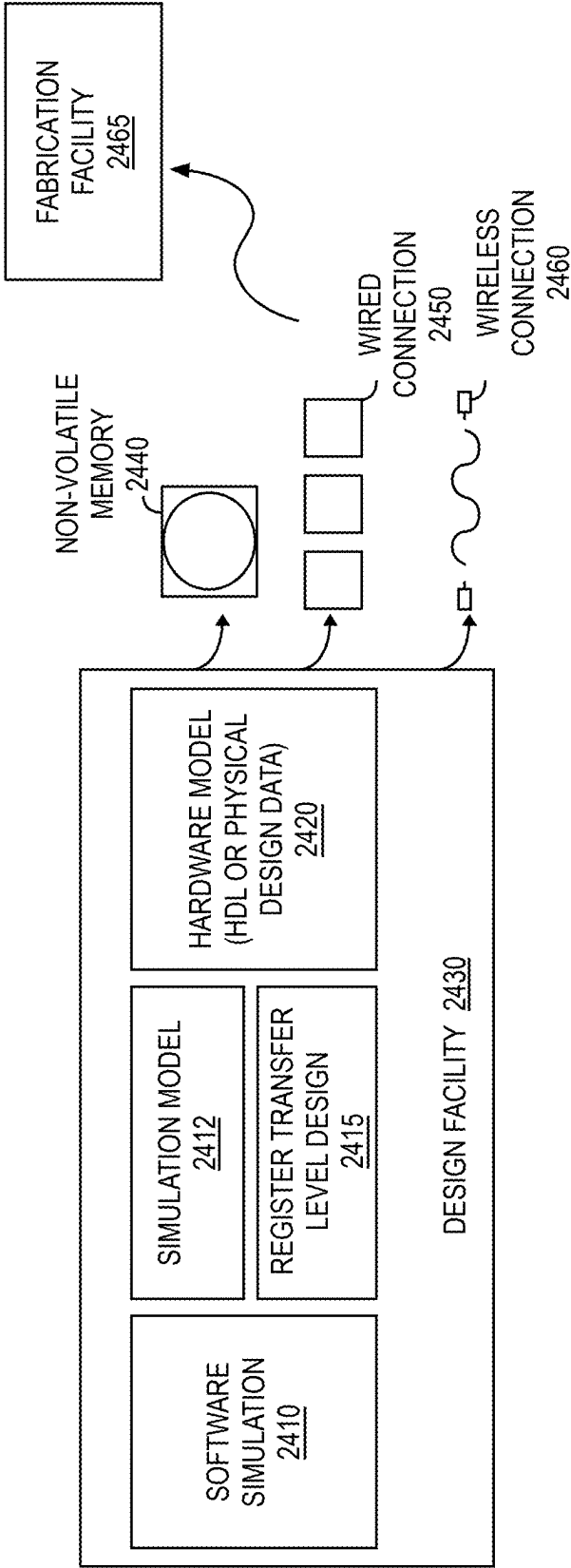


FIG. 24

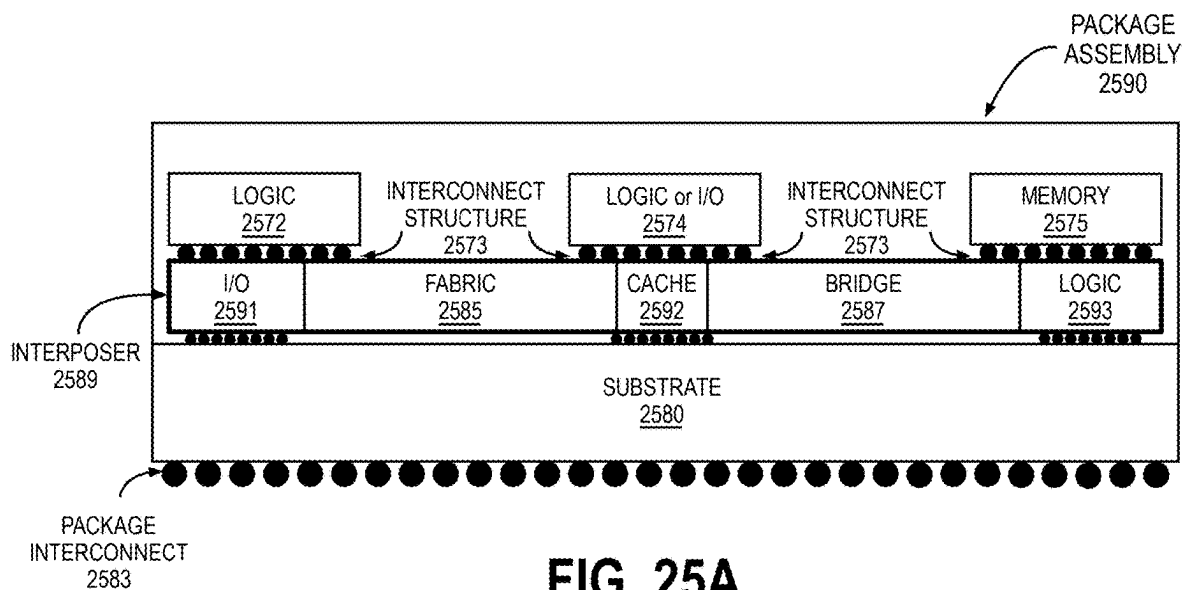


FIG. 25A

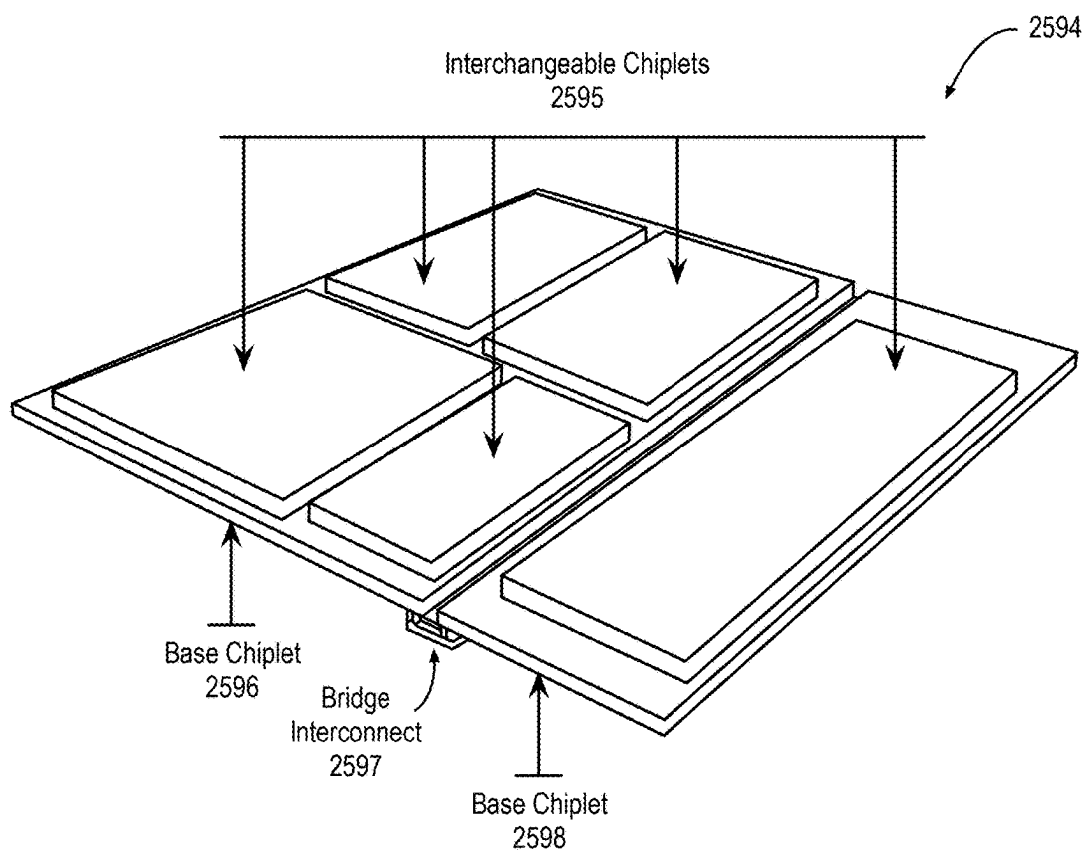


FIG. 25B

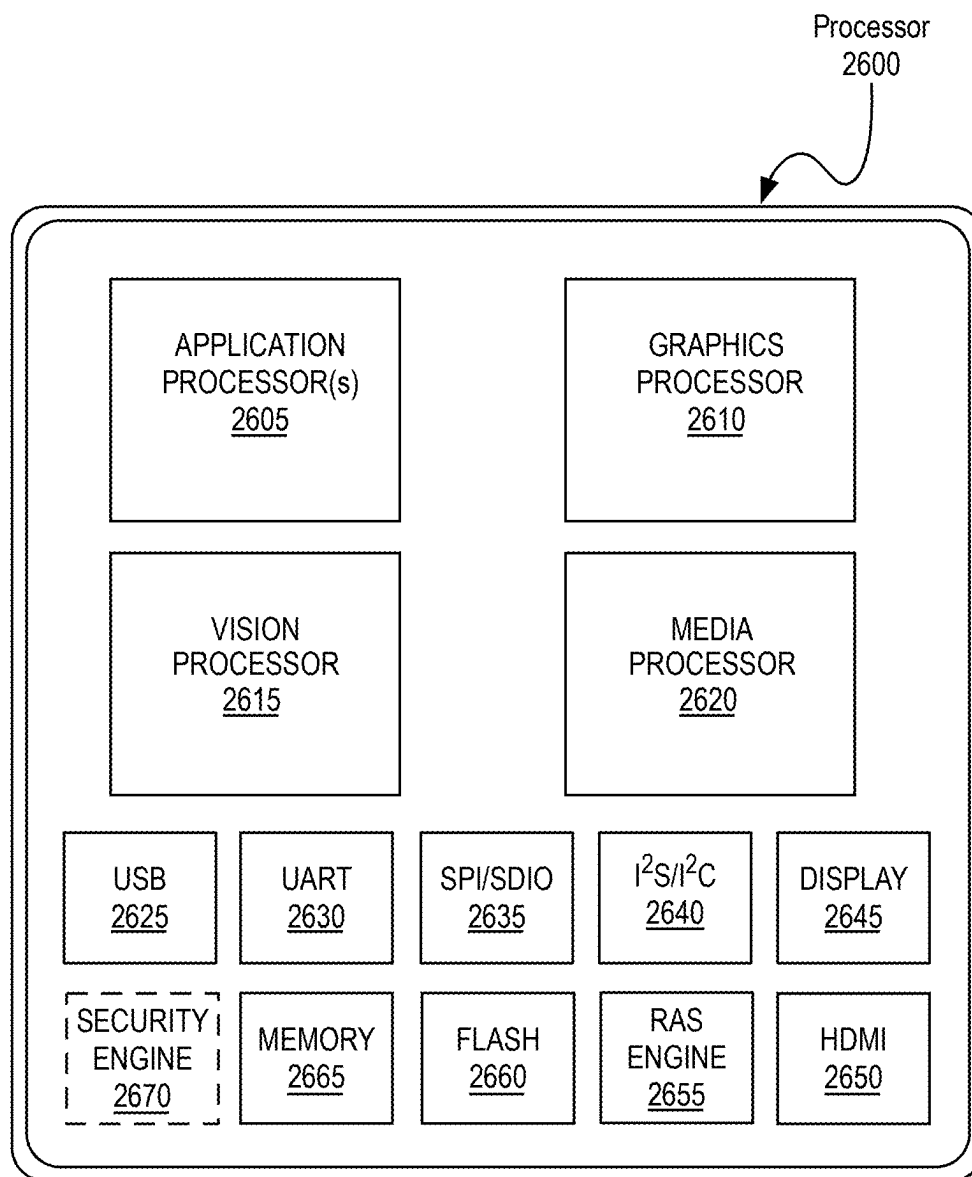


FIG. 26

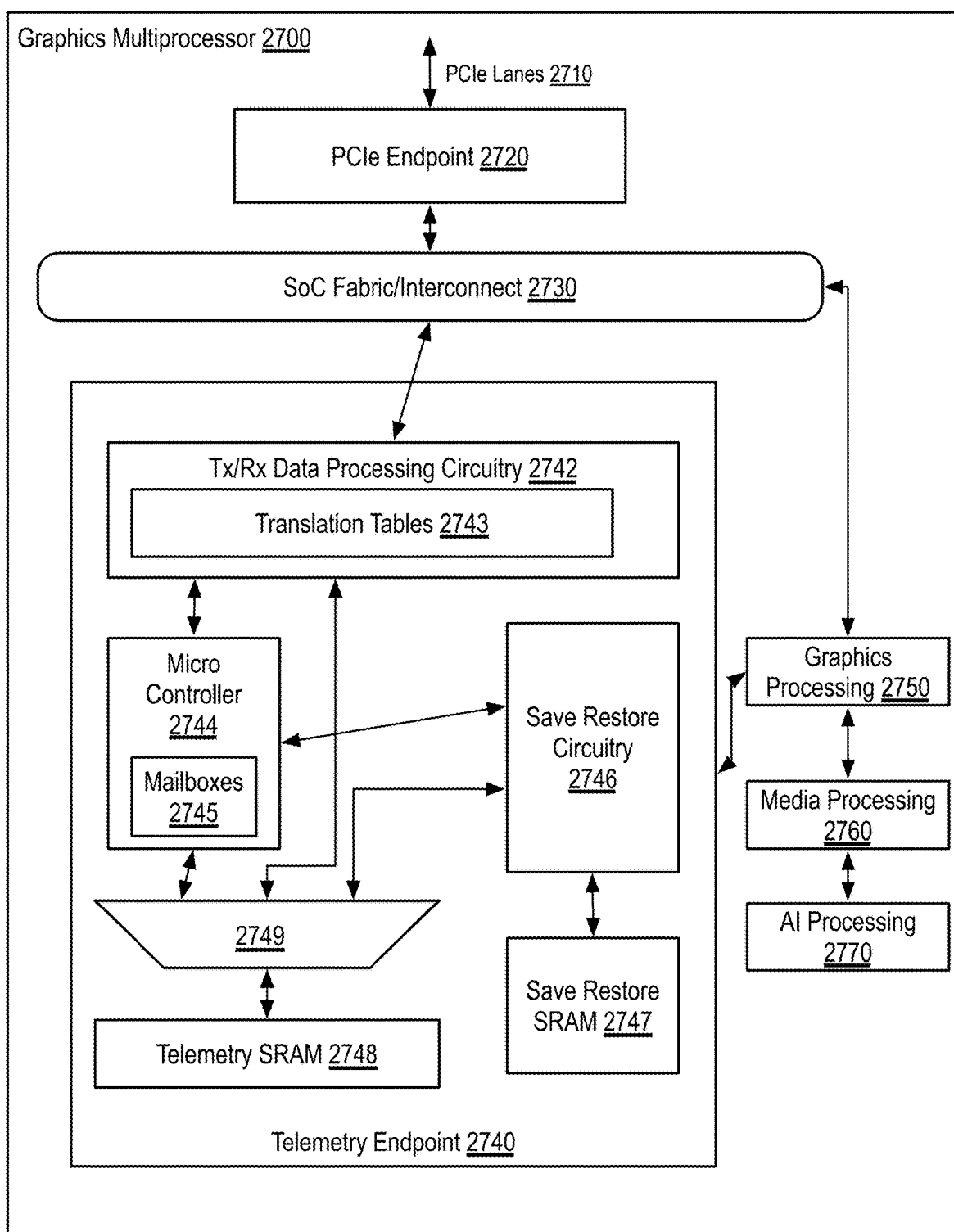


FIG. 27

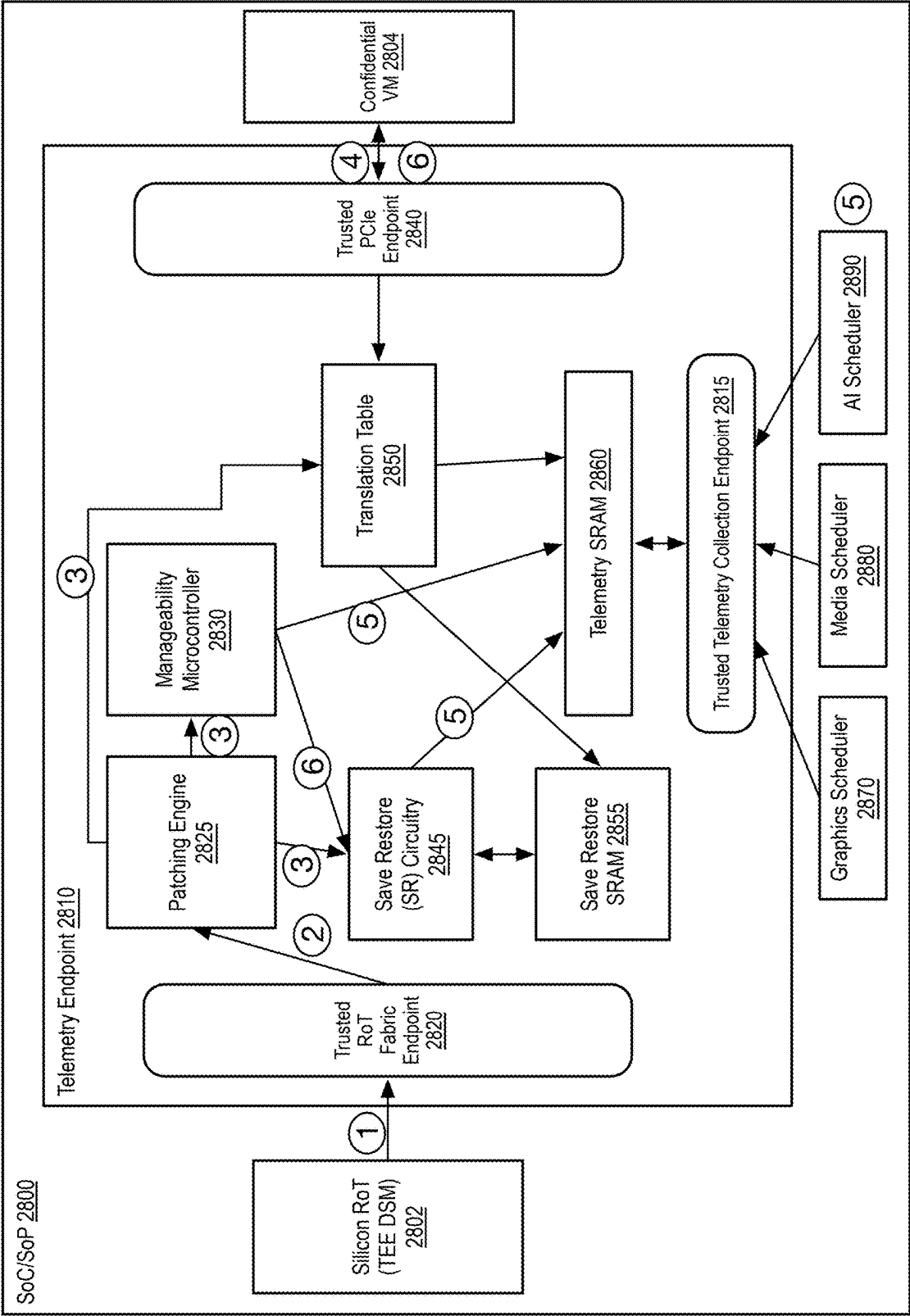


FIG. 28

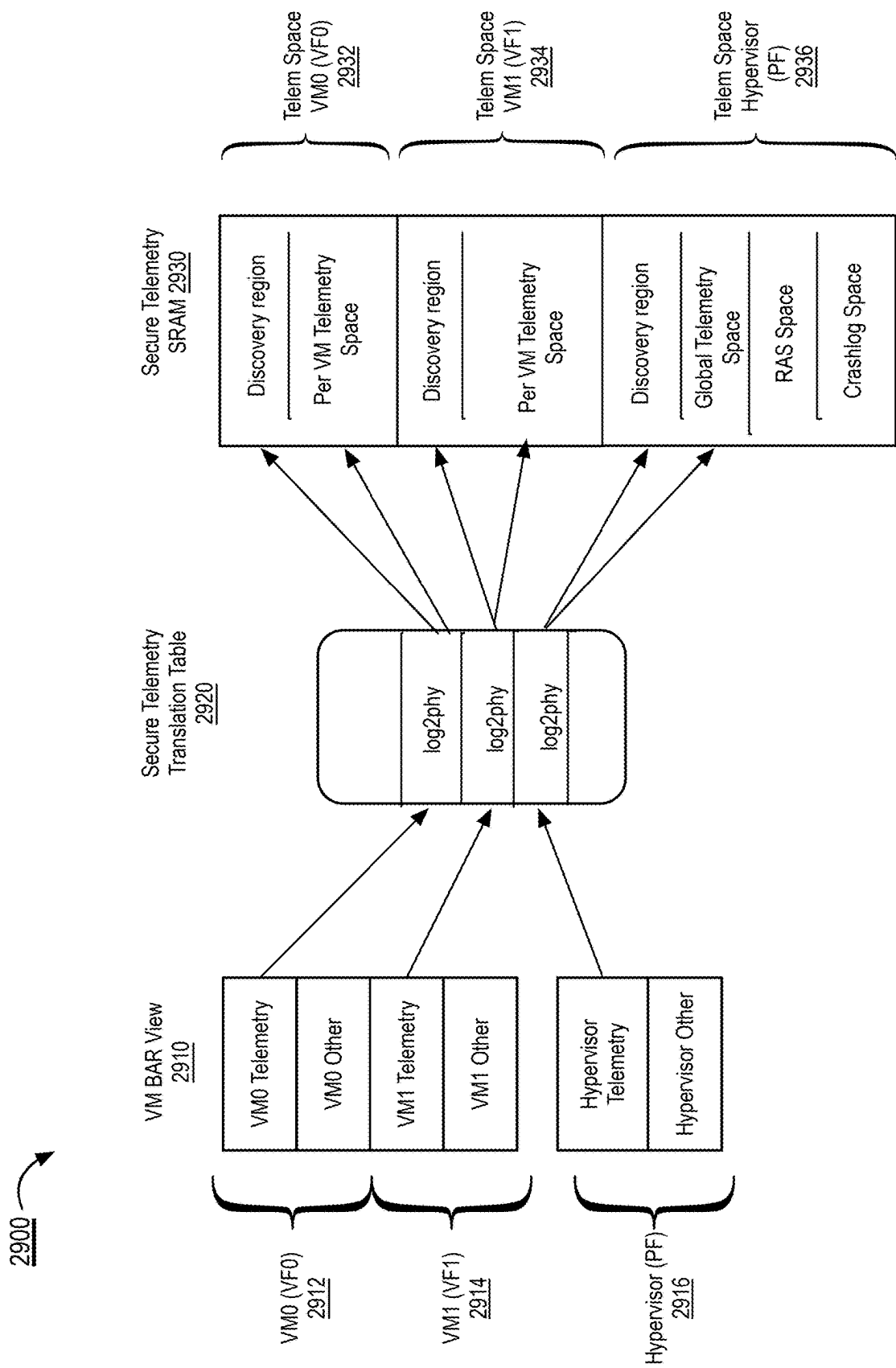


FIG. 29

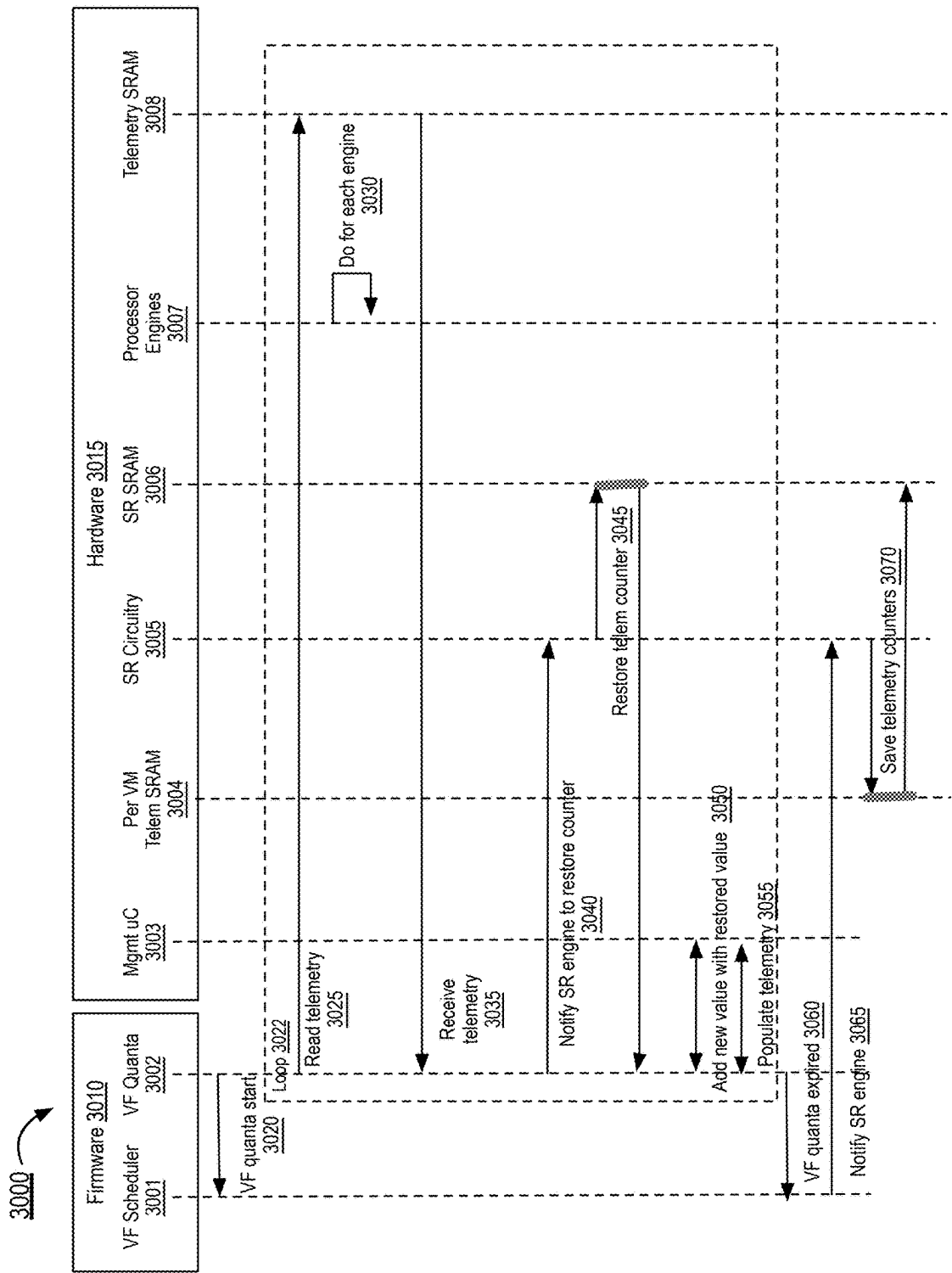
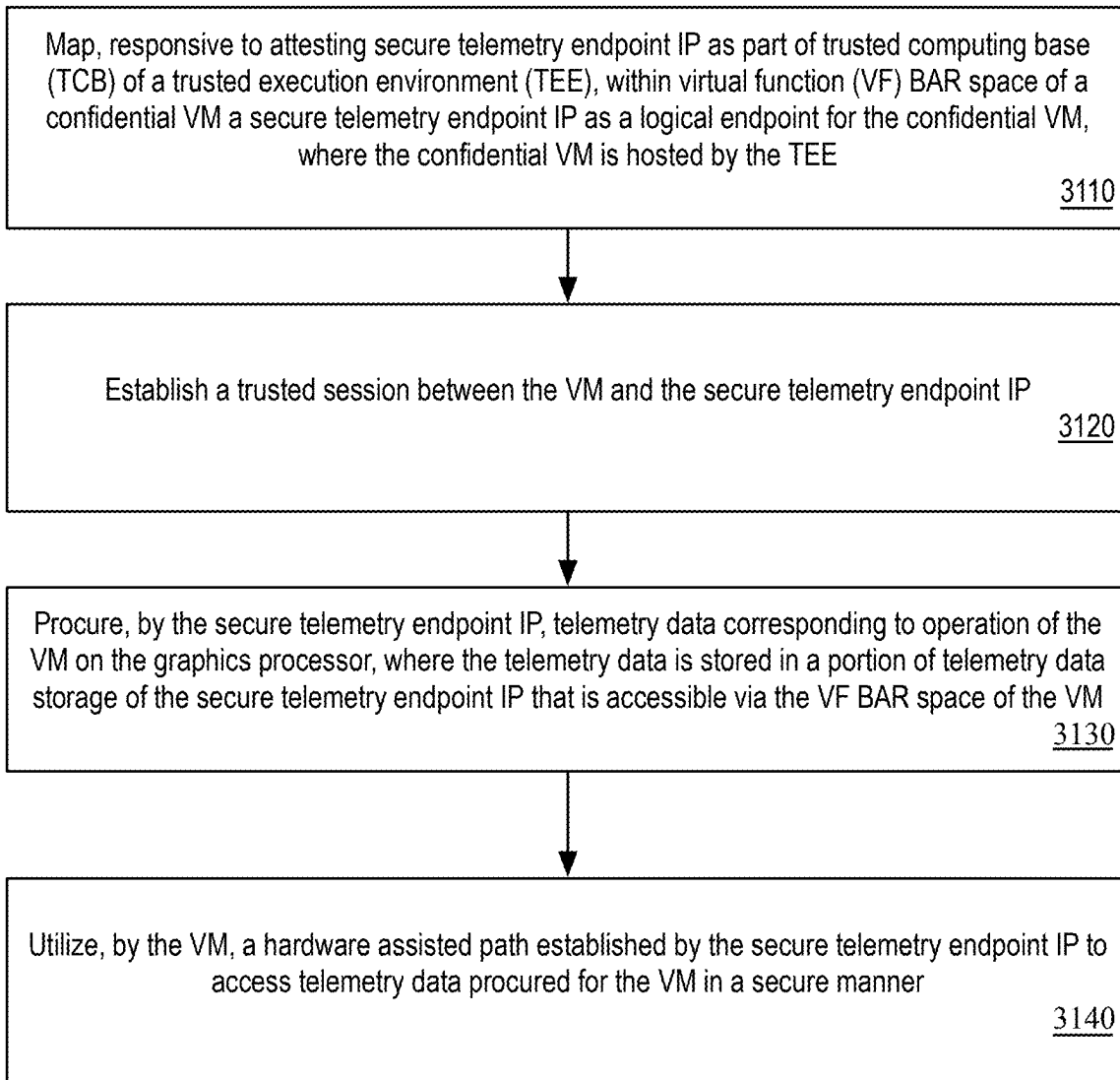


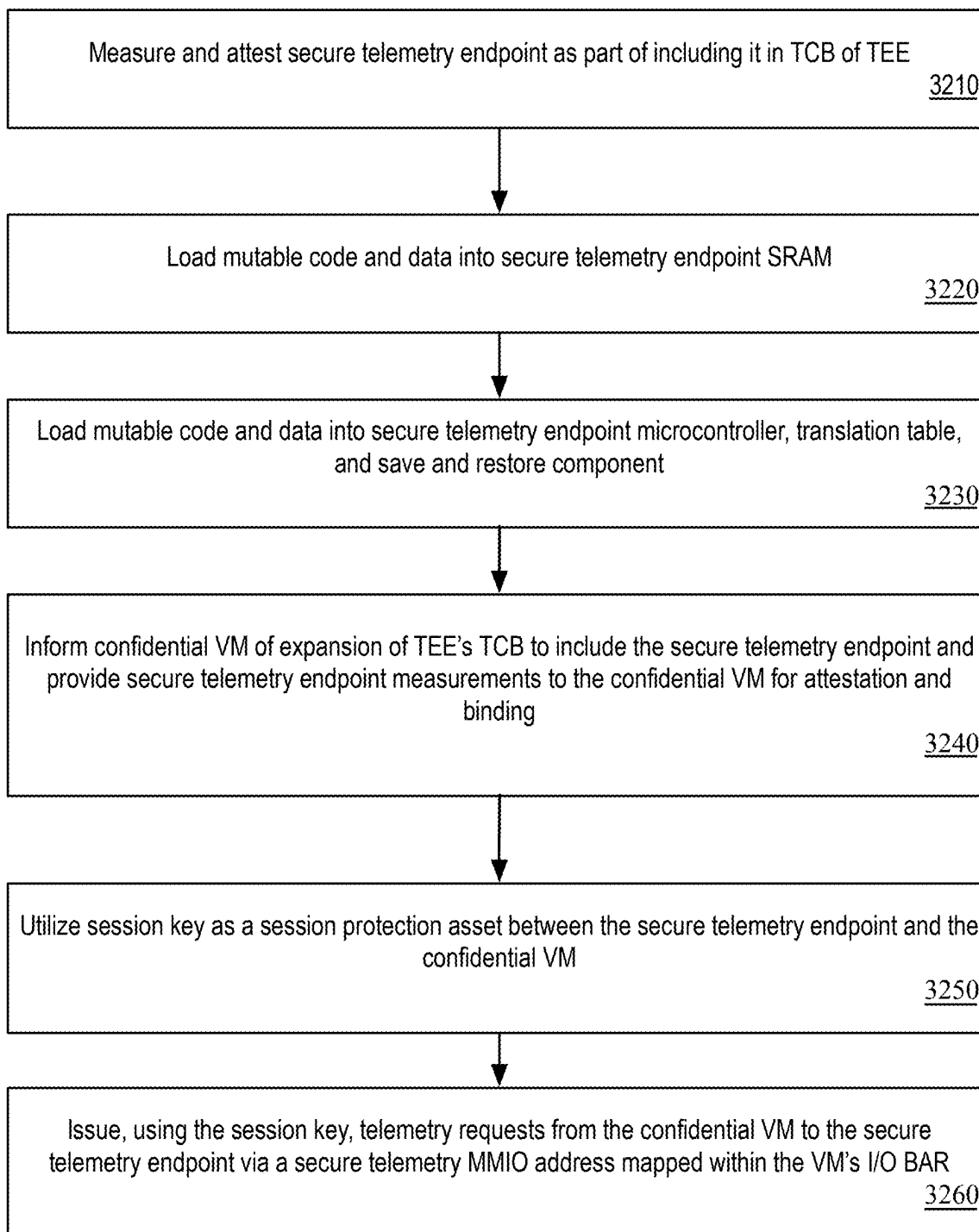
FIG. 30

3100



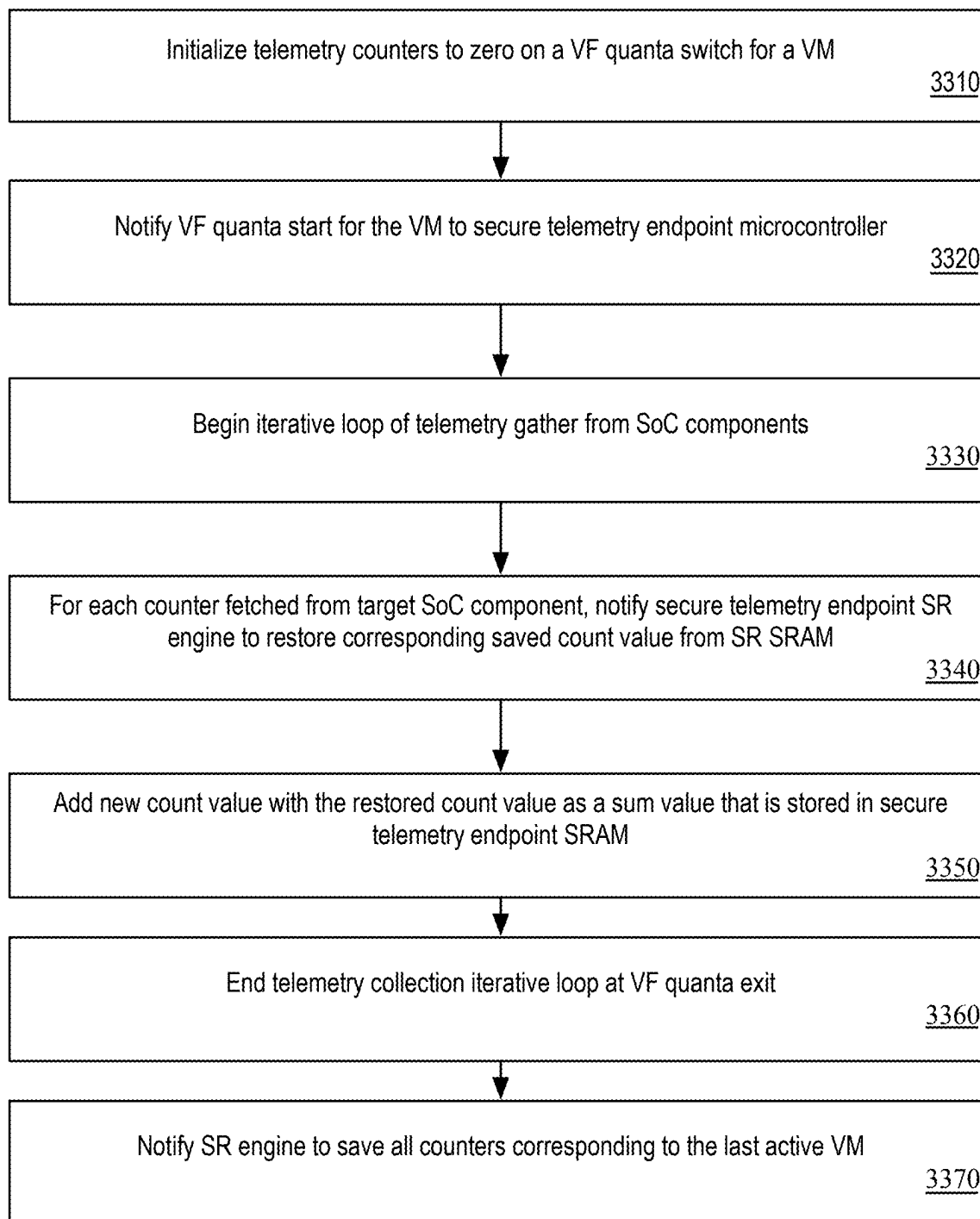
**FIG. 31**

3200



**FIG. 32**

3300



**FIG. 33**

## SECURE VIRTUALIZED TELEMETRY WITH SAVE AND RESTORE OVER GLOBAL COUNTERS

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a non-provisional application of U.S. Provisional Patent Application No. 63/563,560 filed Mar. 11, 2024, which is incorporated herein by reference.

### BACKGROUND OF THE DISCLOSURE

[0002] Current parallel graphics data processing includes systems and methods developed to perform specific operations on graphics data such as, for example, linear interpolation, tessellation, rasterization, texture mapping, depth testing, etc. Traditionally, graphics processors used fixed function computational units to process graphics data. However, more recently, portions of graphics processors have been made programmable, enabling such processors to support a wider variety of operations for processing vertex and fragment data.

[0003] To further increase performance, graphics processors typically implement processing techniques such as pipelining that attempt to process, in parallel, as much graphics data as possible throughout the different parts of the graphics pipeline. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In a SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. A general overview of software and hardware for SIMT architectures can be found in Shane Cook, *CUDA Programming* Chapter 3, pages 37-51 (2013).

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Embodiments described herein are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements, and in which:

[0005] FIG. 1 is a block diagram illustrating a computer system configured to implement one or more aspects of the embodiments described herein;

[0006] FIG. 2A-2E illustrate parallel processor components, including graphics multiprocessors;

[0007] FIG. 3 illustrates a graphics processing unit that includes dedicated sets of graphics processing resources arranged into multi-core groups;

[0008] FIG. 4A-4E illustrate an example architecture in which a plurality of GPUs is communicatively coupled to a plurality of multi-core processors;

[0009] FIG. 5 illustrates a graphics processing pipeline;

[0010] FIG. 6 illustrates a machine learning software stack;

[0011] FIG. 7 illustrates a general-purpose graphics processing unit;

[0012] FIG. 8 illustrates a multi-GPU computing system;

[0013] FIG. 9A-9B illustrate layers of example deep neural networks;

[0014] FIG. 10A-10B illustrate example language models;

[0015] FIG. 11 illustrates training and deployment of a deep neural network;

[0016] FIG. 12 is a block diagram illustrating distributed learning;

[0017] FIG. 13 is a block diagram illustrating a programmable network interface and data processing unit;

[0018] FIG. 14 is a block diagram of a processing system;

[0019] FIG. 15A-15C illustrate computing systems and graphics processors;

[0020] FIG. 16 is a block diagram of a graphics processor, which may be a discrete or integrated graphics processing unit;

[0021] FIG. 17A-17B illustrate block diagrams of additional graphics processor and compute accelerator architectures;

[0022] FIG. 18A-18C illustrate thread execution logic including an array of processing elements employed in a graphics processor core;

[0023] FIG. 19 illustrates a tile of a multi-tile processor, according to an embodiment;

[0024] FIG. 20 is a block diagram illustrating graphics processor instruction formats;

[0025] FIG. 21 is a block diagram of an additional graphics processor architecture;

[0026] FIG. 22A-22B illustrate a graphics processor command format and command sequence;

[0027] FIG. 23 illustrates example graphics software architecture for a data processing system;

[0028] FIG. 24 is a block diagram illustrating an IP core development system;

[0029] FIG. 25A illustrates a cross-section side view of an integrated circuit package assembly that includes multiple units of hardware logic chiplets connected to a substrate (e.g., base die);

[0030] FIG. 25B illustrates a package assembly including interchangeable chiplets;

[0031] FIG. 26 is a block diagram illustrating a system on a chip integrated circuit;

[0032] FIG. 27 illustrates an example of a graphics multiprocessor providing a secure telemetry endpoint for secure virtualized telemetry collection with save and restore over global counters, in accordance with implementations herein;

[0033] FIG. 28 is a block diagram illustrating an example SOC/SOP implementing secure virtualized telemetry with SR schemes over global counters, in accordance with implementations herein;

[0034] FIG. 29 is a block diagram depicting an example of telemetry read and translation, in accordance with implementations herein;

[0035] FIG. 30 is an example sequence diagram illustrating a sequence of events for an example SR cycle, in accordance with implementations herein;

[0036] FIG. 31 is a flow diagram illustrating an embodiment of a method for providing implementation of secure virtualized telemetry with save and restore over global counters;

[0037] FIG. 32 is a flow diagram illustrating an embodiment of a method for providing establishing a secure connection for secure virtualized telemetry with save and restore over global counters; and

[0038] FIG. 33 is a flow diagram illustrating an embodiment of a method for providing save and restore cycles for secure virtualized telemetry with save and restore over global counters.

## DETAILED DESCRIPTION

[0039] Current parallel graphics data processing includes systems and methods developed to perform specific operations on graphics data such as, for example, linear interpolation, tessellation, rasterization, texture mapping, depth testing, etc. Traditionally, graphics processors used fixed function computational units to process graphics data. However, more recently, portions of graphics processors have been made programmable, enabling such processors to support a wider variety of operations for processing vertex and fragment data.

[0040] To further increase performance, graphics processors typically implement processing techniques such as pipelining that attempt to process, in parallel, as much graphics data as possible throughout the different parts of the graphics pipeline. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In a SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency.

[0041] A graphics processing unit (GPU) is communicatively coupled to host/processor cores to accelerate, for example, graphics operations, machine-learning operations, pattern analysis operations, and/or various general-purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or another interconnect (e.g., a high-speed interconnect such as PCIe or NVLink). Alternatively, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0042] While the techniques described herein are primarily discussed in the context of a GPU, techniques may also be implemented in other types of processors, including but not limited to general-purpose processors and accelerator devices, such as artificial intelligence accelerators, vision processors, and neural processing units.

[0043] In the following description, numerous specific details are set forth to provide a more thorough understanding. However, it will be apparent to one of skill in the art that the embodiments described herein may be practiced without one or more of these specific details. In other instances, well-known features have not been described to avoid obscuring the details of the present embodiments.

## System Overview

[0044] FIG. 1 is a block diagram illustrating a computing system 100 configured to implement one or more aspects of the embodiments described herein. The computing system 100 includes a processing subsystem 101 having one or more processor(s) 102 and a system memory 104 communicating via an interconnection path that may include a memory hub 105. The memory hub 105 may be a separate component within a chipset component or may be integrated within the one or more processor(s) 102. The memory hub 105 couples with an I/O subsystem 111 via a communication

link 106. The I/O subsystem 111 includes an I/O hub 107 that can enable the computing system 100 to receive input from one or more input device(s) 108. Additionally, the I/O hub 107 can enable a display controller, which may be included in the one or more processor(s) 102, to provide outputs to one or more display device(s) 110A. In one embodiment the one or more display device(s) 110A coupled with the I/O hub 107 can include a local, internal, or embedded display device.

[0045] The processing subsystem 101, for example, includes one or more parallel processor(s) 112 coupled to memory hub 105 via a communication link 113, such as a bus or fabric. The communication link 113 may be one of any number of standards-based communication link technologies or protocols, such as, but not limited to PCI Express, or may be a vendor specific communications interface or communications fabric. The one or more parallel processor(s) 112 may form a computationally focused parallel or vector processing system that can include a large number of processing cores and/or processing clusters, such as a many integrated core (MIC) processor. For example, the one or more parallel processor(s) 112 form a graphics processing subsystem that can output pixels to one of the one or more display device(s) 110A coupled via the I/O hub 107. The one or more parallel processor(s) 112 can also include a display controller and display interface (not shown) to enable a direct connection to one or more display device(s) 110B.

[0046] Within the I/O subsystem 111, a system storage unit 114 can connect to the I/O hub 107 to provide a storage mechanism for the computing system 100. An I/O switch 116 can be used to provide an interface mechanism to enable connections between the I/O hub 107 and other components, such as a network adapter 118 and/or wireless network adapter 119 that may be integrated into the platform, and various other devices that can be added via one or more add-in device(s) 120. The add-in device(s) 120 may also include, for example, one or more external graphics processor devices, graphics cards, and/or compute accelerators. The network adapter 118 can be an Ethernet adapter or another wired network adapter. The wireless network adapter 119 can include one or more of a Wi-Fi, Bluetooth, near field communication (NFC), or other network device that includes one or more wireless radios.

[0047] The computing system 100 can include other components not explicitly shown, including USB or other port connections, optical storage drives, video capture devices, and the like, which may also be connected to the I/O hub 107. Communication paths interconnecting the various components in FIG. 1 may be implemented using any suitable protocols, such as PCI (Peripheral Component Interconnect) based protocols (e.g., PCI-Express), or any other bus or point-to-point communication interfaces and/or protocol(s), such as the NVLink high-speed interconnect, Compute Express Link™ (CXL™) (e.g., CXL.mem), Infinity Fabric (IF), Ethernet (IEEE 802.3), remote direct memory access (RDMA), InfiniBand, Internet Wide Area RDMA Protocol (iWARP), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), quick UDP Internet Connections (QUIC), RDMA over Converged Ethernet (ROCE), Intel QuickPath Interconnect (QPI), Intel Ultra Path Interconnect (UPI), Intel On-Chip System Fabric (IOSF), Omni-Path, HyperTransport, Advanced Microcontroller Bus Architecture (AMBA) interconnect, OpenCAPI, Gen-Z, Cache

Coherent Interconnect for Accelerators (CCIX), 3GPP Long Term Evolution (LTE) (4G), 3GPP 5G, and variations thereof, or wired or wireless interconnect protocols known in the art. In some examples, data can be copied or stored to virtualized storage nodes using a protocol such as non-volatile memory express (NVMe) over Fabrics (NVMe-oF) or NVMe.

**[0048]** The one or more parallel processor(s) **112** may incorporate circuitry optimized for graphics and video processing, including, for example, video output circuitry, and constitutes a graphics processing unit (GPU). Alternatively or additionally, the one or more parallel processor(s) **112** can incorporate circuitry optimized for general purpose processing, while preserving the underlying computational architecture, described in greater detail herein. Components of the computing system **100** may be integrated with one or more other system elements on a single integrated circuit. For example, the one or more parallel processor(s) **112**, memory hub **105**, processor(s) **102**, and I/O hub **107** can be integrated into a system on chip (SoC) integrated circuit. Alternatively, the components of the computing system **100** can be integrated into a single package to form a system in package (SIP) configuration. In one embodiment at least a portion of the components of the computing system **100** can be integrated into a multi-chip module (MCM), which can be interconnected with other multi-chip modules into a modular computing system. In some configurations, the computing system **100** includes one or more accelerator device(s) **130** coupled with the memory hub **105**, in addition to the processor(s) **102** and the one or more parallel processor(s) **112**. The accelerator device(s) **130** are configured to perform domain specific acceleration of workloads to handle tasks that are computationally intensive or utilize high throughput. The accelerator device(s) **130** can reduce the burden placed on the processor(s) **102** and/or parallel processor(s) **112** of the computing system **100**. The accelerator device(s) **130** can include but are not limited to smart network interface cards, data processing units, cryptographic accelerators, storage accelerators, artificial intelligence (AI) accelerators, neural processing units (NPU), storage accelerators, and/or video transcoding accelerators.

**[0049]** It will be appreciated that the computing system **100** shown herein is illustrative and that variations and modifications are possible. The connection topology, including the number and arrangement of bridges, the number of processor(s) **102**, and the number of parallel processor(s) **112**, may be modified as desired. For instance, system memory **104** can be connected to the processor(s) **102** directly rather than through a bridge, while other devices communicate with system memory **104** via the memory hub **105** and the processor(s) **102**. In other alternative topologies, the parallel processor(s) **112** are connected to the I/O hub **107** or directly to one of the one or more processor(s) **102**, rather than to the memory hub **105**. In other embodiments, the I/O hub **107** and memory hub **105** may be integrated into a single chip. It is also possible that two or more sets of processor(s) **102** are attached via multiple sockets, which can couple with two or more instances of the parallel processor(s) **112**.

**[0050]** Some of the particular components shown herein are optional and may not be included in all implementations of the computing system **100**. For example, any number of add-in cards or peripherals may be supported, or some components may be eliminated. Furthermore, some archi-

tectures may use different terminology for components similar to those illustrated in FIG. 1. For example, the memory hub **105** may be referred to as a Northbridge in some architectures, while the I/O hub **107** may be referred to as a Southbridge.

**[0051]** FIG. 2A illustrates a parallel processor **200**. The parallel processor **200** may be a GPU, GPGPU or the like as described herein. The various components of the parallel processor **200** may be implemented using one or more integrated circuit devices, such as programmable processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGA). The parallel processor **200** may be one or more of the parallel processor(s) **112** shown in FIG. 1.

**[0052]** The parallel processor **200** includes a parallel processing unit **202**. The parallel processing unit includes an I/O unit **204** that enables communication with other devices, including other instances of the parallel processing unit **202**. The I/O unit **204** may be directly connected to other devices. For instance, the I/O unit **204** connects with other devices via the use of a hub or switch interface, such as memory hub **105**. The connections between the memory hub **105** and the I/O unit **204** form a communication link **113**. Within the parallel processing unit **202**, the I/O unit **204** connects with a host interface **206** and a memory crossbar **216**, where the host interface **206** receives commands directed to performing processing operations and the memory crossbar **216** receives commands directed to performing memory operations. In one embodiment, the I/O unit **204** is configured to enable secure I/O operations via Trusted Execution Environment (TEE)-I/O support. TEE-I/O enables trusted I/O virtualization, in which a trust relationship can be established directly between a secure virtual environment, such as trusted virtual machine, and the parallel processor **200** or secure partitions of the parallel processor.

**[0053]** When the host interface **206** receives a command buffer via the I/O unit **204**, the host interface **206** can direct work operations to perform those commands to a front end **208**. In one embodiment the front end **208** couples with a scheduler **210**, which is configured to distribute commands or other work items to a processing cluster array **212**. The scheduler **210** ensures that the processing cluster array **212** is properly configured and in a valid state before tasks are distributed to the processing clusters of the processing cluster array **212**. The scheduler **210** may be implemented via firmware logic executing on a microcontroller. The microcontroller implemented scheduler **210** is configurable to perform complex scheduling and work distribution operations at coarse and fine granularity, enabling rapid preemption and context switching of threads executing on the processing cluster array **212**. Preferably, the host software can prove workloads for scheduling on the processing cluster array **212** via one of multiple graphics processing doorbells. In other examples, polling for new workloads or interrupts can be used to identify or indicate availability of work to perform. The workloads can then be automatically distributed across the processing cluster array **212** by the scheduler **210** logic within the scheduler microcontroller.

**[0054]** The processing cluster array **212** can include up to “N” processing clusters (e.g., cluster **214A**, cluster **214B**, through cluster **214N**). Each cluster **214A-214N** of the processing cluster array **212** can execute a large number of concurrent threads. The scheduler **210** can allocate work to the clusters **214A-214N** of the processing cluster array **212**

using various scheduling and/or work distribution algorithms, which may vary depending on the workload arising for each type of program or computation. The scheduling can be handled dynamically by the scheduler **210** or can be assisted in part by compiler logic during compilation of program logic configured for execution by the processing cluster array **212**. Optionally, different clusters **214A-214N** of the processing cluster array **212** can be allocated for processing different types of programs or for performing different types of computations.

**[0055]** The processing cluster array **212** can be configured to perform various types of parallel processing operations. For example, the processing cluster array **212** is configured to perform general-purpose parallel compute operations. For example, the processing cluster array **212** can include logic to execute processing tasks including filtering of video and/or audio data, performing modeling operations, including physics operations, and performing data transformations.

**[0056]** The processing cluster array **212** is configured to perform parallel graphics processing operations. In such embodiments in which the parallel processor **200** is configured to perform graphics processing operations, the processing cluster array **212** can include additional logic to support the execution of such graphics processing operations, including, but not limited to texture sampling logic to perform texture operations, as well as tessellation logic and other vertex processing logic. Additionally, the processing cluster array **212** can be configured to execute graphics processing related shader programs such as, but not limited to vertex shaders, tessellation shaders, geometry shaders, and pixel shaders. The parallel processing unit **202** can transfer data from system memory via the I/O unit **204** for processing. During processing the transferred data can be stored to on-chip memory (e.g., parallel processor memory **222**) during processing, then written back to system memory.

**[0057]** In embodiments in which the parallel processing unit **202** is used to perform graphics processing, the scheduler **210** may be configured to divide the processing workload into approximately equal sized tasks, to better enable distribution of the graphics processing operations to multiple clusters **214A-214N** of the processing cluster array **212**. In some of these embodiments, portions of the processing cluster array **212** can be configured to perform different types of processing. For example, a first portion may be configured to perform vertex shading and topology generation, a second portion may be configured to perform tessellation and geometry shading, and a third portion may be configured to perform pixel shading or other screen space operations, to produce a rendered image for display. Intermediate data produced by one or more of the clusters **214A-214N** may be stored in buffers to allow the intermediate data to be transmitted between clusters **214A-214N** for further processing.

**[0058]** During operation, the processing cluster array **212** can receive processing tasks to be executed via the scheduler **210**, which receives commands defining processing tasks from front end **208**. For graphics processing operations, processing tasks can include indices of data to be processed, e.g., surface (patch) data, primitive data, vertex data, and/or pixel data, as well as state parameters and commands defining how the data is to be processed (e.g., what program is to be executed). The scheduler **210** may be configured to

fetch the indices corresponding to the tasks or may receive the indices from the front end **208**. The front end **208** can be configured to ensure the processing cluster array **212** is configured to a valid state before the workload specified by incoming command buffers (e.g., batch-buffers, push buffers, etc.) is initiated.

**[0059]** Each of the one or more instances of the parallel processing unit **202** can couple with parallel processor memory **222**. The parallel processor memory **222** can be accessed via the memory crossbar **216**, which can receive memory requests from the processing cluster array **212** as well as the I/O unit **204**. The memory crossbar **216** can access the parallel processor memory **222** via a memory interface **218**. The memory interface **218** can include multiple partition units (e.g., partition unit **220A**, partition unit **220B**, through partition unit **220N**) that can each couple to a portion (e.g., memory unit) of parallel processor memory **222**. The number of partition units **220A-220N** may be configured to be equal to the number of memory units, such that each partition unit **220A-220N** has a corresponding memory unit **224A-224N**. In other embodiments, the number of partition units **220A-220N** may not be equal to the number of memory units.

**[0060]** The memory units **224A-224N** can include various types of memory devices, including dynamic random-access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. Optionally, the memory units **224A-224N** may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM). Persons skilled in the art will appreciate that the specific implementation of the memory units **224A-224N** can vary and can be selected from one of various conventional designs. Render targets, such as frame buffers or texture maps may be stored across the memory units **224A-224N**, allowing partition units **220A-220N** to write portions of each render target in parallel to efficiently use the available bandwidth of parallel processor memory **222**. In some embodiments, a local instance of the parallel processor memory **222** may be excluded in favor of a unified memory design that utilizes system memory in conjunction with local cache memory.

**[0061]** Optionally, any one of the clusters **214A-214N** of the processing cluster array **212** has the ability to process data that will be written to any of the memory units **224A-224N** within parallel processor memory **222**. The memory crossbar **216** can be configured to transfer the output of each cluster **214A-214N** to any partition unit **220A-220N** or to another cluster **214A-214N**, which can perform additional processing operations on the output. Each cluster **214A-214N** can communicate with the memory interface **218** through the memory crossbar **216** to read from or write to various external memory devices. In one of the embodiments with the memory crossbar **216** the memory crossbar **216** has a connection to the memory interface **218** to communicate with the I/O unit **204**, as well as a connection to a local instance of the parallel processor memory **222**, enabling the processing units within the different processing clusters **214A-214N** to communicate with system memory or other memory that is not local to the parallel processing unit **202**. Generally, the memory crossbar **216** may, for example, be able to use virtual channels to separate traffic streams between the clusters **214A-214N** and the partition units **220A-220N**.

**[0062]** While a single instance of the parallel processing unit **202** is illustrated within the parallel processor **200**, any number of instances of the parallel processing unit **202** can be included. For example, multiple instances of the parallel processing unit **202** can be provided on a single add-in card, or multiple add-in cards can be interconnected. For example, the parallel processor **200** can be an add-in device, included in the add-in device(s) **120** of FIG. 1, which may be a graphics card such as a discrete graphics card that includes one or more GPUs, one or more memory devices, and device-to-device or network or fabric interfaces. The different instances of the parallel processing unit **202** can be configured to inter-operate even if the different instances have different numbers of processing cores, different amounts of local parallel processor memory, and/or other configuration differences. Optionally, some instances of the parallel processing unit **202** can include higher precision floating-point units relative to other instances. Systems incorporating one or more instances of the parallel processing unit **202** or the parallel processor **200** can be implemented in a variety of configurations and form factors, including but not limited to desktop, laptop, or handheld personal computers, servers, workstations, game consoles, and/or embedded systems. An orchestrator can form composite nodes for workload performance using one or more of: disaggregated processor resources, cache resources, memory resources, storage resources, and networking resources.

**[0063]** In one embodiment, the parallel processing unit **202** can be partitioned into multiple instances. Those multiple instances can be configured to execute workloads associated with different clients in an isolated manner, enabling a pre-determined quality of service to be provided for each client. For example, each cluster **214A-214N** can be compartmentalized and isolated from other clusters, allowing the processing cluster array **212** to be divided into multiple compute partitions or instances. In such configuration, workloads that execute on an isolated partition are protected from faults or errors associated with a different workload that executes on a different partition. The partition units **220A-220N** can be configured to enable a dedicated and/or isolated path to memory for the clusters **214A-214N** associated with the respective compute partitions. This data-path isolation enables the compute resources within a partition can communicate with one or more assigned memory units **224A-224N** without being subjected to interference by the activities of other partitions. In one embodiment, data-path isolation can be enhanced by encrypting data in memory of the various partitions with encryption keys that are specialized to an associated partition, such that data is secured on a per-partition basis while at rest and in transit.

**[0064]** FIG. 2B is a block diagram of a partition unit **220**. The partition unit **220** may be an instance of one of the partition units **220A-220N** of FIG. 2A. As illustrated, the partition unit **220** includes an L2 cache **221**, a frame buffer interface **225**, and a ROP **226** (raster operations unit). The L2 cache **221** is a read/write cache that is configured to perform load and store operations received from the memory crossbar **216** and ROP **226**. Read misses and urgent write-back requests are output by L2 cache **221** to frame buffer interface **225** for processing. Updates can also be sent to the frame buffer via the frame buffer interface **225** for processing. In one embodiment, the frame buffer interface **225** interfaces with a memory unit **224** of the memory units

**224A-224N** within parallel processor memory **222** of FIG. 2A. The partition unit **220** may additionally or alternatively also interface with one of the memory units in parallel processor memory via a memory controller (not shown).

**[0065]** In graphics applications, the ROP **226** is a processing unit that performs raster operations such as stencil, z test, blending, and the like. The ROP **226** then outputs processed graphics data that is stored in graphics memory. In some embodiments the ROP **226** includes or couples with a CODEC **227** that includes compression logic to compress depth or color data that is written to memory or the L2 cache **221** and decompress depth or color data that is read from memory or the L2 cache **221**. The compression logic can be lossless compression logic that makes use of one or more of multiple compression algorithms. The type of compression that is performed by the CODEC **227** can vary based on the statistical characteristics of the data to be compressed. For example, in one embodiment, delta color compression is performed on depth and color data on a per-tile basis. In one embodiment the CODEC **227** includes compression and decompression logic that can compress and decompress compute data associated with machine learning operations. The CODEC **227** can, for example, compress sparse matrix data for sparse machine learning operations. The CODEC **227** can also compress sparse matrix data that is encoded in a sparse matrix format (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compress sparse column (CSC), etc.) to generate compressed and encoded sparse matrix data. The compressed and encoded sparse matrix data can be decompressed and/or decoded before being processed by processing elements or the processing elements can be configured to consume compressed, encoded, or compressed and encoded data for processing. In one embodiment, the CODEC **227** can be configured as a general-purpose data compression engine for use in GPU database acceleration and large volume data analytics.

**[0066]** The ROP **226** may be included within each processing cluster (e.g., cluster **214A-214N** of FIG. 2A) instead of within the partition unit **220**. In such embodiment, read and write requests for pixel data are transmitted over the memory crossbar **216** instead of pixel fragment data. The processed graphics data may be displayed on a display device, such as one of the one or more display device(s) **110A-110B** of FIG. 1, routed for further processing by the processor(s) **102**, or routed for further processing by one of the processing entities within the parallel processor **200** of FIG. 2A.

**[0067]** FIG. 2C is a block diagram of a processing cluster **214** within a parallel processing unit. For example, the processing cluster **214** represents an instance of one of the processing clusters **214A-214N** of FIG. 2A. The processing cluster **214** can be configured to execute many threads in parallel, where the term “thread” refers to an instance of a particular program executing on a particular set of input data. Optionally, single-instruction, multiple-data (SIMD) instruction issue techniques may be used to support parallel execution of a large number of threads without providing multiple independent instruction units. Alternatively, single-instruction, multiple-thread (SIMT) techniques may be used to support parallel execution of a large number of generally synchronized threads, using a common instruction unit configured to issue instructions to a set of processing engines within each one of the processing clusters. Unlike a SIMD execution regime, where all processing engines typi-

cally execute identical instructions, SIMT execution allows different threads to follow divergent execution paths more readily through a given thread program. Persons skilled in the art will understand that a SIMD processing regime represents a functional subset of a SIMT processing regime.

**[0068]** Operation of the processing cluster 214 can be controlled via a pipeline manager 232 that distributes processing tasks to SIMT parallel processors. The pipeline manager 232 receives instructions from the scheduler 210 of FIG. 2A and manages execution of those instructions via a graphics multiprocessor 234 and/or a texture unit 236. The graphics multiprocessor 234 is an example instance of a SIMT parallel processor. However, various types of SIMT parallel processors of differing architectures may be included within the processing cluster 214. One or more instances of the graphics multiprocessor 234 can be included within a processing cluster 214. The graphics multiprocessor 234 may also be referred to as a streaming multiprocessors (SM) and is capable of simultaneous execution of a large number of execution threads.

**[0069]** The graphics multiprocessor 234 can process data and a data crossbar 240 can be used to distribute the processed data to one of multiple possible destinations, including instances of the graphics multiprocessor 234 within the processing cluster 214. The pipeline manager 232 can facilitate the distribution of processed data by specifying destinations for processed data to be distributed via the data crossbar 240. Each graphics multiprocessor 234 within the processing cluster 214 can include an identical set of functional execution logic (e.g., arithmetic logic units, load-store units, etc.). The functional execution logic can be configured in a pipelined manner in which new instructions can be issued before previous instructions are complete. The functional execution logic supports a variety of operations including integer and floating-point arithmetic, comparison operations, Boolean operations, bit-shifting, and computation of various algebraic functions. The same functional-unit hardware could be leveraged to perform different operations and any combination of functional units may be present.

**[0070]** The instructions transmitted to the processing cluster 214 constitute a thread. A set of threads executing across the set of parallel processing engines is a thread group. A thread group executes the same program on different input data. Each thread within a thread group can be assigned to a different processing engine within a graphics multiprocessor 234. A thread group may include fewer threads than the number of processing engines within the graphics multiprocessor 234. When a thread group includes fewer threads than the number of processing engines, one or more of the processing engines may be idle during cycles in which that thread group is being processed. A thread group may also include more threads than the number of processing engines within the graphics multiprocessor 234. When the thread group includes more threads than the number of processing engines within the graphics multiprocessor 234, processing can be performed over consecutive clock cycles. Optionally, multiple thread groups can be executed concurrently on the graphics multiprocessor 234.

**[0071]** The graphics multiprocessor 234 may include an internal cache memory to perform load and store operations. Optionally, the graphics multiprocessor 234 can forego an internal cache and use a cache memory (e.g., level 1 (L1) cache 248) within the processing cluster 214. Each graphics multiprocessor 234 also has access to level 2 (L2) caches

within the partition units (e.g., partition units 220A-220N of FIG. 2A) that are shared among all instances of the processing cluster 214 and may be used to transfer data between threads. The graphics multiprocessor 234 may also access off-chip global memory, which can include one or more of local parallel processor memory and/or system memory. Any memory external to the parallel processing unit 202 may be used as global memory. Embodiments in which the processing cluster 214 includes multiple instances of the graphics multiprocessor 234 can share common instructions and data, which may be stored in the L1 cache 248.

**[0072]** Each processing cluster 214 may include an MMU 245 (memory management unit) that is configured to map virtual addresses into physical addresses. In other embodiments, one or more instances of the MMU 245 may reside within the memory interface 218 of FIG. 2A. The MMU 245 includes a set of page table entries (PTEs) used to map a virtual address to a physical address of a tile and optionally a cache line index. The MMU 245 may include address translation lookaside buffers (TLB) or caches that may reside within the graphics multiprocessor 234 or the L1 cache 248 of processing cluster 214. The physical address is processed to distribute surface data access locality to allow efficient request interleaving among partition units. The cache line index may be used to determine whether a request for a cache line is a hit or miss.

**[0073]** In graphics and computing applications, a processing cluster 214 may be configured such that each graphics multiprocessor 234 is coupled to a texture unit 236 for performing texture mapping operations, e.g., determining texture sample positions, reading texture data, and filtering the texture data. Texture data is read from an internal texture L1 cache (not shown) or in some embodiments from the L1 cache within graphics multiprocessor 234 and is fetched from an L2 cache, local parallel processor memory, or system memory. Each graphics multiprocessor 234 outputs processed tasks to the data crossbar 240 to provide the processed task to another processing cluster 214 for further processing or to store the processed task in an L2 cache, local parallel processor memory, or system memory via the memory crossbar 216. A preROP 242 (pre-raster operations unit) is configured to receive data from graphics multiprocessor 234, direct data to ROP units, which may be located with partition units as described herein (e.g., partition units 220A-220N of FIG. 2A). The preROP 242 unit can perform optimizations for color blending, organize pixel color data, and perform address translations.

**[0074]** It will be appreciated that the core architecture described herein is illustrative and that variations and modifications are possible. Any number of processing units, e.g., graphics multiprocessor 234, texture units 236, preROPs 242, etc., may be included within a processing cluster 214. A parallel processing unit as described herein may include any number of instances of the processing cluster 214. Optionally, each processing cluster 214 can be configured to operate independently of other instances of the processing cluster 214 using, for example, separate and distinct processing units, L1 caches, L2 caches to facilitate data and fault isolation.

**[0075]** FIG. 2D shows an example of the graphics multiprocessor 234 in which the graphics multiprocessor 234 couples with the pipeline manager 232 of the processing cluster 214. The graphics multiprocessor 234 has an execution pipeline including but not limited to an instruction

cache **252**, an instruction unit **254**, an address mapping unit **256**, a register file **258**, one or more general purpose graphics processing unit cores (GPGPU cores **262**), and one or more load/store units **266**. The GPGPU cores **262** and load/store units **266** are coupled with cache memory **272** and shared memory **270** via a memory and cache interconnect **268**. The graphics multiprocessor **234** may additionally include ray-tracing cores **263** that include hardware logic to accelerate ray-tracing operations and tensor cores **264** that include hardware logic to accelerate tensor (e.g., matrix) operations. The instruction cache **252** may receive a stream of instructions to execute from the pipeline manager **232**. The instructions are cached in the instruction cache **252** and dispatched for execution by the instruction unit **254**. The instruction unit **254** can dispatch instructions as thread groups (e.g., warps), with each thread of the thread group assigned to a different execution unit within the GPGPU cores **262**. An instruction can access any of a local, shared, or global address space by specifying an address within a unified address space. The address mapping unit **256** can be used to translate addresses in the unified address space into a distinct memory address that can be accessed by the load/store units **266**.

**[0076]** The register file **258** provides a set of registers for the functional units of the graphics multiprocessor **234**. The register file **258** provides temporary storage for operands connected to the data paths of the functional units (e.g., GPGPU cores **262**, load/store units **266**) of the graphics multiprocessor **234**. The register file **258** may be divided between each of the functional units such that each functional unit is allocated a dedicated portion of the register file **258**. For example, the register file **258** may be divided between the different warps being executed by the graphics multiprocessor **234**.

**[0077]** The GPGPU cores **262** can each include floating-point units (FPUs) and/or integer arithmetic logic units (ALUs) that are used to execute instructions of the graphics multiprocessor **234**. In some implementations, the GPGPU cores **262** can include hardware logic that may otherwise reside within the tensor cores **264** and/or ray-tracing cores **263**. The GPGPU cores **262** can be similar in architecture or can differ in architecture. For example and in one embodiment, a first portion of the GPGPU cores **262** include a single precision FPU and an integer ALU while a second portion of the GPGPU cores include a double precision FPU. Optionally, the FPUs can implement the IEEE 754-2008 standard for floating-point arithmetic or enable variable precision floating-point arithmetic. In one embodiment, the single precision FPUs, or a separate set of FPUs, are configurable to perform operations on 16-bit floating-point operands, such as operands in a half precision format or a bfloat16 format (e.g., Brain floating-point), which is a 16-bit floating-point format with one sign bit, eight exponent bits, and eight significant bits, of which seven are explicitly stored. The FPUs within one or more the GPGPU cores **262** may also support one or more 8-bit floating-point formats. Supported 8-bit floating-point formats include the E4M3 format having a 4-bit exponent and a 3-bit mantissa and the E5M2 format having a 5-bit exponent and 2-bit mantissa. The graphics multiprocessor **234** can additionally include one or more fixed function or special function units to perform specific functions such as copy rectangle or pixel blending operations. One or more of the GPGPU cores can also include fixed or special function logic.

**[0078]** The GPGPU cores **262** may include SIMD logic capable of performing a single instruction on multiple sets of data. Optionally, GPGPU cores **262** can physically execute SIMD4, SIMD8, and SIMD16 instructions and logically execute SIMD1, SIMD2, and SIMD32 instructions. The SIMD instructions for the GPGPU cores can be generated at compile time by a shader compiler or automatically generated when executing programs written and compiled for single program multiple data (SPMD) or SIMT architectures. Multiple threads of a program configured for the SIMT execution model can be executed via a single SIMD instruction. For example and in one embodiment, eight SIMT threads that perform the same or similar operations can be executed in parallel as a SIMD8 instruction. In one embodiment, a warp of 32 SIMT threads can be executed as a single SIMD32 instruction. Warp divergence can be handled via multiple SIMD instructions.

**[0079]** The memory and cache interconnect **268** is an interconnect network that connects each of the functional units of the graphics multiprocessor **234** to the register file **258** and to the shared memory **270**. For example, the memory and cache interconnect **268** is a crossbar interconnect that allows the load/store unit **266** to implement load and store operations between the shared memory **270** and the register file **258**. The register file **258** can operate at the same frequency as the GPGPU cores **262**, thus data transfer between the GPGPU cores **262** and the register file **258** is very low latency. The shared memory **270** can be used to enable communication between threads that execute on the functional units within the graphics multiprocessor **234**. The shared memory **270** can also be used as a program managed cache. The cache memory **272** can be used as an automatically managed data cache for example, to cache texture data communicated between the functional units and the texture unit **236**. The shared memory **270** and the cache memory **272** can couple with the data crossbar **240** to enable communication with other components of the processing cluster, facilitating the cooperative execution of cluster workgroups via multiple graphics multiprocessors within a processing cluster. Threads executing on the GPGPU cores **262** can programmatically store data within the shared memory in addition to the automatically cached data that is stored within the cache memory **272**. In one embodiment, the shared memory **270** and the cache memory **272** may be combined into a single configurable memory unit that can be selectively configured as the cache memory **272** or the shared memory **270**.

**[0080]** FIG. 2E illustrates a graphics multiprocessor **235** having an alternate configuration relative to the graphics multiprocessor **234** of FIG. 2D. The disclosure of any features in combination with the graphics multiprocessor **235** described herein also discloses a corresponding combination with the graphics multiprocessor **234** of FIG. 2D, but is not limited as such. The graphics multiprocessor **235** of FIG. 2E includes multiple additional instances of execution resources **286A-286D** relative to the graphics multiprocessor **234** of FIG. 2D. For example, the graphics multiprocessor **235** can include multiple instruction units **254A-254D**, register files **258A-258D**, and texture units **280A-280D**. The graphics multiprocessor **235** also includes multiple sets of graphics or compute execution units (e.g., GPGPU cores **262A-262D**, ray-tracing cores **263A-263D**, and tensor cores **264A-264D**) and multiple sets of load/store units **266A-266D**. The execution resources **286A-286D** work in concert

with texture unit(s) 280A-280D for texture operations, while sharing an instruction cache 252, shared memory 270, and cache memories 272A-272B. In one embodiment, the execution resources 286A-286D additionally include multi-function units (MUFU) and/or special-function units (SFU) (e.g., MFU 267A-267D) that are used to perform specialized mathematical operations, such as transcendental operations including exponential, logarithmic, and trigonometric functions.

[0081] The various components can communicate via an interconnect fabric 290. The interconnect fabric 290 may include one or more crossbar switches to enable communication between the various components of the graphics multiprocessor 235. The GPGPU cores 262A-262D, ray tracing cores 263A-263B, and tensor cores 264A-264D can each communicate with shared memory 270 via the interconnect fabric 290. The interconnect fabric 290 can arbitrate communication within the graphics multiprocessor 235 to ensure a fair bandwidth allocation between components. In one embodiment, the interconnect fabric 290 may be a separate, high-speed network fabric layer upon which each component of the graphics multiprocessor 235 is stacked. The components of the graphics multiprocessor 235 can also communicate with remote components via the interconnect fabric 290.

[0082] In one embodiment, the graphics multiprocessor 235 includes a tensor transfer engine 292, which is a copy engine that is configurable to accelerate the movement of tensor data into and out of the graphics multiprocessor 235. The tensor transfer engine 292 can accelerate tensor memory operations by asynchronously performing address generation and data movement operations for N-dimensional blocks of tensor data, which offloads operations that would otherwise be performed manually by program code executed by the graphics multiprocessor 235. The tensor transfer engine 292 is configurable to copy data between, for example, the shared memory 270 and/or cache memory 272A-272B of the graphics multiprocessor 235 and memory external to the graphics multiprocessor 235, such as graphics processor global memory (e.g., parallel processor memory 222). In one embodiment, data transfers performed by the tensor transfer engine 292 can be configured to selectively bypass various levels of intermediate data storage between the source and destination memories. For example, a transfer between global memory and shared memory 270 can bypass the register files 258A-258D. In one embodiment, threads can synchronize on asynchronous tensor transfers via a non-blocking barrier synchronization mechanism.

[0083] In various embodiments, the graphics multiprocessor 235 can be tailored for specific use cases via the inclusion or exclusion of certain components, allowing various implementations of the graphics multiprocessor 235 that are tailored to target power, performance, and area characteristic. For example, compute oriented variants of the graphics multiprocessor 235 that will not perform graphics operations can exclude the ray tracing cores 263A-263D. Fully graphics oriented variants can exclude the tensor transfer engine 292, while graphics oriented variants additionally configured to accelerate neural network inference may include at least a version of the tensor transfer engine 292.

[0084] Persons skilled in the art will understand that the architecture described in FIG. 1 and FIG. 2A-2E are descriptive and not limiting as to the scope of the present embodi-

ments. Thus, the techniques described herein may be implemented on any properly configured processing unit, including, without limitation, one or more mobile application processors, one or more desktop or server central processing units (CPUs) including multi-core CPUs, one or more parallel processing units, such as the parallel processing unit 202 of FIG. 2A, as well as one or more graphics processors or special purpose processing units, without departure from the scope of the embodiments described herein.

[0085] The parallel processor or GPGPU as described herein may be communicatively coupled to host/processor cores to accelerate graphics operations, machine-learning operations, pattern analysis operations, and various general-purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or other interconnect (e.g., a high-speed interconnect such as PCIe, NVLink, or other known protocols, standardized protocols, or proprietary protocols). In other embodiments, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0086] FIG. 3 illustrates a graphics processing unit (GPU 380) which includes dedicated sets of graphics processing resources arranged into multi-core groups 365A-365N. The multi-core groups 365A-365N correspond with the graphics multiprocessor 234 of FIG. 2D or the graphics multiprocessor 235 of FIG. 2E. While the details of a single example of the multi-core groups 365A-365N (e.g., multi-core group 365A) are provided, it will be appreciated that the other multi-core groups 365B-365N may be equipped with the same or similar sets of graphics processing resources. Details described with respect to the multi-core groups 365A-365N may also apply to graphics multiprocessor 234 or graphics multiprocessor 235, as described herein.

[0087] As illustrated, a multi-core group 365A may include graphics cores 370, tensor cores 371, and ray tracing cores 372. The graphics cores 370 are analogous to the GPGPU cores 262A-262D and are configurable to execute instructions to perform graphics and/or general-purpose compute operations. A scheduler/dispatcher 368 schedules and dispatches the graphics threads for execution on the various cores within the multi-core group 365A. Register files 369 are included that store operand values used by the cores when performing graphics or general-purpose compute operations for executed threads. These register files 369 may include, for example, registers configurable to store integer values or floating-point values, including vector registers for storing packed integer and/or floating-point data elements and tile registers for storing tensor/matrix values. The tile registers may be implemented as multi-dimensional registers that include combined sets of vector registers.

[0088] One or more combined level 1 (L1) caches and shared memory units 373 store graphics data such as texture data, vertex data, pixel data, ray data, bounding volume data, etc., locally within each multi-core group 365A. One or more texture units 374 can also be used to perform texturing operations, such as texture mapping and sampling. A Level

2 (L2) cache **375** shared by all or a subset of the multi-core groups **365A-365N** stores graphics data and/or instructions for multiple concurrent graphics threads. As illustrated, the L2 cache **375** may be shared across a plurality of multi-core groups **365A-365N**. One or more memory controllers **367** couple the GPU **380** to a memory **366** which may be a system memory (e.g., DRAM) and/or a dedicated graphics memory (e.g., GDDR6 memory).

**[0089]** Input/output circuitry (I/O circuitry **363**) couples the GPU **380** to one or more I/O devices **362** such as digital signal processors (DSPs), network controllers, or user input devices. An on-chip interconnect may be used to couple the I/O devices **362** to the GPU **380** and memory **366**. At least one I/O memory management units (IOMMU **364**) of the I/O circuitry **363** couples the I/O devices **362** directly to the memory **366**. Optionally, the IOMMU **364** manages multiple sets of page tables to map virtual addresses to physical addresses in memory **366**. The I/O devices **362**, CPU(s) **361**, and GPU **380** may then share the same virtual address space.

**[0090]** In one implementation of the IOMMU **364**, the IOMMU **364** supports virtualization. In this case, it may manage a first set of page tables to map guest/graphics virtual addresses to guest/graphics physical addresses and a second set of page tables to map the guest/graphics physical addresses to system/host physical addresses (e.g., within memory **366**). The base addresses of each of the first and second sets of page tables may be stored in control registers and swapped out on a context switch (e.g., so that the new context is provided with access to the relevant set of page tables). While not illustrated in FIG. 3, each of the cores within the multi-core groups **365A-365N** may include translation lookaside buffers (TLBs) to cache guest virtual to guest physical translations, guest physical to host physical translations, and guest virtual to host physical translations.

**[0091]** The CPU(s) **361**, GPU **380**, and I/O devices **362** may be integrated on a single semiconductor chip and/or chip package. The memory **366** may be integrated on the same chip or may be coupled to the one or more memory controllers **367** via an off-chip interface. In one implementation, the memory **366** comprises GDDR6 memory which shares the same virtual address space as other physical system-level memories, although the underlying principles described herein are not limited to this specific implementation.

**[0092]** The tensor cores **371** may include a plurality of execution units specifically designed to perform matrix operations, which are the compute operation used to perform deep learning operations. For example, simultaneous matrix multiplication operations may be used for neural network training and inferencing. The tensor cores **371** may perform matrix processing using a variety of operand precisions including single precision floating-point (e.g., 32 bits), half-precision floating-point (e.g., 16 bits), integer words (16 bits), bytes (8 bits), and half-bytes (4 bits). For example, a neural network implementation extracts features of each rendered scene, potentially combining details from multiple frames, to construct a high-quality final image.

**[0093]** In deep learning implementations, parallel matrix multiplication work may be scheduled for execution on the tensor cores **371**. The training of neural networks, in particular, utilizes a significant number of matrix dot product operations. In order to process an inner-product formulation of an  $N \times N \times N$  matrix multiply, the tensor cores **371** may include at least  $N$  dot-product processing elements. Before

the matrix multiply begins, one entire matrix is loaded into tile registers and at least one column of a second matrix is loaded each cycle for  $N$  cycles. Each cycle, there are  $N$  dot products that are processed.

**[0094]** Matrix elements may be stored at different precisions depending on the particular implementation, including 16-bit words, 8-bit bytes (e.g., INT8) and 4-bit half-bytes (e.g., INT4). Different precision modes may be specified for the tensor cores **371** to ensure that the most efficient precision is used for different workloads (e.g., such as inferencing workloads which can tolerate quantization to bytes and half-bytes). Supported formats additionally include 64-bit floating-point (FP64) and non-IEEE floating-point formats such as the bfloat16 format. One embodiment includes support for a reduced precision tensor-float (TF32) mode, which performs computations using the range of FP32 (8-bits) and the precision of FP16 (10-bits). Reduced precision TF32 operations can be performed on FP32 inputs and produce FP32 outputs at higher performance relative to FP32 and increased precision relative to FP16. In one embodiment, one or more 8-bit floating-point (FP8), 6-bit floating-point (FP6), and 4-bit floating-point (FP4) formats are supported, including floating-point formats represented as microscaling (MX) formats.

**[0095]** In one embodiment the tensor cores **371** support a sparse mode of operation for matrices in which the vast majority of values are zero. The tensor cores **371** include support for sparse input matrices that are encoded in a sparse matrix representation (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compressed sparse column (CSC), etc.). The tensor cores **371** also include support for compressed sparse matrix representations in the event that the sparse matrix representation may be further compressed. Compressed, encoded, and/or compressed and encoded matrix data, along with associated compression and/or encoding metadata, can be read by the tensor cores **371** and the non-zero values can be extracted. For example, for a given input matrix A, a non-zero value can be loaded from the compressed and/or encoded representation of at least a portion of matrix A. Based on the location in matrix A for the non-zero value, which may be determined from index or coordinate metadata associated with the non-zero value, a corresponding value in input matrix B may be loaded. Depending on the operation to be performed (e.g., multiply), the load of the value from input matrix B may be bypassed if the corresponding value is a zero value. In one embodiment, the pairings of values for certain operations, such as multiply operations, may be pre-scanned by scheduler logic and operations between non-zero inputs are scheduled. Depending on the dimensions of matrix A and matrix B and the operation to be performed, output matrix C may be dense or sparse. Where output matrix C is sparse and depending on the configuration of the tensor cores **371**, output matrix C may be output in a compressed format, a sparse encoding, or a compressed sparse encoding.

**[0096]** The ray tracing cores **372** may accelerate ray tracing operations for both real-time ray tracing and non-real-time ray tracing implementations. In particular, the ray tracing cores **372** may include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. The ray tracing cores **372** may also include circuitry for performing depth testing and culling (e.g., using a

Z buffer or similar arrangement). In one implementation, the ray tracing cores 372 perform traversal and intersection operations in concert with the image denoising techniques described herein, at least a portion of which may be executed on the tensor cores 371. For example, the tensor cores 371 may implement a deep learning neural network to perform denoising of frames generated by the ray tracing cores 372. However, the CPU(s) 361, graphics cores 370, and/or ray tracing cores 372 may also implement all or a portion of the denoising and/or deep learning algorithms.

[0097] In addition, as described above, a distributed approach to denoising may be employed in which the GPU 380 is in a computing device coupled to other computing devices over a network or high-speed interconnect. In this distributed approach, the interconnected computing devices may share neural network learning/training data to improve the speed with which the overall system learns to perform denoising for different types of image frames and/or different graphics applications.

[0098] The ray tracing cores 372 may process all BVH traversal and/or ray-primitive intersections, saving the graphics cores 370 from being overloaded with thousands of instructions per ray. For example, each ray tracing core 372 includes a first set of specialized circuitry for performing bounding box tests (e.g., for traversal operations) and/or a second set of specialized circuitry for performing the ray-triangle intersection tests (e.g., intersecting rays which have been traversed). Thus, for example, the multi-core group 365A can simply launch a ray probe, and the ray tracing cores 372 independently perform ray traversal and intersection and return hit data (e.g., a hit, no hit, multiple hits, etc.) to the thread context. The graphics cores 370 and tensor cores 371 are then freed to perform other graphics or compute work while the ray tracing cores 372 perform the traversal and intersection operations. Optionally, each ray tracing core 372 may include a traversal unit to perform BVH testing operations and/or an intersection unit which performs ray-primitive intersection tests. The intersection unit generates a “hit”, “no hit”, or “multiple hit” response, which it provides to the appropriate thread. During the traversal and intersection operations, the execution resources of the other cores (e.g., graphics cores 370 and tensor cores 371) are freed to perform other forms of graphics work. In one optional embodiment described below, a hybrid rasterization/ray tracing approach is used in which rendering operations are distributed between the graphics cores 370 and ray tracing cores 372.

[0099] The ray tracing cores 372 may include hardware support for a ray tracing instruction set such as Microsoft’s DirectX Ray Tracing (DXR) which includes a DispatchRays command, as well as ray-generation, closest-hit, any-hit, and miss shaders, which enable the assignment of sets of shaders and textures for each object. Another ray tracing platform which may be supported by the ray tracing cores 372, graphics cores 370 and tensor cores 371 is Vulkan API (e.g., Vulkan version 1.1.85 and later). Note, however, that the underlying principles described herein are not limited to any particular ray tracing ISA. In general, the ray tracing cores 372, tensor cores 371, and graphics cores 370 may support a ray tracing instruction set that includes instructions/functions for one or more of ray generation, closest hit, any hit, ray-primitive intersection, per-primitive and hierarchical bounding box construction, miss, visit, and exceptions.

More specifically, an embodiment includes ray tracing instructions to perform one or more of the following functions:

- [0100] Ray Generation—Ray generation instructions may be executed for each pixel, sample, or other user-defined work assignment.
- [0101] Closest Hit—A closest hit instruction may be executed to locate the closest intersection point of a ray with primitives within a scene.
- [0102] Any Hit—An any hit instruction identifies multiple intersections between a ray and primitives within a scene, potentially to identify a new closest intersection point.
- [0103] Intersection—An intersection instruction performs a ray-primitive intersection test and outputs a result.
- [0104] Per-primitive Bounding box Construction—This instruction builds a bounding box around a given primitive or group of primitives (e.g., when building a new BVH or other acceleration data structure).
- [0105] Miss—Indicates that a ray misses all geometry within a scene, or specified region of a scene.
- [0106] Visit—Indicates the child volumes a ray will traverse.
- [0107] Exceptions—Includes various types of exception handlers (e.g., invoked for various error conditions).

[0108] In one embodiment the ray tracing cores 372 may be adapted to accelerate general-purpose compute operations using computational techniques that are analogous to ray intersection tests. A compute framework can be provided that enables shader programs to be compiled into low level instructions and/or primitives that perform general-purpose compute operations via the ray tracing cores. Example computational problems that can benefit from compute operations performed on the ray tracing cores 372 include computations involving beam, wave, ray, or particle propagation within a coordinate space. Interactions associated with that propagation can be computed relative to a geometry or mesh within the coordinate space. For example, computations associated with electromagnetic signal propagation through an environment can be accelerated via the use of instructions or primitives that are executed via the ray tracing cores. Diffraction and reflection of the signals by objects in the environment can be computed as direct ray-tracing analogies.

[0109] Ray tracing cores 372 can also be used to perform computations that are not directly analogous to ray tracing. For example, mesh projection, mesh refinement, and volume sampling computations can be accelerated using the ray tracing cores 372. Generic coordinate space calculations, such as nearest neighbor calculations can also be performed. For example, the set of points near a given point can be discovered by defining a bounding box in the coordinate space around the point. BVH and ray probe logic within the ray tracing cores 372 can then be used to determine the set of point intersections within the bounding box. The intersections constitute the origin point and the nearest neighbors to that origin point. Computations that are performed using the ray tracing cores 372 can be performed in parallel with computations performed on the graphics cores 370 and tensor cores 371. A shader compiler can be configured to compile a compute shader or other general-purpose graphics

processing program into low level primitives that can be parallelized across the graphics cores 370, tensor cores 371, and ray tracing cores 372.

#### Techniques for GPU to Host Processor Interconnection

[0110] FIG. 4A illustrates an example architecture in which a plurality of GPUs 410-413, e.g., such as the parallel processor 200 shown in FIG. 2A, are communicatively coupled to a plurality of multi-core processors 405-406 over high-speed links 440A-440D (e.g., buses, point-to-point interconnects, etc.). The high-speed links 440A-440D may support a communication throughput of 4 GB/s, 30 GB/s, 80 GB/s or higher, depending on the implementation. Various interconnect protocols may be used including, but not limited to, PCIe 4.0, PCIe 5.0, PCIe 6.0, and various NVLink and NVlink-C2C (Chip-to-Chip) interconnect protocols (e.g., NVLink v5). However, the underlying principles described herein are not limited to any particular communication protocol or throughput.

[0111] Two or more of the GPUs 410-413 may be interconnected over high-speed links 442A-442B, which may be implemented using the same or different protocols/links than those used for high-speed links 440A-440D. Similarly, two or more of the multi-core processors 405-406 may be connected over high-speed link 443 which may be symmetric multi-processor (SMP) buses operating at 20 GB/s, 30 GB/s, 120 GB/s or lower or higher speeds. Alternatively, all communication between the various system components shown in FIG. 4A may be accomplished using the same protocols/links (e.g., over a common interconnection fabric). As mentioned, however, the underlying principles described herein are not limited to any particular type of interconnect technology.

[0112] Each of multi-core processor 405 and multi-core processor 406 may be communicatively coupled to a processor memory 401-402, via memory interconnects 430A-430B, respectively, and each GPU 410-413 is communicatively coupled to GPU memory 420-423 over GPU memory interconnects 450A-450D, respectively. The memory interconnects 430A-430B and 450A-450D may utilize the same or different memory access technologies. By way of example, and not limitation, the processor memories 401-402 and GPU memories 420-423 may be volatile memories such as dynamic random-access memories (DRAMs) (including stacked DRAMs), Graphics DDR SDRAM (GDDR) (e.g., GDDR5, GDDR6, GDDR7), or High Bandwidth Memory (HBM) and/or may be non-volatile memories such as 3D XPoint/Optane or Nano-Ram. For example, some portion of the memories may be volatile memory and another portion may be non-volatile memory (e.g., using a two-level memory (2LM) hierarchy). A memory subsystem as described herein may be compatible with a number of memory technologies, such as Double Data Rate versions released by JEDEC (Joint Electronic Device Engineering Council).

[0113] As described below, although the various processors 405-406 and GPUs 410-413 may be physically coupled to particular processor memory 401-402 and GPU memory 420-423, respectively, a unified memory architecture may be implemented in which the same virtual system address space (also referred to as the “effective address” space) is distributed among all of the various physical memories. For example, processor memories 401-402 may each comprise 64 GB of the system memory address space and GPU

memories 420-423 may each comprise 32 GB of the system memory address space (resulting in a total of 256 GB addressable memory in this example).

[0114] FIG. 4B illustrates additional optional details for an interconnection between a processor 407 and a graphics accelerator 446. The graphics accelerator 446 may include one or more GPU chips integrated on a line card which is coupled to the processor 407 via the high-speed link 440. Alternatively, the graphics accelerator 446 may be integrated on the same package or chip as the processor 407. The processor 407 includes a plurality of cores 460A-460D, each with a translation lookaside buffer 461A-461D and one or more caches 462A-462D. The cores may include various other components for executing instructions and processing data which are not illustrated to avoid obscuring the underlying principles of the components described herein (e.g., instruction fetch units, branch prediction units, decoders, execution units, reorder buffers, etc.). The caches 462A-462D may comprise level 1 (L1) and level 2 (L2) caches. In addition, one or more shared cache(s) 456 may be included in the caching hierarchy and shared by sets of the cores 460A-460D. For example, one embodiment of the processor 407 includes 24 cores, each with its own L1 cache, twelve shared L2 caches, and twelve shared L3 caches. In this embodiment, one of the L2 and L3 caches are shared by two adjacent cores. The processor 407 connects with system memory 441, which may include processor memories 401-402.

[0115] Coherency is maintained for data and instructions stored in the various caches 462A-462D, shared cache(s) 456 and system memory 441 via inter-core communication over a coherence bus 464. For example, each cache may have cache coherency logic/circuitry associated therewith to communicate to over the coherence bus 464 in response to detected reads or writes to particular cache lines. In one implementation, a cache snooping protocol is implemented over the coherence bus 464 to snoop cache accesses. Cache snooping/coherency techniques are well understood by those of skill in the art and will not be described in detail here to avoid obscuring the underlying principles described herein. A proxy circuit 425 may be provided that communicatively couples the graphics accelerator 446 to the coherence bus 464, allowing the graphics accelerator 446 to participate in the cache coherence protocol as a peer of the cores. In particular, an interface 435 provides connectivity to the proxy circuit 425 over high-speed link 440 (e.g., a PCIe bus, NVLink, etc.) and an interface 437 connects the graphics accelerator 446 to the high-speed link 440.

[0116] In one implementation, the interface 437 couples with an accelerator integration circuit 436 that provides cache management, memory access, context management, and interrupt management services on behalf of graphics processing engines 431, 432, . . . , N of the graphics accelerator 446. The graphics processing engines 431, 432, . . . , N may each comprise a separate graphics processing unit (GPU). Alternatively, the graphics processing engines 431, 432, . . . , N may comprise different types of graphics processing engines within a GPU such as graphics execution units, media processing engines (e.g., video encoders/decoders), samplers, and block image transfer (BLIT) engines. In other words, the graphics accelerator may include graphics processing engines 431-432, . . . , N of a single GPU or the graphics processing engines 431-432, . . . , N may be associated with multiple GPUs integrated on a common

package, line card, or chip. The graphics processing engines **431-432**, . . . , **N** may be configured with any graphics processor or compute accelerator architecture described herein. Work to be performed by the graphics processing engines **431**, **432** can be specified via work descriptors that provide an indication of the work to be done by the graphics accelerator **446**.

[0117] The accelerator integration circuit **436** may include a memory management unit (MMU **439**) for performing various memory management functions such as virtual-to-physical memory translations (also referred to as effective-to-real memory translations) and memory access protocols for accessing the system memory **441**. The MMU **439** may also include a translation lookaside buffer (TLB) (not shown) for caching the virtual/effective to physical/real address translations. In one implementation, a cache **438** stores commands and data for efficient access by the graphics processing engines **431**, **432**, . . . , **N**. The data stored in cache **438** and graphics memories **433-434**, . . . , **M** may be kept coherent with the core caches **462A-462D**, shared cache(s) **456** and system memory **441**. As mentioned, this may be accomplished via a proxy circuit **425** which takes part in the cache coherency mechanism on behalf of the cache **438** and graphics memories **433-434**, . . . , **M** (e.g., sending updates to the cache **438** related to modifications/accesses of cache lines on processor caches **462A-462D**, shared cache(s) **456** and receiving updates from the cache **438**).

[0118] Registers **445** store context data for threads executed by the graphics processing engines **431-432**, . . . , **N** and a context management circuit **448** manages the thread contexts. For example, the context management circuit **448** may perform save and restore operations to save and restore contexts of the various threads during contexts switches (e.g., where a first thread is saved and a second thread is restored so that the second thread can be executed by a graphics processing engine). For example, on a context switch, the context management circuit **448** may store current register values to a designated region in memory (e.g., identified by a context pointer). It may then restore the register values when returning to the context. An interrupt management circuit **447**, for example, may receive and processes interrupts received from system devices.

[0119] In one implementation, virtual/effective addresses from a graphics processing engine are translated to real/physical addresses in system memory **441** by the MMU **439**. Optionally, the accelerator integration circuit **436** supports multiple (e.g., 4, 8, 16) graphics accelerators **446** and/or other accelerator devices. The graphics accelerator **446** may be dedicated to a single application executed on the processor **407** or may be shared between multiple applications. Optionally, a virtualized graphics execution environment is provided in which the resources of the graphics processing engines **431-432**, . . . , **N** are shared with multiple applications, virtual machines (VMs), or containers. The resources may be subdivided into “slices” which are allocated to different VMs and/or applications based on the processing requirements and priorities associated with the VMs and/or applications, or based on a pre-determined partitioning profile for a graphics accelerator **446**. VMs and containers can be used interchangeably herein.

[0120] A virtual machine (VM) can be software that runs an operating system and one or more applications. A VM can be defined by specification, configuration files, virtual disk

file, non-volatile random-access memory (NVRAM) setting file, and the log file and is backed by the physical resources of a host computing platform. A VM can include an operating system (OS) or application environment that is installed on software, which imitates dedicated hardware. The end user has the same experience on a virtual machine as they would have on dedicated hardware. Specialized software, called a hypervisor, emulates the PC client or server’s CPU, memory, hard disk, network, and other hardware resources completely, enabling virtual machines to share the resources. The hypervisor can emulate multiple virtual hardware platforms that are isolated from each other, allowing virtual machines to run Linux®, Windows® Server, VMware ESXi, and other operating systems on the same underlying physical host.

[0121] A container can be a software package of applications, configurations, and dependencies so the applications run reliably on one computing environment to another. Containers can share an operating system installed on the server platform and run as isolated processes. A container can be a software package that contains components that the software can utilize when running, such as system tools, libraries, and settings. Containers are not installed like traditional software programs, which allows them to be isolated from the other software and the operating system itself. The isolated nature of containers provides several benefits. First, the software in a container will run the same in different environments. For example, a container that includes PHP and MySQL can run identically on both a Linux® computer and a Windows® machine. Second, containers provide added security since the software will not affect the host operating system. While an installed application may alter system settings and modify resources, such as the Windows registry, a container can modify settings within the container.

[0122] Thus, the accelerator integration circuit **436** acts as a bridge to the system for the graphics accelerator **446** and provides address translation and system memory cache services. In one embodiment, to facilitate the bridging functionality, the accelerator integration circuit **436** may also include shared I/O **497** (e.g., PCIe, USB, or others) and hardware to enable system control of voltage, clocking, performance, thermals, and security. The shared I/O **497** may utilize separate physical connections or may traverse the high-speed link **440**. In addition, the accelerator integration circuit **436** may provide virtualization facilities for the host processor to manage virtualization of the graphics processing engines, interrupts, and memory management.

[0123] Because hardware resources of the graphics processing engines **431-432**, . . . , **N** are mapped explicitly to the real address space seen by the processor **407**, any host processor can address these resources directly using an effective address value. One optional function of the accelerator integration circuit **436** is the physical separation of the graphics processing engines **431-432**, . . . , **N** so that they appear to the system as independent units. In one embodiment, the accelerator integration circuit **436** includes security circuitry **444** that enables configurable cryptographic isolation of data associated with each slice of resources. Different slices can be associated with different security domains, such that data associated with the various security domains are encrypted using different cryptographic keys. In one embodiment, the security domains of the graphics accelerator **446** can be integrated into trusted execution

environments supported by the processor 407. In one embodiment, secure I/O capabilities can be enabled that allow each security domain to be presented as a separate trusted I/O device, which enables support for trusted DMA and MMIO operations.

[0124] One or more graphics memories 433-434, . . . , M may be coupled to each of the graphics processing engines 431-432, . . . , N, respectively. The graphics memories 433-434, . . . , M store instructions and data being processed by each of the graphics processing engines 431-432, . . . , N. The graphics memories 433-434, . . . , M may be volatile memories such as DRAMs (including stacked DRAMs), GDDR memory (e.g., GDDR5, GDDR6), or HBM, and/or may be non-volatile memories such as 3D XPoint/Optane, Samsung Z-NAND, or Nano-Ram.

[0125] To reduce data traffic over the high-speed link 440, biasing techniques may be used to ensure that the data stored in graphics memories 433-434, . . . , M is data which will be used most frequently by the graphics processing engines 431-432, . . . , N and preferably not used by the cores 460A-460D (at least not frequently). Similarly, the biasing mechanism attempts to keep data utilized by the cores (and preferably not the graphics processing engines 431-432, . . . , N) within the caches 462A-462D, shared cache(s) 456 and the system memory 441.

[0126] In an alternative variant, the accelerator integration circuit 436 is integrated within the processor 407 and the graphics processing engines 431-432, . . . , N communicate over the high-speed link 440 to the accelerator integration circuit 436 via interface 437 and interface 435 (which, again, may utilize any form of bus, fabric, or interface protocol). In this variant, the accelerator integration circuit 436 performs the same operations as those described above.

[0127] The embodiments described may support different programming models including a dedicated-process programming model (no graphics accelerator virtualization) and shared programming models (with virtualization). The latter may include programming models which are controlled by the accelerator integration circuit 436 and programming models which are controlled by the graphics accelerator 446. In the embodiments of the dedicated-process model, graphics processing engines 431, 432, . . . , N may be dedicated to a single application or process under a single operating system. The single application can funnel other application requests to the graphics processing engines 431, 432, . . . , N, providing virtualization within a VM/partition. In the dedicated-process programming models, the graphics processing engines 431, 432, . . . , N, may be shared by multiple VM/application partitions. The shared models utilize a system hypervisor to virtualize the graphics processing engines 431-432, . . . , N to allow access by each operating system. For single-partition systems without a hypervisor, the graphics processing engines 431-432, . . . , N are owned by the operating system. In both cases, the operating system can virtualize the graphics processing engines 431-432, . . . , N to provide access to each process or application. For the shared programming model, the graphics accelerator 446 or individual graphics processing engines 431-432, . . . , N selects a process element using a process handle. The process elements may be stored in system memory 441 and be addressable using the effective address to real address translation techniques described herein. The process handle may be an implementation-specific value provided to the host process when registering

its context with the graphics processing engines 431-432, . . . , N, which can be performed by the host process by calling system software to add the process element to the process element linked list. The lower 16-bits of the process handle may be the offset of the process element within the process element linked list.

[0128] FIG. 4C illustrates an accelerator integration slice 490. As used herein, a “slice” comprises a specified portion of the processing resources of the accelerator integration circuit 436. The application’s address space 482 within system memory 441 stores process elements 483. The process elements 483 may be stored in response to GPU invocations 481 from an application 480 executed on the processor 407. A process element 483 contains the process state for the application 480. A work descriptor (WD 484) contained in the process element 483 can be a single job requested by an application or may contain a pointer to a queue of jobs. In the latter case, the WD 484 is a pointer to the job request queue in the application’s address space 482.

[0129] The graphics accelerator 446 and/or the individual graphics processing engines 431-432, . . . , N can be shared by all or a subset of the processes in the system. For example, the technologies described herein may include an infrastructure for setting up the process state and sending a WD 484 to a graphics accelerator 446 to start a job in a virtualized environment.

[0130] In one implementation, the dedicated-process programming model is implementation-specific. In this model, a single process owns the graphics accelerator 446 or an individual graphics processing engine (e.g., graphics processing engine 431). Because the graphics accelerator 446 is owned by a single process, the hypervisor initializes the accelerator integration circuit 436 for the owning partition and the operating system initializes the accelerator integration circuit 436 for the owning process at the time when the graphics accelerator 446 is assigned.

[0131] In operation, the fetch unit 491 in the accelerator integration slice 490 fetches a WD 484 to be processed. The WD 484 includes the indication of work to be done by one or more graphics processing engines of the graphics accelerator 446. Data from the WD 484 may be stored in registers 445 and used by the MMU 439, interrupt management circuit 447 and/or context management circuit 448 as illustrated. For example, the MMU 439 may include segment/page walk circuitry for accessing segment/page tables 486 within the OS virtual address space 485. The interrupt management circuit 447 may process interrupt events 492 received from the graphics accelerator 446. When performing graphics operations, an effective address 493 generated by a graphics processing engine 431-432, . . . , N is translated to a real address by the MMU 439.

[0132] The registers 445 may be duplicated for each graphics processing engine 431-432, . . . , N and/or graphics accelerator 446 and may be initialized by the hypervisor or operating system. Each of these duplicated registers may be included in an accelerator integration slice 490. In one embodiment, each graphics processing engine 431-432, . . . , N may be presented to the hypervisor 496 as a distinct graphics processor device. Quality of Service (QoS) settings can be configured for clients of a specific graphics processing engine 431-432, . . . , N. Cryptographic and physical data isolation between the clients of each engine can be enabled via isolated memory access paths and

automatic data encryption for each client. Example registers that may be initialized by the hypervisor are shown in Table 1.

TABLE 1

| Hypervisor Initialized Registers |                                                                     |
|----------------------------------|---------------------------------------------------------------------|
| 1                                | Slice Control Register                                              |
| 2                                | Real Address (RA) Scheduled Processes Area Pointer                  |
| 3                                | Authority Mask Override Register                                    |
| 4                                | Interrupt Vector Table Entry Offset                                 |
| 5                                | Interrupt Vector Table Entry Limit                                  |
| 6                                | State Register                                                      |
| 7                                | Logical Partition ID                                                |
| 8                                | Real address (RA) Hypervisor Accelerator Utilization Record Pointer |
| 9                                | Storage Description Register                                        |

[0133] Example registers that may be initialized by the operating system are shown in Table 2.

TABLE 2

| Operating System Initialized Registers |                                                             |
|----------------------------------------|-------------------------------------------------------------|
| 1                                      | Process and Thread Identification                           |
| 2                                      | Effective Address (EA) Context Save/Restore Pointer         |
| 3                                      | Virtual Address (VA) Accelerator Utilization Record Pointer |
| 4                                      | Virtual Address (VA) Storage Segment Table Pointer          |
| 5                                      | Authority Mask                                              |
| 6                                      | Work descriptor                                             |

[0134] Each WD **484** may be specific to a particular graphics accelerator **446** and/or graphics processing engine **431-432**, . . . , **N**. It contains all the information a graphics processing engine utilizes to do its work, or it can be a pointer to a memory location where the application has set up a command queue of work to be completed.

[0135] FIG. 4D illustrates additional optional details of a shared model. It includes a hypervisor real address space **498** in which a process element list **499** is stored. The hypervisor real address space **498** is accessible via a hypervisor **496** which virtualizes the graphics processing engines for the operating system **495**.

[0136] The shared programming models allow for all or a subset of processes from all or a subset of partitions in the system to use a graphics accelerator **446**. There are two programming models where the graphics accelerator **446** is shared by multiple processes and partitions: time-sliced shared and graphics directed shared.

[0137] In this model, the hypervisor **496** owns the graphics accelerator **446** and makes its function available to all operating systems **495**. For a graphics accelerator **446** to support virtualization by the hypervisor **496**, the graphics accelerator **446** may adhere to the following requirements: 1) An application's job request should be autonomous (that is, the state does not have to be maintained between jobs), or the graphics accelerator **446** should provide a context save and restore mechanism. 2) An application's job request is guaranteed by the graphics accelerator **446** to complete in a specified amount of time, including any translation faults, or the graphics accelerator **446** provides the ability to preempt the processing of the job. 3) The graphics accelerator **446** should be guaranteed fairness between processes when operating in the directed shared programming model.

[0138] For the directed shared model, the application **480** may have to make an operating system **495** system call with

a graphics accelerator **446** type, a work descriptor (WD), an authority mask register (AMR) value, and a context save/restore area pointer (CSRP). The graphics accelerator **446** type describes the targeted acceleration function for the system call. The graphics accelerator **446** type may be a system-specific value. The WD is formatted specifically for the graphics accelerator **446** and can be in the form of a graphics accelerator **446** command, an effective address pointer to a user-defined structure, an effective address pointer to a queue of commands, or any other data structure to describe the work to be done by the graphics accelerator **446**. In one embodiment, the AMR value is the AMR state to use for the current process. The value passed to the operating system is similar to an application setting the AMR. If the accelerator integration circuit **436** and graphics accelerator **446** implementations do not support a User Authority Mask Override Register (UAMOR), the operating system may apply the current UAMOR value to the AMR value before passing the AMR in the hypervisor call. The hypervisor **496** may optionally apply the current Authority Mask Override Register (AMOR) value before placing the AMR into the process element **483**. The CSRP may be one of the registers **445** containing the effective address of an area in the application's address space **482** for the graphics accelerator **446** to save and restore the context state. This pointer is optional if no state is to be saved between jobs or when a job is preempted. The context save/restore area may be pinned system memory.

[0139] Upon receiving the system call, the operating system **495** may verify that the application **480** has registered and been given the authority to use the graphics accelerator **446**. The operating system **495** then calls the hypervisor **496** with the information shown in Table 3.

TABLE 3

| OS to Hypervisor Call Parameters |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| 1                                | A work descriptor (WD)                                               |
| 2                                | An Authority Mask Register (AMR) value (potentially masked).         |
| 3                                | An effective address (EA) Context Save/Restore Area Pointer (CSRP)   |
| 4                                | A process ID (PID) and optional thread ID (TID)                      |
| 5                                | A virtual address (VA) accelerator utilization record pointer (AURP) |
| 6                                | The virtual address of the storage segment table pointer (SSTP)      |
| 7                                | A logical interrupt service number (LISN)                            |

[0140] Upon receiving the hypervisor call, the hypervisor **496** verifies that the operating system **495** has registered and been given the authority to use the graphics accelerator **446**. The hypervisor **496** then puts the process element **483** into the process element linked list for the corresponding graphics accelerator **446** type. The process element may include the information shown in Table 4.

TABLE 4

| Process Element Information |                                                                    |
|-----------------------------|--------------------------------------------------------------------|
| 1                           | A work descriptor (WD)                                             |
| 2                           | An Authority Mask Register (AMR) value (potentially masked).       |
| 3                           | An effective address (EA) Context Save/Restore Area Pointer (CSRP) |

TABLE 4-continued

| Process Element Information |                                                                       |  |
|-----------------------------|-----------------------------------------------------------------------|--|
| 4                           | A process ID (PID) and optional thread ID (TID)                       |  |
| 5                           | A virtual address (VA) accelerator utilization record pointer (AURP)  |  |
| 6                           | The virtual address of the storage segment table pointer (SSTP)       |  |
| 7                           | A logical interrupt service number (LISN)                             |  |
| 8                           | Interrupt vector table, derived from the hypervisor call parameters.  |  |
| 9                           | A state register (SR) value                                           |  |
| 10                          | A logical partition ID (LPID)                                         |  |
| 11                          | A real address (RA) hypervisor accelerator utilization record pointer |  |
| 12                          | The Storage Descriptor Register (SDR)                                 |  |

[0141] The hypervisor may initialize the registers 445 of accelerator integration slice 490.

[0142] As illustrated in FIG. 4E, in one optional implementation a unified memory addressable via a common virtual memory address space used to access the physical processor memories 401-402 and GPU memories 420-423 is employed. In this implementation, operations executed on the GPUs 410-413 utilize the same virtual/effective memory address space to access the processors memories 401-402 and vice versa, thereby simplifying programmability. A first portion of the virtual/effective address space may be allocated to the processor memory 401, a second portion to the second processor memory 402, a third portion to the GPU memory 420, and so on. The entire virtual/effective memory space (sometimes referred to as the effective address space) may thereby be distributed across each of the processor memories 401-402 and GPU memories 420-423, allowing any processor or GPU to access any physical memory with a virtual address mapped to that memory.

[0143] Bias/coherence management circuitry 494A-494E within one or more of the MMUs 439A-439E may be provided that ensures cache coherence between the caches of the host processors (e.g., multi-core processor 405) and the GPUs 410-413 and implements biasing techniques indicating the physical memories in which certain types of data should be stored. While multiple instances of bias/coherence management circuitry 494A-494E are illustrated in FIG. 4E, the bias/coherence circuitry may be implemented within the MMU of one or more host processors and/or within the accelerator integration circuit 436.

[0144] The GPU-attached memory 420-423 may be mapped as part of system memory and accessed using shared virtual memory (SVM) technology, but without suffering the typical performance drawbacks associated with full system cache coherence. The ability to GPU-attached memory 420-423 to be accessed as system memory without onerous cache coherence overhead provides a beneficial operating environment for GPU offload. This arrangement allows the host processor software to setup operands and access computation results, without the overhead of tradition I/O DMA data copies. Such traditional copies involve driver calls, interrupts and memory mapped I/O (MMIO) accesses that are all inefficient relative to simple memory accesses. At the same time, the ability to access GPU attached memory 420-423 without cache coherence overheads can affect the execution time of an offloaded computation. In cases with substantial streaming write memory traffic, for example, cache coherence overhead can significantly reduce the effective write bandwidth seen by a GPU 410-413. The efficiency

of operand setup, the efficiency of results access, and the efficiency of GPU computation all play a role in determining the effectiveness of GPU offload.

[0145] A selection between GPU bias and host processor bias may be driven by a bias tracker data structure. A bias table may be used, for example, which may be a page-granular structure (i.e., controlled at the granularity of a memory page) that includes 1 or 2 bits per GPU-attached memory page. The bias table may be implemented in a stolen memory range of one or more GPU-attached memories 420-423, with or without a bias cache in the GPU 410-413 (e.g., to cache frequently/recently used entries of the bias table). Alternatively, the entire bias table may be maintained within the GPU.

[0146] In one implementation, the bias table entry associated with each access to the GPU-attached memory 420-423 is accessed prior the actual access to the GPU memory, causing the following operations. First, local requests from the GPU 410-413 that find their page in GPU bias are forwarded directly to a corresponding GPU memory 420-423. Local requests from the GPU that find their page in host bias are forwarded to the processor (e.g., over a high-speed link as discussed above). Optionally, requests from the host processor that find the requested page in host processor bias complete the request like a normal memory read. Alternatively, requests directed to a GPU-biased page may be forwarded to the GPU 410-413. The GPU may then transition the page to a host processor bias if it is not currently using the page. The bias state of a page can be changed either by a software-based mechanism, a hardware-assisted software-based mechanism, or, for a limited set of cases, a purely hardware-based mechanism. One mechanism for changing the bias state employs an API call (e.g., OpenCL), which, in turn, calls the GPU's device driver which, in turn, sends a message (or enqueues a command descriptor) to the GPU directing it to change the bias state and, for some transitions, perform a cache flushing operation in the host. The cache flushing operation is utilized for a transition from host processor bias to GPU bias but is not utilized for the opposite transition.

[0147] Cache coherency may be maintained by temporarily rendering GPU-biased pages uncacheable by the host processor. To access these pages, the host processor may request access from the GPU 410 which may or may not grant access right away, depending on the implementation. Thus, to reduce communication between the host processor and GPU 410 it is beneficial to ensure that GPU-biased pages are those which are utilized by the GPU but not the host processor and vice versa.

#### Graphics Processing Pipeline

[0148] FIG. 5 illustrates a graphics processing pipeline 500. A graphics multiprocessor, such as graphics multiprocessor 234 as in FIG. 2D or graphics multiprocessor 235 of FIG. 2E can implement the graphics processing pipeline 500. The graphics multiprocessor can be included within the parallel processing subsystems as described herein, such as the parallel processor 200 of FIG. 2A, which may be related to the parallel processor(s) 112 of FIG. 1 and may be used in place of one of those. The various parallel processing systems can implement the graphics processing pipeline 500 via one or more instances of the parallel processing unit (e.g., parallel processing unit 202 of FIG. 2A) as described herein. For example, a shader unit (e.g., graphics multipro-

cessor 234 of FIG. 2C) may be configured to perform the functions of one or more of a vertex processing unit 504, a tessellation control processing unit 508, a tessellation evaluation processing unit 512, a geometry processing unit 516, and a fragment/pixel processing unit 524. The functions of data assembler 502, primitive assemblers 506, 514, 518, tessellation unit 510, rasterizer 522, and raster operations unit 526 may also be performed by other processing engines within a processing cluster (e.g., processing cluster 214 of FIG. 2A) and a corresponding partition unit (e.g., partition unit 220A-220N of FIG. 2A). The graphics processing pipeline 500 may also be implemented using dedicated processing units for one or more functions. It is also possible that one or more portions of the graphics processing pipeline 500 are performed by parallel processing logic within a general-purpose processor (e.g., CPU). Optionally, one or more portions of the graphics processing pipeline 500 can access on-chip memory (e.g., parallel processor memory 222 as in FIG. 2A) via a memory interface 528, which may be an instance of the memory interface 218 of FIG. 2A. The graphics processing pipeline 500 may also be implemented via a multi-core group 365A as in FIG. 3.

[0149] The data assembler 502 is a processing unit that may collect vertex data for surfaces and primitives. The data assembler 502 then outputs the vertex data, including the vertex attributes, to the vertex processing unit 504. The vertex processing unit 504 is a programmable execution unit that executes vertex shader programs, lighting and transforming vertex data as specified by the vertex shader programs. The vertex processing unit 504 reads data that is stored in cache, local or system memory for use in processing the vertex data and may be programmed to transform the vertex data from an object-based coordinate representation to a world space coordinate space or a normalized device coordinate space.

[0150] A first instance of a primitive assembler 506 receives vertex attributes from the vertex processing unit 504. The primitive assembler 506 readings stored vertex attributes and constructs graphics primitives for processing by tessellation control processing unit 508. The graphics primitives include triangles, line segments, points, patches, and so forth, as supported by various graphics processing application programming interfaces (APIs).

[0151] The tessellation control processing unit 508 treats the input vertices as control points for a geometric patch. The control points are transformed from an input representation from the patch (e.g., the patch's bases) to a representation that is suitable for use in surface evaluation by the tessellation evaluation processing unit 512. The tessellation control processing unit 508 can also compute tessellation factors for edges of geometric patches. A tessellation factor applies to a single edge and quantifies a view-dependent level of detail associated with the edge. A tessellation unit 510 is configured to receive the tessellation factors for edges of a patch and to tessellate the patch into multiple geometric primitives such as line, triangle, or quadrilateral primitives, which are transmitted to a tessellation evaluation processing unit 512. The tessellation evaluation processing unit 512 operates on parameterized coordinates of the subdivided patch to generate a surface representation and vertex attributes for each vertex associated with the geometric primitives.

[0152] A second instance of a primitive assembler 514 receives vertex attributes from the tessellation evaluation

processing unit 512, reading stored vertex attributes, and constructs graphics primitives for processing by the geometry processing unit 516. The geometry processing unit 516 is a programmable execution unit that executes geometry shader programs to transform graphics primitives received from primitive assembler 514 as specified by the geometry shader programs. The geometry processing unit 516 may be programmed to subdivide the graphics primitives into one or more new graphics primitives and calculate parameters used to rasterize the new graphics primitives.

[0153] The geometry processing unit 516 may be able to add or delete elements in the geometry stream. The geometry processing unit 516 outputs the parameters and vertices specifying new graphics primitives to primitive assembler 518. The primitive assembler 518 receives the parameters and vertices from the geometry processing unit 516 and constructs graphics primitives for processing by a viewport scale, cull, and clip unit 520. The geometry processing unit 516 reads data that is stored in parallel processor memory or system memory for use in processing the geometry data. The viewport scale, cull, and clip unit 520 performs clipping, culling, and viewport scaling and outputs processed graphics primitives to a rasterizer 522.

[0154] The rasterizer 522 can perform depth culling and other depth-based optimizations. The rasterizer 522 also performs scan conversion on the new graphics primitives to generate fragments and output those fragments and associated coverage data to the fragment/pixel processing unit 524. The fragment/pixel processing unit 524 is a programmable execution unit that is configured to execute fragment shader programs or pixel shader programs. The fragment/pixel processing unit 524 transforming fragments or pixels received from rasterizer 522, as specified by the fragment or pixel shader programs. For example, the fragment/pixel processing unit 524 may be programmed to perform operations included but not limited to texture mapping, shading, blending, texture correction and perspective correction to produce shaded fragments or pixels that are output to a raster operations unit 526. The fragment/pixel processing unit 524 can read data that is stored in either the parallel processor memory or the system memory for use when processing the fragment data. Fragment or pixel shader programs may be configured to shade at sample, pixel, tile, or other granularities depending on the sampling rate configured for the processing units.

[0155] The raster operations unit 526 is a processing unit that performs raster operations including, but not limited to stencil, z-test, blending, and the like, and outputs pixel data as processed graphics data to be stored in graphics memory (e.g., parallel processor memory 222 as in FIG. 2A, and/or system memory 104 as in FIG. 1), to be displayed on the one or more display device(s) 110A-110B or for further processing by one of the one or more processor(s) 102 or parallel processor(s) 112. The raster operations unit 526 may be configured to compress z or color data that is written to memory and decompress z or color data that is read from memory.

#### Machine Learning Overview

[0156] The architecture described above can be applied to perform training and inference operations using machine learning models. Machine learning has been successful at solving many kinds of tasks. The computations that arise when training and using machine learning algorithms (e.g.,

neural networks) lend themselves naturally to efficient parallel implementations. Accordingly, parallel processors such as general-purpose graphics processing units (GPGPUs) have played a significant role in the practical implementation of deep neural networks. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In an SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. The efficiency provided by parallel machine learning algorithm implementations allows the use of high-capacity networks and enables those networks to be trained on larger datasets.

**[0157]** A machine learning algorithm is an algorithm that can learn based on a set of data. For example, machine learning algorithms can be designed to model high-level abstractions within a data set. For example, image recognition algorithms can be used to determine which of several categories to which a given input belong; regression algorithms can output a numerical value given an input; and pattern recognition algorithms can be used to generate translated text or perform text to speech and/or speech recognition.

**[0158]** An example type of machine learning algorithm is a neural network. There are many types of neural networks; a simple type of neural network is a feedforward network. A feedforward network may be implemented as an acyclic graph in which the nodes are arranged in layers. Typically, a feedforward network topology includes an input layer and an output layer that are separated by at least one hidden layer. The hidden layer transforms input received by the input layer into a representation that is useful for generating output in the output layer. The network nodes are fully connected via edges to the nodes in adjacent layers, but there are no edges between nodes within each layer. Data received at the nodes of an input layer of a feedforward network are propagated (i.e., “fed forward”) to the nodes of the output layer via an activation function that calculates the states of the nodes of each successive layer in the network based on coefficients (“weights”) respectively associated with each of the edges connecting the layers. Depending on the specific model being represented by the algorithm being executed, the output from the neural network algorithm can take various forms.

**[0159]** Before a machine learning algorithm can be used to model a particular problem, the algorithm is trained using a training data set. Training a neural network involves selecting a network topology, using a set of training data representing a problem being modeled by the network, and adjusting the weights until the network model performs with a minimal error for all instances of the training data set. For example, during a supervised learning training process for a neural network, the output produced by the network in response to the input representing an instance in a training data set is compared to the “correct” labeled output for that instance, an error signal representing the difference between the output and the labeled output is calculated, and the weights associated with the connections are adjusted to minimize that error as the error signal is backward propagated through the layers of the network. The network is considered “trained” when the errors for each of the outputs generated from the instances of the training data set are minimized.

**[0160]** The accuracy of a machine learning algorithm can be affected significantly by the quality of the data set used to train the algorithm. The training process can be computationally intensive and may utilize a significant amount of time on a conventional general-purpose processor. Accordingly, parallel processing hardware is used to train many types of machine learning algorithms. This is particularly useful for optimizing the training of neural networks, as the computations performed in adjusting the coefficients in neural networks lend themselves naturally to parallel implementations. Specifically, many machine learning algorithms and software applications have been adapted to make use of the parallel processing hardware within general-purpose graphics processing devices.

**[0161]** FIG. 6 is a generalized diagram of a machine learning software stack 600. A machine learning application 602 is any logic that can be configured to train a neural network using a training dataset or to use a trained deep neural network to implement machine intelligence. The machine learning application 602 can include training and inference functionality for a neural network and/or specialized software that can be used to train a neural network before deployment. The machine learning application 602 can implement any type of machine intelligence including but not limited to image recognition, mapping and localization, autonomous navigation, speech synthesis, medical imaging, or language translation. Example machine learning applications 602 include, but are not limited to, voice-based virtual assistants, image or facial recognition algorithms, autonomous navigation, and the software tools that are used to train the machine learning models used by the machine learning applications 602.

**[0162]** Hardware acceleration for the machine learning application 602 can be enabled via a machine learning framework 604. The machine learning framework 604 can provide a library of machine learning primitives. Machine learning primitives are basic operations that are commonly performed by machine learning algorithms. Without the machine learning framework 604, developers of machine learning algorithms would have to create and optimize the main computational logic associated with the machine learning algorithm, then re-optimize the computational logic as new parallel processors are developed. Instead, the machine learning application can be configured to perform the computations using the primitives provided by the machine learning framework 604. Example primitives include tensor convolutions, activation functions, and pooling, which are computational operations that are performed while training a convolutional neural network (CNN). The machine learning framework 604 can also provide primitives to implement basic linear algebra subprograms performed by many machine-learning algorithms, such as matrix and vector operations. Examples of a machine learning framework 604 include, but are not limited to, TensorFlow, TensorRT, PyTorch, MXNet, Caffe, and other high-level machine learning frameworks.

**[0163]** The machine learning framework 604 can process input data received from the machine learning application 602 and generate the appropriate input to a compute framework 606. The compute framework 606 can abstract the underlying instructions provided to the GPGPU driver 608 to enable the machine learning framework 604 to take advantage of hardware acceleration via the GPGPU hardware 610 without the machine learning framework 604

having to have intimate knowledge of the architecture of the GPGPU hardware 610. Additionally, the compute framework 606 can enable hardware acceleration for the machine learning framework 604 across a variety of types and generations of the GPGPU hardware 610. In one example, compute frameworks 606 can include the CUDA compute framework and associated machine learning libraries, such as the CUDA Deep Neural Network (cuDNN) library. The machine learning software stack 600 can also include communication libraries or frameworks to facilitate multi-GPU and multi-node compute.

#### GPGPU Machine Learning Acceleration

[0164] FIG. 7 illustrates a general-purpose graphics processing unit (GPGPU 700), which may be the parallel processor 200 of FIG. 2A or the parallel processor(s) 112 of FIG. 1. The general-purpose processing unit may be configured to provide support for hardware acceleration of primitives provided by a machine learning framework to accelerate the processing the type of computational workloads associated with training deep neural networks. Additionally, the GPGPU 700 can be linked directly to other instances of the GPGPU to create a multi-GPU cluster to improve training speed for particularly deep neural networks. Primitives are also supported to accelerate inference operations for deployed neural networks.

[0165] The GPGPU 700 includes a host interface 702 to enable a connection with a host processor. The host interface 702 may be a PCI Express interface. However, the host interface can also be a vendor specific communications interface or communications fabric. The GPGPU 700 receives commands from the host processor and uses a global scheduler 704 to distribute execution threads associated with those commands to a set of processing clusters 706A-706H. The processing clusters 706A-706H share a cache memory 708. The cache memory 708 can serve as a higher-level cache for cache memories within the processing clusters 706A-706H. The illustrated processing clusters 706A-706H may correspond with processing clusters 214A-214N as in FIG. 2A.

[0166] The GPGPU 700 includes memory 714A-714B coupled with the processing clusters 706A-706H via a set of memory controllers 712A-712B. The memory 714A-714B can include various types of memory devices including dynamic random-access memory (DRAM) or graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. The memory 714A-714B may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM).

[0167] Each of the processing clusters 706A-706H may include a set of graphics multiprocessors, such as the graphics multiprocessor 234 of FIG. 2D, graphics multiprocessor 235 of FIG. 2E, or may include a multi-core group 365A-365N as in FIG. 3. The graphics multiprocessors of the compute cluster include multiple types of integer and floating-point logic units that can perform computational operations at a range of precisions including suited for machine learning computations. For example, at least a subset of the floating-point units in each of the processing clusters 706A-706H can be configured to perform 16-bit or 32-bit floating-point operations, while a different subset of the floating-point units can be configured to perform 64-bit floating-point operations.

[0168] Multiple instances of the GPGPU 700 can be configured to operate as a compute cluster. The communication mechanism used by the compute cluster for synchronization and data exchange varies across embodiments. For example, the multiple instances of the GPGPU 700 communicate over the host interface 702. In one embodiment the GPGPU 700 includes an I/O hub 709 that couples the GPGPU 700 with a GPU link 710 that enables a direct connection to other instances of the GPGPU. The GPU link 710 may be coupled to a dedicated GPU-to-GPU bridge that enables communication and synchronization between multiple instances of the GPGPU 700. Optionally, the GPU link 710 couples with a high-speed interconnect to transmit and receive data to other GPGPUs or parallel processors. The multiple instances of the GPGPU 700 may be located in separate data processing systems and communicate via a network device that is accessible via the host interface 702. The GPU link 710 may be configured to enable a connection to a host processor in addition to or as an alternative to the host interface 702.

[0169] While the illustrated configuration of the GPGPU 700 can be configured to train neural networks, an alternate configuration of the GPGPU 700 can be configured for deployment within a high performance or low power inferencing platform. In an inferencing configuration, the GPGPU 700 includes fewer of the processing clusters 706A-706H relative to the training configuration. Additionally, memory technology associated with the memory 714A-714B may differ between inferencing and training configurations. In one embodiment, the inferencing configuration of the GPGPU 700 can support inferencing specific instructions. For example, an inferencing configuration can provide support for one or more 8-bit integer or floating-point dot product instructions, which are commonly used during inferencing operations for deployed neural networks.

[0170] FIG. 8 illustrates a multi-GPU computing system 800. The multi-GPU computing system 800 can include a processor 802 coupled to multiple GPGPUs 806A-806D via a host interface switch 804. The host interface switch 804 may be a PCI express switch device that couples the processor 802 to a PCI express bus over which the processor 802 can communicate with the set of GPGPUs 806A-806D. Each of the multiple GPGPUs 806A-806D can be an instance of the GPGPU 700 of FIG. 7. The GPGPUs 806A-806D can interconnect via a set of high-speed point to point GPU to GPU links (P2P GPU links 816). The high-speed GPU to GPU links can connect to each of the GPGPUs 806A-806D via a dedicated GPU link, such as the GPU link 710 as in FIG. 7. The P2P GPU links 816 enable direct communication between each of the GPGPUs 806A-806D without utilizing communication over the host interface bus to which the processor 802 is connected. With GPU-to-GPU traffic directed to the P2P GPU links, the host interface bus remains available for system memory access or to communicate with other instances of the multi-GPU computing system 800, for example, via one or more network devices. While in FIG. 8 the GPGPUs 806A-806D connect to the processor 802 via the host interface switch 804, the processor 802 may alternatively include direct support for the P2P GPU links 816 and connect directly to the GPGPUs 806A-806D. In one embodiment the P2P GPU link 816 enable the multi-GPU computing system 800 to operate as a single logical GPU.

## Machine Learning Neural Network Implementations

**[0171]** The computing architecture described herein can be configured to perform the types of parallel processing that is particularly suited for training and deploying neural networks for machine learning. A neural network can be generalized as a network of functions having a graph relationship. As is well-known in the art, there are a variety of types of neural network implementations used in machine learning. One example type of neural network is the feed-forward network, as previously described. A second example type of neural network is the Convolutional Neural Network (CNN), while a third example type of neural network are recurrent neural networks (RNNs).

**[0172]** A CNN is a specialized feedforward neural network for processing data having a known, grid-like topology, such as image data. Accordingly, CNNs are commonly used for compute vision and image recognition applications, but they also may be used for other types of pattern recognition such as speech and language processing. The nodes in the CNN input layer are organized into a set of “filters” (feature detectors inspired by the receptive fields found in the retina), and the output of each set of filters is propagated to nodes in successive layers of the network. The computations for a CNN include applying the convolution mathematical operation to each filter to produce the output of that filter. Convolution is a specialized kind of mathematical operation performed by two functions to produce a third function that is a modified version of one of the two original functions. In convolutional network terminology, the first function to the convolution can be referred to as the input, while the second function can be referred to as the convolution kernel. The output may be referred to as the feature map. The input to a convolution layer can be a multidimensional array of data that defines the various color components of an input image. The convolution kernel can be a multidimensional array of parameters, where the parameters are adapted by the training process for the neural network.

**[0173]** RNNs are a family of feedforward neural networks that include feedback connections between layers. RNNs enable the modeling of sequential data by sharing parameter data across different parts of the neural network. The architecture for an RNN includes cycles that represent the influence of a present value of a variable on its own value at a future time, as at least a portion of the output data from the RNN is used as feedback for processing subsequent input in a sequence. This feature makes RNNs particularly useful for language processing due to the variable nature in which language data can be composed.

**[0174]** The figures described below present example feed-forward, CNN, and RNN networks, as well as describe a general process for respectively training and deploying each of those types of networks. It will be understood that these descriptions are example and non-limiting as to any specific embodiment described herein and the concepts illustrated can be applied generally to deep neural networks and machine learning techniques in general.

**[0175]** Deep neural networks used in deep learning typically include a front-end network to perform feature recognition coupled to a back-end network which represents a mathematical model that can perform operations (e.g., object classification, speech recognition, etc.) based on the feature representation provided to the model. Deep learning enables machine learning to be performed without utilizing

hand crafted feature engineering to be performed for the model. Instead, deep neural networks can learn features based on statistical structure or correlation within the input data. The learned features can be provided to a mathematical model that can map detected features to an output. The mathematical model used by the network is generally specialized for the specific task to be performed, and different models will be used to perform different task.

**[0176]** Once the neural network is structured, a learning model can be applied to the network to train the network to perform specific tasks. The learning model describes how to adjust the weights within the model to reduce the output error of the network. Backpropagation of errors is a common method used to train neural networks. An input vector is presented to the network for processing. The output of the network is compared to the desired output using a loss function and an error value is calculated for each of the neurons in the output layer. The error values are then propagated backwards until each neuron has an associated error value which roughly represents its contribution to the original output. The network can then learn from those errors using an algorithm, such as the stochastic gradient descent algorithm, to update the weights of the of the neural network.

**[0177]** FIG. 9A-9B illustrate an example convolutional neural network. FIG. 9A illustrates various layers within a CNN. As shown in FIG. 9A, an example CNN used to model image processing can receive input **902** describing the red, green, and blue (RGB) components of an input image. The input **902** can be processed by multiple convolutional layers (e.g., first convolutional layer **904**, second convolutional layer **906**). The output from the multiple convolutional layers may optionally be processed by fully connected layers **908**. Neurons in a fully connected layer have full connections to all activations in the previous layer, as previously described for a feedforward network. The output from the fully connected layers **908** can be used to generate an output result from the network. The activations within the fully connected layers **908** can be computed using matrix multiplication instead of convolution. Not all CNN implementations make use of fully connected layers **908**. For example, in some implementations the second convolutional layer **906** can generate output for the CNN.

**[0178]** The convolutional layers are sparsely connected, which differs from traditional neural network configuration found in the fully connected layers **908**. Traditional neural network layers are fully connected, such that every output unit interacts with every input unit. However, the convolutional layers are sparsely connected because the output of the convolution of a field is input (instead of the respective state value of each of the nodes in the field) to the nodes of the subsequent layer, as illustrated. The kernels associated with the convolutional layers perform convolution operations, the output of which is sent to the next layer. The dimensionality reduction performed within the convolutional layers is one aspect that enables the CNN to scale to process large images.

**[0179]** FIG. 9B illustrates example computation stages within a convolutional layer of a CNN. Input to a convolutional layer **912** of a CNN can be processed in three stages of a convolutional layer **914**. The three stages can include a convolution stage **916**, a detector stage **918**, and a pooling stage **920**. The convolutional layer **914** can then output data to a successive convolutional layer. The final convolutional

layer of the network can generate output feature map data or provide input to a fully connected layer, for example, to generate a classification value for the input to the CNN.

[0180] In the convolution stage 916 performs several convolutions in parallel to produce a set of linear activations. The convolution stage 916 can include an affine transformation, which is any transformation that can be specified as a linear transformation plus a translation. Affine transformations include rotations, translations, scaling, and combinations of these transformations. The convolution stage computes the output of functions (e.g., neurons) that are connected to specific regions in the input, which can be determined as the local region associated with the neuron. The neurons compute a dot product between the weights of the neurons and the region in the local input to which the neurons are connected. The output from the convolution stage 916 defines a set of linear activations that are processed by successive stages of the convolutional layer 914.

[0181] The linear activations can be processed by a detector stage 918. In the detector stage 918, each linear activation is processed by a non-linear activation function. The non-linear activation function increases the nonlinear properties of the overall network without affecting the receptive fields of the convolution layer. Several types of non-linear activation functions may be used. One particular type is the rectified linear unit (ReLU), which uses an activation function defined as  $f(x) = \max(0, x)$ , such that the activation is thresholded at zero.

[0182] The pooling stage 920 uses a pooling function that replaces the output of the second convolutional layer 906 with a summary statistic of the nearby outputs. The pooling function can be used to introduce translation invariance into the neural network, such that small translations to the input do not change the pooled outputs. Invariance to local translation can be useful in scenarios where the presence of a feature in the input data is more important than the precise location of the feature. Various types of pooling functions can be used during the pooling stage 920, including max pooling, average pooling, and l2-norm pooling. Additionally, some CNN implementations do not include a pooling stage. Instead, such implementations substitute and additional convolution stage having an increased stride relative to previous convolution stages.

[0183] The output from the convolutional layer 914 can then be processed by the next layer 922. The next layer 922 can be an additional convolutional layer or one of the fully connected layers 908. For example, the first convolutional layer 904 of FIG. 9A can output to the second convolutional layer 906, while the second convolutional layer can output to a first layer of the fully connected layers 908.

[0184] A variant on the CNN is a convolutional deep belief network, which has a structure similar to a CNN and is trained in a manner similar to a deep belief network. A deep belief network (DBN) is a generative neural network that is composed of multiple layers of stochastic (random) variables. DBNs can be trained layer-by-layer using greedy unsupervised learning. The learned weights of the DBN can then be used to provide pre-train neural networks by determining an initial set of weights for the neural network.

[0185] FIG. 10A-10B illustrate an example language models. FIG. 10A illustrates a recurrent neural network (RNN 1000). In a recurrent neural network (RNN), the previous state of the network influences the output of the current state of the network. RNNs can be built in a variety of ways using

a variety of functions. The use of RNNs generally revolves around using mathematical models to predict the future based on a prior sequence of inputs. For example, an RNN may be used to perform statistical language modeling to predict an upcoming word given a previous sequence of words. The RNN 1000 can be described as having an input layer 1002 that receives an input vector, hidden layers 1004 to implement a recurrent function, a feedback mechanism 1005 to enable a 'memory' of previous states, and an output layer 1006 to output a result. The RNN 1000 operates based on time-steps. The state of the RNN at a given time step is influenced based on the previous time step via the feedback mechanism 1005. For a given time step, the state of the hidden layers 1004 is defined by the previous state and the input at the current time step. An initial input ( $x_1$ ) at a first-time step can be processed by the hidden layer 1004. A second input ( $x_2$ ) can be processed by the hidden layer 1004 using state information that is determined during the processing of the initial input ( $x_1$ ). A given state can be computed as  $s_t = f(Ux_t + Ws_{t-1})$ , where U and W are parameter matrices. The function f is generally a nonlinearity, such as the hyperbolic tangent function (Tanh) or a variant of the rectifier function  $f(x) = \max(0, x)$ . However, the specific mathematical function used in the hidden layers 1004 can vary depending on the specific implementation details of the RNN 1000. Acceleration for variations on RNN networks may also be enabled. One example RNN variant is the long short term memory (LSTM) RNN. LSTM RNNs are capable of learning long-term dependencies that may be utilized for processing longer sequences of language.

[0186] FIG. 10B illustrates baseline components of a transformer model 1010. The transformer model 1010 address issues found in RNN models with long input sequences as well as enabling enable increased parallelization. The transformer model 1010, and variants thereof, are used to build Language Models (LLMs) that perform various tasks such as machine translation, automatic summarization, and dialogue management. A transformer model 1010 may also be configured to perform image generation tasks, such as text-to-image generation.

[0187] The transformer model 1010 includes multiple instances of an encoder 1016 and a decoder 1026. The encoders are stacked end-to-end, with the output of the final encoder being routed as input to a multiheaded attention layer of each decoder 1026. Input that enters the bottommost instance of encoder 1016 is processed by input embeddings 1012 that convert input tokens into vectors that can be processed by the encoder 1016. The output dictionary is vectorized by output embeddings 1022 before entry into the bottommost instance of decoder 1026. Positional encodings 1014, 1024 are added to input vectors and output vectors at the bottoms of the encoder 1016 and decoder 1026 stacks. The positional encodings 1014, 1024, inject information about the relative or absolute position of tokens in a sequence of tokens to be processed as the transformer model 1010 does not naturally encode the order of the tokens.

[0188] The encoder 1016 of the transformer model 1010 analyses the input text and creates a number of hidden states that protect the context and meaning of the text data. Encoder 1016 layers make up part of the core of the transformer architecture, although decoder 1026 variants of the transformer model 1010 are also possible. The encoder 1016 includes two sublayers, a multi-head attention (MHA) sublayer and a feedforward network (FFN) sublayer. The

MHA sublayer performs multiple concurrent self-attention operations to compute attention scores, which enable the transformer model **1010** weigh the importance and relative relationships of different tokens in the input sequence in a context-aware manner. The FFN sublayer is a position-wise fully connected feedforward neural network. The output of each sublayer is processed by an add and normalize (A&N) operation, defined as  $\text{LayerNorm}(x + \text{Sublayer}(x))$ , where the output of the sublayer is added to the input of the sublayer and normalized. In some implementations, the normalization operation may be performed before the sublayer instead of afterwards.

[0189] The decoder **1026** includes three sublayers, a masked MHA sublayer, an MHA sublayer, and an FFN sublayer. The masked MHA sublayer is similar to the MHA layer, excepting that masking is applied to prevent query position from attending to the keys of future positions. The MHA sublayer of the decoder **1026** is similar to the MHA sublayer of the encoder, with an additional input from the encoder **1016**. The FFN of the decoder **1026** is the same as the FFN of the encoder **1016**. The Linear and Softmax block takes the output of the last instance of decoder **1026** of the decoder stack and generates a probability distribution that represent output probabilities.

[0190] GPU acceleration is also available for variants of the transformer model **1010** that replace some or all FFN sublayers with sparse mixture of experts (MoE) layers. MoE layers each include a number of experts, where each expert is a neural network. The MoE layers may themselves be FFNs, or may also be MoEs, allowing hierarchical MoE layers.

[0191] The training of a transformer model **1010** can be optimized via the use of adaptive precision logic that adjusts the computation precision that is applied during training. The adaptive precision logic can attempt to use the smallest datatype possible during training without significantly reducing training accuracy. For example, to minimize data loss due to the use of 16-bit, 8-bit, and 4-bit floating-point format, dynamic scaling and casting can be applied during training based on the statistical analysis of the tensor data that is generated during training. Tensor data that is generated during training can be statistically analyzed using a variety of analysis techniques to determine a set of scaling factors to apply to blocks of data, such as the input to each layer of the transformer model **1010**. For example, absolute minimum and/or maximum values can be used to determine scaling factors that can prevent underflow or overflow of low-precision floating-point data types.

[0192] FIG. 11 illustrates training and deployment of a deep neural network. Once a given network has been structured for a task the neural network is trained using a training dataset **1102**. Training frameworks **1104** have been developed to enable hardware acceleration of the training process. For example, the machine learning framework **604** of FIG. 6 may be configured as a training framework **1104**. The training framework **1104** can hook into an untrained neural network **1106** and enable the untrained neural net to be trained using the parallel processing resources described herein to generate a trained neural network **1108**. To start the training process the initial weights may be chosen randomly or by pre-training using a deep belief network. The training cycle then be performed in either a supervised or unsupervised manner.

[0193] Supervised learning is a learning method in which training is performed as a mediated operation, such as when the training dataset **1102** includes input paired with the desired output for the input, or where the training dataset includes input having known output and the output of the neural network is manually graded. The network processes the inputs and compares the resulting outputs against a set of expected or desired outputs. Errors are then propagated back through the system. The training framework **1104** can adjust to adjust the weights that control the untrained neural network **1106**. The training framework **1104** can provide tools to monitor how well the untrained neural network **1106** is converging towards a model suitable to generating correct answers based on known input data. The training process occurs repeatedly as the weights of the network are adjusted to refine the output generated by the neural network. The training process can continue until the neural network reaches a statistically desired accuracy associated with a trained neural network **1108**. The trained neural network **1108** can then be deployed to implement any number of machine learning operations to generate an inference result **1114** based on input of new data **1112**.

[0194] Unsupervised learning is a learning method in which the network attempts to train itself using unlabeled data. Thus, for unsupervised learning the training dataset **1102** will include input data without any associated output data. The untrained neural network **1106** can learn groupings within the unlabeled input and can determine how individual inputs are related to the overall dataset. Unsupervised training can be used to generate a self-organizing map, which is a type of trained neural network **1108** capable of performing operations useful in reducing the dimensionality of data. Unsupervised training can also be used to perform anomaly detection, which allows the identification of data points in an input dataset that deviate from the normal patterns of the data.

[0195] Variations on supervised and unsupervised training may also be employed. Semi-supervised learning is a technique in which in the training dataset **1102** includes a mix of labeled and unlabeled data of the same distribution. Incremental learning is a variant of supervised learning in which input data is continuously used to further train the model. Incremental learning enables the trained neural network **1108** to adapt to the new data **1112** without forgetting the knowledge instilled within the network during initial training. Whether supervised or unsupervised, the training process for particularly deep neural networks may be too computationally intensive for a single compute node. Instead of using a single compute node, a distributed network of computational nodes can be used to accelerate the training process.

[0196] FIG. 12 is a block diagram illustrating distributed learning. Distributed learning is a training model that uses multiple distributed computing nodes to perform supervised or unsupervised training of a neural network. The distributed computational nodes can each include one or more host processors and one or more of the general-purpose processing nodes, such as the GPGPU **700** as in FIG. 7. As illustrated, distributed learning can be performed with model parallelism **1202**, data parallelism **1204**, or a combination of model and data parallelism **1206**.

[0197] In model parallelism **1202**, different computational nodes in a distributed system can perform training computations for different parts of a single network. For example,

each layer of a neural network can be trained by a different processing node of the distributed system. The benefits of model parallelism include the ability to scale to particularly large models. Splitting the computations associated with different layers of the neural network enables the training of very large neural networks in which the weights of all layers would not fit into the memory of a single computational node. In some instances, model parallelism can be particularly useful in performing unsupervised training of large neural networks. Some implementations of model parallelism **1202** may also be referred to as tensor parallelism.

**[0198]** In data parallelism **1204**, the different nodes of the distributed network have a complete instance of the model and each node receives a different portion of the data. The results from the different nodes are then combined. While different approaches to data parallelism are possible, data parallel training approaches utilize a technique of combining results and synchronizing the model parameters between each node. Example approaches to combining data include parameter averaging and update-based data parallelism. Parameter averaging trains each node on a subset of the training data and sets the global parameters (e.g., weights, biases) to the average of the parameters from each node. Parameter averaging uses a central parameter server that maintains the parameter data. Update based data parallelism is similar to parameter averaging except that instead of transferring parameters from the nodes to the parameter server, the updates to the model are transferred. Additionally, update-based data parallelism can be performed in a decentralized manner, where the updates are compressed and transferred between nodes.

**[0199]** Combined model and data parallelism **1206** can be implemented, for example, in a distributed system in which each computational node includes multiple GPUs. Combined model and data parallelism **1206** may also be referred to as hybrid parallelism. Each node can have a complete instance of the model, with separate GPUs within each node being used to train different portions of the model. Distributed training has increased overhead relative to training on a single machine. However, the parallel processors and GPGPUs described herein can each implement various techniques to reduce the overhead of distributed training, including techniques to enable high bandwidth GPU-to-GPU data transfer and accelerated remote data synchronization. Pipeline parallelism is a variant of combined model and data parallelism **1206**, in which different nodes contain less than the entire model, but more than a single layer of the model. In pipeline parallelism, different groups of layers or sub-models are distributed across the different processing nodes. Another variant is expert parallelism, which routes requests to specific experts within the model to different GPUs. Expert parallelism can be used, for example, with MoE transformer models.

**[0200]** FIG. 13 is a block diagram illustrating a programmable network interface **1300** and data processing unit. The programmable network interface **1300** is a programmable network engine that can be used to accelerate network-based compute tasks within a distributed environment. The programmable network interface **1300** can couple with a host system via host interface **1370**. The programmable network interface **1300** can be used to accelerate network or storage operations for CPUs or GPUs of the host system. The host system can be, for example, a node of a distributed learning

system used to perform distributed training, for example, as shown in FIG. 12. The host system can also be a data center node within a data center.

**[0201]** In one embodiment, access to remote storage containing model data can be accelerated by the programmable network interface **1300**. For example, the programmable network interface **1300** can be configured to present remote storage devices as local storage devices to the host system. The programmable network interface **1300** can also accelerate remote direct memory access (RDMA) operations performed between GPUs of the host system with GPUs of remote systems. In one embodiment, the programmable network interface **1300** can enable storage functionality such as, but not limited to NVME-oF. The programmable network interface **1300** can also accelerate encryption, data integrity, compression, and other operations for remote storage on behalf of the host system, allowing remote storage to approach the latencies of storage devices that are directly attached to the host system.

**[0202]** The programmable network interface **1300** can also perform resource allocation and management on behalf of the host system. Storage security operations can be offloaded to the programmable network interface **1300** and performed in concert with the allocation and management of remote storage resources. Network-based operations to manage access to the remote storage that would otherwise be performed by a processor of the host system can instead be performed by the programmable network interface **1300**.

**[0203]** In one embodiment, network and/or data security operations can be offloaded from the host system to the programmable network interface **1300**. Data center security policies for a data center node can be handled by the programmable network interface **1300** instead of the processors of the host system. For example, the programmable network interface **1300** can detect and mitigate against an attempted network-based attack (e.g., DDoS) on the host system, preventing the attack from compromising the availability of the host system.

**[0204]** The programmable network interface **1300** can include a system on a chip (SoC **1320**) that executes an operating system via multiple processor cores **1322**. The processor cores **1322** can include general-purpose processor (e.g., CPU) cores. In one embodiment the processor cores **1322** can also include one or more GPU cores. The SoC **1320** can execute instructions stored in a memory device **1340**. A storage device **1350** can store local operating system data. The storage device **1350** and memory device **1340** can also be used to cache remote data for the host system. Network ports **1360A-1360B** enable a connection to a network or fabric and facilitate network access for the SoC **1320** and, via the host interface **1370**, for the host system. The programmable network interface **1300** can also include an I/O interface **1375**, such as a USB interface. The I/O interface **1375** can be used to couple external devices to the programmable network interface **1300** or as a debug interface. The programmable network interface **1300** also includes a management interface **1330** that enables software on the host device to manage and configure the programmable network interface **1300** and/or SoC **1320**. In one embodiment the programmable network interface **1300** may also include one or more accelerators or GPUs **1345** to accept offload of parallel compute tasks from the SoC **1320**, host system, or remote systems coupled via the network ports **1360A-1360B**.

### Example Machine Learning Applications

**[0205]** Machine learning can be applied to solve a variety of technological problems, including but not limited to computer vision, autonomous driving and navigation, speech recognition, and language processing. Computer vision has traditionally been one of the most active research areas for machine learning applications. Applications of computer vision range from reproducing human visual abilities, such as recognizing faces, to creating new categories of visual abilities. For example, computer vision applications can be configured to recognize sound waves from the vibrations induced in objects visible in a video. Parallel processor accelerated machine learning enables computer vision applications to be trained using significantly larger training dataset than previously feasible and enables inferencing systems to be deployed using low power parallel processors.

**[0206]** Parallel processor accelerated machine learning has autonomous driving applications including lane and road sign recognition, obstacle avoidance, navigation, and driving control. Accelerated machine learning techniques can be used to train driving models based on datasets that define the appropriate responses to specific training input. The parallel processors described herein can enable rapid training of the increasingly complex neural networks used for autonomous driving solutions and enables the deployment of low power inferencing processors in a mobile platform suitable for integration into autonomous vehicles.

**[0207]** Parallel processor accelerated deep neural networks have enabled machine learning approaches to automatic speech recognition (ASR). ASR includes the creation of a function that computes the most probable linguistic sequence given an input acoustic sequence. Accelerated machine learning using deep neural networks have enabled the replacement of the hidden Markov models (HMMs) and Gaussian mixture models (GMMs) previously used for ASR.

**[0208]** Parallel processor accelerated machine learning can also be used to accelerate natural language processing. Automatic learning procedures can make use of statistical inference algorithms to produce models that are robust to erroneous or unfamiliar input. Example natural language processor applications include automatic machine translation between human languages.

**[0209]** The parallel processing platforms used for machine learning can be divided into training platforms and deployment platforms. Training platforms are generally highly parallel and include optimizations to accelerate multi-GPU single node training and multi-node, multi-GPU training. Example parallel processors suited for training include the GPGPU 700 of FIG. 7 and the multi-GPU computing system 800 of FIG. 8. On the contrary, deployed machine learning platforms generally include lower power parallel processors suitable for use in products such as cameras, autonomous robots, and autonomous vehicles.

**[0210]** Additionally, machine learning techniques can be applied to accelerate or enhance graphics processing activities. For example, a machine learning model can be trained to recognize output generated by a GPU accelerated application and generate an upscaled version of that output. Such techniques can be applied to accelerate the generation of high-resolution images for a gaming application. Various other graphics pipeline activities can benefit from the use of machine learning. For example, machine learning models can be trained to perform tessellation operations on geom-

etry data to increase the complexity of geometric models, allowing fine-detailed geometry to be automatically generated from geometry of relatively lower detail.

### Additional System Overview

**[0211]** FIG. 14 is a block diagram of a processing system 1400. The elements of FIG. 14 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. The processing system 1400 may be used in a single processor desktop system, a multiprocessor workstation system, or a server system having processor(s) 1402 or processor cores 1407. The processing system 1400 may be a processing platform incorporated within a system-on-a-chip (SoC) integrated circuit for use in mobile, handheld, or embedded devices such as within Internet-of-things (IoT) devices with wired or wireless connectivity to a local or wide area network.

**[0212]** The processing system 1400 may be a processing system having components that correspond with those of FIG. 1. For example, in different configurations, processor(s) 1402 or processor cores 1407 may correspond with processor(s) 102 of FIG. 1. The graphics processor(s) 1408 may correspond with parallel processor(s) 112 of FIG. 1. The external graphics processor 1418 may be one of the add-in device(s) 120 of FIG. 1.

**[0213]** The processing system 1400 can include, couple with, or be integrated within: a server-based gaming platform; a game console, including a game and media console; a mobile gaming console, a handheld game console, or an online game console. The processing system 1400 may be part of a mobile phone, smart phone, tablet computing device or mobile Internet-connected device such as a laptop with low internal storage capacity. The processing system 1400 can also include, couple with, or be integrated within: a wearable device, such as a smart watch wearable device; smart eyewear or clothing enhanced with augmented reality (AR) or virtual reality (VR) features to provide visual, audio or tactile outputs to supplement real world visual, audio or tactile experiences or otherwise provide text, audio, graphics, video, holographic images or video, or tactile feedback. The processing system 1400 may include or be part of a television or set top box device. The processing system 1400 can include, couple with, or be integrated within a self-driving vehicle such as a bus, tractor trailer, car, motor or electric power cycle, plane or glider (or any combination thereof). The self-driving vehicle may use the processing system 1400 to process the environment sensed around the vehicle.

**[0214]** The processor(s) 1402 may include one or more instances of processor cores 1407 to process instructions which, when executed, perform operations for system or user software. The least one of the processor cores 1407 may be configured to process a specific instruction set 1409. The instruction set 1409 may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). In one embodiment, one of the processor cores 1407 may process a different instruction set 1409, which may include instructions to facilitate the emulation of other

instruction sets. Processor core **1407** may also include other processing devices, such as a Digital Signal Processor (DSP).

**[0215]** The processor(s) **1402** may include cache memory **1404**. Depending on the architecture, the processor(s) **1402** can have a single internal cache or multiple levels of internal cache. In some embodiments, the cache memory is shared among various components of the processor(s) **1402**. In some embodiments, the processor(s) **1402** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **1407** using known cache coherency techniques. A register file **1406** can be additionally included in the processor(s) **1402** and may include different types of registers for storing different types of data (e.g., integer registers, floating-point registers, status registers, and an instruction pointer register). Some registers may be general-purpose registers, while other registers may be specific to the design of the processor(s) **1402**.

**[0216]** The processor(s) **1402** may be coupled with one or more interface bus(es) **1410** to transmit communication signals such as address, data, or control signals between the processor(s) **1402** and other components in the processing system **1400**. The interface bus(es) **1410**, in one of these embodiments, can be a processor bus, such as a version of the Direct Media Interface (DMI) bus. However, processor busses are not limited to the DMI bus, and may include one or more Peripheral Component Interconnect buses (e.g., PCI, PCI express), memory busses, or other types of interface busses. For example, the processor(s) **1402** may include a memory controller **1416** and a platform controller hub **1430**. The memory controller **1416** facilitates communication between a memory device and other components of the processing system **1400**, while the platform controller hub **1430** provides connections to I/O devices via a local I/O bus.

**[0217]** The memory device **1420** can be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. The memory device **1420** can, for example, operate as system memory for the processing system **1400**, to store data **1422** and instructions **1421** for use when the processor(s) **1402** executes an application or process. The memory controller **1416** can optionally couple with an external graphics processor **1418**, which may communicate with the graphics processor(s) **1408** in the processor(s) **1402** to perform graphics and media operations. In some embodiments, graphics, media, and/or compute operations may be assisted by an accelerator **1412** which is a coprocessor that can be configured to perform a specialized set of graphics, media, or compute operations. For example, the accelerator **1412** may be a matrix multiplication accelerator used to optimize machine learning or compute operations. The accelerator **1412** can be a ray-tracing accelerator that can be used to perform ray-tracing operations in concert with the graphics processor(s) **1408**. The accelerator **1412** may also be an AI accelerator or NPU to accelerate neural network training or inference operations. In one embodiment, an external accelerator **1419** may be used in place of or in concert with the accelerator **1412**. The accelerator **1412** and/or external accelerator **1419** may have functionality similar to that of the accelerator device(s) **130** of FIG. 1.

**[0218]** A display device **1411** may be provided that can connect to the processor(s) **1402**. The display device **1411** can be one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). The display device **1411** can be a head mounted display (HMD) such as a stereoscopic display device for use in VR or AR applications.

**[0219]** The platform controller hub **1430** may enable peripherals to connect to memory device **1420** and processor(s) **1402** via a high-speed I/O bus. The I/O peripherals include, but are not limited to, an audio controller **1446**, a network controller **1434**, a firmware interface **1428**, a wireless transceiver **1426**, touch sensors **1425**, a data storage device **1424** (e.g., non-volatile memory, volatile memory, hard disk drive, flash memory, NAND, 3D NAND, 3D XPoint/Optane, etc.). The data storage device **1424** can connect via a storage interface (e.g., SATA) or via a peripheral bus, such as a Peripheral Component Interconnect bus (e.g., PCI, PCI express). The touch sensors **1425** can include touch screen sensors, pressure sensors, or fingerprint sensors. The wireless transceiver **1426** can be a Wi-Fi transceiver, a Bluetooth transceiver, or a mobile network transceiver such as a 3G, 4G, 5G, or Long-Term Evolution (LTE) transceiver. The firmware interface **1428** enables communication with system firmware, and can be, for example, a unified extensible firmware interface (UEFI). The network controller **1434** can enable a network connection to a wired network. In some embodiments, a high-performance network controller (not shown) couples with the interface bus(es) **1410**. The audio controller **1446** may be a multi-channel high-definition audio controller. In some of these embodiments the processing system **1400** includes an optional legacy I/O controller **1440** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to the system. The platform controller hub **1430** can also connect to one or more Universal Serial Bus (USB) controllers **1442** to connect to input devices, such as keyboard and mouse **1443** combinations, a camera **1444**, or other USB input devices.

**[0220]** It will be appreciated that the processing system **1400** shown is example and not limiting, as other types of data processing systems that are differently configured may also be used. For example, an instance of the memory controller **1416** and platform controller hub **1430** may be integrated into a discrete external graphics processor, such as the external graphics processor **1418**. The platform controller hub **1430** and/or memory controller **1416** may be external to the processor(s) **1402**. For example, the memory controller **1416** and platform controller hub **1430** may be external to the processing system **1400** and configured as a memory controller hub and peripheral controller hub within a system chipset that is in communication with the processor(s) **1402**.

**[0221]** For example, circuit boards (“sleds”) can be used on which components such as CPUs, memory, and other components are placed and are designed for increased thermal performance. Processing components such as the processors may be located on a top side of a sled while near memory, such as DIMMs, are located on a bottom side of the sled. As a result of the enhanced airflow provided by this design, the components may operate at higher frequencies and power levels than in typical systems, thereby increasing performance. Furthermore, the sleds are configured to blindly mate with power and data communication cables in

a rack, thereby enhancing their ability to be quickly removed, upgraded, reinstalled, and/or replaced. Similarly, individual components located on the sleds, such as processors, accelerators, memory, and data storage drives, are configured to be easily upgraded due to their increased spacing from each other. In the illustrative embodiment, the components additionally include hardware attestation features to prove their authenticity.

[0222] A data center can utilize a single network architecture (“fabric”) that supports multiple other network architectures including Ethernet and Omni-Path. The sleds can be coupled to switches via optical fibers, which provide higher bandwidth and lower latency than typical twisted pair cabling (e.g., Category 5, Category 5e, Category 6, Category 7, Category 8, etc.). Due to the high bandwidth, low latency interconnections and network architecture, the data center may, in use, pool resources, such as memory, accelerators (e.g., GPUs, graphics accelerators, FPGAs, ASICs, neural network and/or artificial intelligence accelerators, etc.), and data storage drives that are physically disaggregated, and provide them to compute resources (e.g., processors), enabling the compute resources to access the pooled resources as if they were local.

[0223] A power supply or source can provide voltage and/or current to the processing system 1400 or any component or system described herein. In one example, the power supply includes an AC to DC (alternating current to direct current) adapter to plug into a wall outlet. Such AC power can be renewable energy (e.g., solar power) power source. In one example, the power source includes a DC power source, such as an external AC to DC converter. A power source or power supply may also include wireless charging hardware to charge via proximity to a charging field. The power source can include an internal battery, alternating current supply, motion-based power supply, solar power supply, or fuel cell source.

[0224] FIG. 15A-15C illustrate computing systems and graphics processors. The elements of FIG. 15A-15C having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0225] FIG. 15A is a block diagram of a processor 1500, which may be a variant of one of the processor(s) 1402 and may be used in place of one of those. Therefore, the disclosure of any features in combination with the processor 1500 herein also discloses a corresponding combination with the processor(s) 1402 but is not limited to such. The processor 1500 may have one or more processor cores 1502A-1502N, at least one memory controller 1514, and a graphics processor 1508. The graphics processor 1508 may be integrated within the processor 1500, within a system chipset, or coupled via a system bus. The processor 1500 can include additional cores up to and including additional core 1502N represented by the dashed lined boxes. Each of processor cores 1502A-1502N includes one or more internal cache unit(s) 1504A-1504N. In some embodiments each processor core 1502A-1502N also has access to one or more shared cache unit(s) 1506. The internal cache units 1504A-1504N and shared cache unit(s) 1506 represent a cache memory hierarchy within the processor 1500. The cache memory hierarchy may include at least one level of instruction and data cache within each processor core and one or

more levels of shared mid-level cache, such as a Level 2 (L2), Level 3 (L3), Level 4 (L4), or other levels of cache, where the highest level of cache before external memory is classified as the LLC. In some embodiments, cache coherency logic maintains coherency between the various cache units (e.g., shared cache unit(s) 1506 and internal cache unit(s) 1504A-1504N).

[0226] The processor 1500 may also include one or more bus controller units 1516 and a system agent core 1510. The one or more bus controller units 1516 manage a set of peripheral buses, such as one or more PCI or PCI express busses. The one or more bus controller units 1516 can also manage one or more memory buses to various external memory devices (not shown). The system agent core 1510 provides management functionality for the various processor components and can include the at least one memory controller 1514.

[0227] For example, one or more of the processor cores 1502A-1502N may include support for simultaneous multi-threading. The system agent core 1510 includes components for coordinating and operating cores 1502A-1502N during multi-threaded processing. System agent core 1510 may additionally include a power control unit (PCU), which includes logic and components to regulate the power state of processor cores 1502A-1502N and a graphics processor 1508.

[0228] The processor 1500 may additionally include a graphics processor 1508 to execute graphics processing operations. In some of these embodiments, the graphics processor 1508 couples with the one or more shared cache unit(s) 1506, and the system agent core 1510, including the at least one memory controller 1514. The system agent core 1510 may also include a display controller 1511 to drive graphics processor output to one or more coupled displays. The display controller 1511 may also be a separate module coupled with the graphics processor via at least one interconnect or may be integrated within the graphics processor 1508.

[0229] A ring or mesh based interconnect 1512 may be used to couple the internal components of the processor 1500. However, an alternative interconnect unit may be used, such as a point-to-point interconnect, a switched interconnect, or other techniques, including techniques well known in the art. In some of these embodiments with a ring or mesh based interconnect 1512, the graphics processor 1508 couples with the ring or mesh based interconnect 1512 via an I/O link 1513.

[0230] The example I/O link 1513 represents at least one of multiple varieties of I/O interconnects, including an on package I/O interconnect which facilitates communication between various processor components and a high-performance memory module 1518, such as an embedded DRAM module (eDRAM) or a high-bandwidth memory (HBM) module. Optionally, each of the processor cores 1502A-1502N and the graphics processor 1508 can use the high-performance memory module 1518 as unified memory and/or a shared Last Level Cache when a DRAM memory system is also present. Optionally, one or more accelerators 1515 may also be included within the processor 1500, including, for example, an NPU to accelerate certain neural network operations. The NPU can enable lower power inference operations relative to use of the graphics processor 1508, or can operate in concert with the graphics processor 1508 to enable higher inference performance relative to the

graphics processor **1508** alone. In one embodiment, an NPU within the one or more accelerators **1515** can include matrix or tensor acceleration logic and can be used to implement at least some of the computational operations that are described herein as implementable via the graphics processor **1508**.

[0231] The processor cores **1502A-1502N** may, for example, be homogenous cores executing the same instruction set architecture. Alternatively, the processor cores **1502A-1502N** are heterogeneous in terms of instruction set architecture (ISA), where one or more of processor cores **1502A-1502N** execute a first instruction set, while at least one of the other cores executes a subset of the first instruction set or a different instruction set. The processor cores **1502A-1502N** may be heterogeneous in terms of micro-architecture, where one or more cores having a relatively higher power consumption couple with one or more power cores having a lower power consumption. As another example, the processor cores **1502A-1502N** are heterogeneous in terms of computational capability. Additionally, the processor **1500** can be implemented on one or more chips or chiplets, or as an SoC integrated circuit having the illustrated components, in addition to other components. The SoC integrated circuit may be implemented using multiple chiplets.

[0232] FIG. **15B** is a block diagram of hardware logic of a graphics processor core block **1519**, according to some embodiments described herein. In some embodiments, elements of FIG. **15B** having the same reference numbers (or names) as the elements of any other figure herein may operate or function in a manner similar to that described elsewhere herein. In one embodiment, the graphics processor core block **1519** is example of one partition of a graphics processor. The graphics processor core block **1519** can be included within the graphics processor **1508** of FIG. **15A** or a discrete graphics processor, parallel processor, and/or compute accelerator. A graphics processor as described herein may include multiple graphics core blocks based on target power and performance envelopes. Each graphics processor core block **1519** can include a function block **1530** coupled with multiple graphics cores **1521A-1521F** that include modular blocks of fixed function logic and general-purpose programmable logic. The graphics processor core block **1519** also includes shared/cache memory **1536** that is accessible by all graphics cores **1521A-1521F**, rasterizer logic **1537**, and additional fixed function logic **1538**.

[0233] In some embodiments, the function block **1530** includes a geometry/fixed function pipeline **1531** that can be shared by all graphics cores in the graphics processor core block **1519**. In various embodiments, the geometry/fixed function pipeline **1531** includes a 3D geometry pipeline a video front-end unit, a thread spawner and global thread dispatcher, and a unified return buffer manager, which manages unified return buffers. In one embodiment the function block **1530** also includes a graphics SoC interface **1532**, a graphics microcontroller **1533**, and a media pipeline **1534**. The graphics SoC interface **1532** provides an interface between the graphics processor core block **1519** and other core blocks within a graphics processor or compute accelerator SoC. The graphics microcontroller **1533** is a programmable sub-processor that is configurable to manage various functions of the graphics processor core block **1519**, including thread dispatch, scheduling, and pre-emption. The media pipeline **1534** includes logic to facilitate the decoding,

encoding, pre-processing, and/or post-processing of multimedia data, including image and video data. The media pipeline **1534** implement media operations via requests to compute or sampling logic within the graphics cores **1521-1521F**. One or more pixel backends **1535** can also be included within the function block **1530**. The one or more pixel backends **1535** include a cache memory to store pixel color values and can perform blend operations and lossless color compression of rendered pixel data.

[0234] In one embodiment the graphics SoC interface **1532** enables the graphics processor core block **1519** to communicate with general-purpose application processor cores (e.g., CPUs) and/or other components within an SoC or a system host CPU that is coupled with the SoC via a peripheral interface. The graphics SoC interface **1532** also enables communication with off-chip memory hierarchy elements such as a shared last level cache memory, system RAM, and/or embedded on-chip or on-package DRAM. The graphics SoC interface **1532** can also enable communication with fixed function devices within the SoC, such as camera imaging pipelines, and enables the use of and/or implements global memory atomics that may be shared between the graphics processor core block **1519** and CPUs within the SoC. The graphics SoC interface **1532** can also implement power management controls for the graphics processor core block **1519** and enable an interface between a clock domain of the graphics processor core block **1519** and other clock domains within the SoC. In one embodiment the graphics SoC interface **1532** enables receipt of command buffers from a command streamer and global thread dispatcher that are configured to provide commands and instructions to each of one or more graphics cores within a graphics processor. The commands and instructions can be dispatched to the media pipeline **1534** when media operations are to be performed, the geometry and fixed function pipeline **1531** when graphics processing operations are to be performed. When compute operations are to be performed, compute dispatch logic can dispatch the commands to the graphics cores **1521A-1521F**, bypassing the geometry and media pipelines.

[0235] The graphics microcontroller **1533** can be configured to perform various scheduling and management tasks for the graphics processor core block **1519**. In one embodiment the graphics microcontroller **1533** can perform graphics and/or compute workload scheduling on the various vector engines **1522A-1522F**, **1524A-1524F** and matrix engines **1523A-1523F**, **1525A-1525F** within the graphics cores **1521A-1521F**. In this scheduling model, host software executing on a CPU core of an SoC including the graphics processor core block **1519** can submit workloads to one of multiple graphics processor doorbells, which invokes a scheduling operation on the appropriate graphics engine. Scheduling operations include determining which workload to run next, submitting a workload to a command streamer, pre-empting existing workloads running on an engine, monitoring progress of a workload, and notifying host software when a workload is complete. In one embodiment the graphics microcontroller **1533** can also facilitate low-power or idle states for the graphics processor core block **1519**, providing the graphics processor core block **1519** with the ability to save and restore registers within the graphics processor core block **1519** across low-power state transitions independently from the operating system and/or graphics driver software on the system.

[0236] The graphics processor core block 1519 may have greater than or fewer than the illustrated graphics cores 1521A-1521F, up to N modular graphics cores. For each set of N graphics cores, the graphics processor core block 1519 can also include shared/cache memory 1536, which can be configured as shared memory or cache memory, rasterizer logic 1537, and additional fixed function logic 1538 to accelerate various graphics and compute processing operations.

[0237] Within each graphics cores 1521A-1521F is set of execution resources that may be used to perform graphics, media, and compute operations in response to requests by graphics pipeline, media pipeline, or shader programs. The graphics cores 1521A-1521F include multiple vector engines 1522A-1522F, 1524A-1524F, matrix acceleration units 1523A-1523F, 1525A-1525D, cache/shared local memory (SLM), a sampler 1526A-1526F, and a ray tracing unit 1527A-1527F.

[0238] The vector engines 1522A-1522F, 1524A-1524F are general-purpose graphics processing units capable of performing floating-point and integer/fixed-point logic operations in service of a graphics, media, or compute operation, including graphics, media, or compute/GPGPU programs. The vector engines 1522A-1522F, 1524A-1524F can operate at variable vector widths using SIMD, SIMT, or SIMT+SIMD execution modes. The matrix acceleration units 1523A-1523F, 1525A-1525D include matrix-matrix and matrix-vector acceleration logic that improves performance on matrix operations, particularly low and mixed precision (e.g., INT8, FP16, BF16, FP8, FP4) matrix operations used for machine learning. In one embodiment, the matrix acceleration units 1523A-1523F, 1525A-1525D have support for microscaling (MX) formats. In one embodiment, each of the matrix acceleration units 1523A-1523F, 1525A-1525D includes one or more systolic arrays of processing elements that can perform concurrent matrix multiply or dot product operations on matrix elements.

[0239] The sampler 1526A-1526F can read media or texture data into memory and can sample data differently based on a configured sampler state and the texture/media format that is being read. Threads executing on the vector engines 1522A-1522F, 1524A-1524F or matrix acceleration units 1523A-1523F, 1525A-1525D can make use of the cache/SLM 1528A-1528F within each of the graphics cores 1521A-1521F. The cache/SLM 1528A-1528F can be configured as cache memory or as a pool of shared memory that is local to each of the respective graphics cores 1521A-1521F. The ray tracing units 1527A-1527F within the graphics cores 1521A-1521F include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. In one embodiment the ray tracing units 1527A-1527F include circuitry for performing depth testing and culling (e.g., using a depth buffer or similar arrangement). In one implementation, the ray tracing units 1527A-1527F perform traversal and intersection operations in concert with image denoising, at least a portion of which may be performed using an associated matrix acceleration unit 1523A-1523F, 1525A-1525D.

[0240] FIG. 15C is a block diagram of general-purpose graphics processing unit (GPGPU 1570) that can be configured as a graphics processor, e.g., the graphics processor 1508, and/or compute accelerator, according to embodi-

ments described herein. The GPGPU 1570 can interconnect with host processors (e.g., one or more CPU(s) 1546) and memory 1571, 1572 via one or more system and/or memory busses. Memory 1571 may be system memory that can be shared with the one or more CPU(s) 1546, while memory 1572 is device memory that is dedicated to the GPGPU 1570. For example, components within the GPGPU 1570 and memory 1572 may be mapped into memory addresses that are accessible to the one or more CPU(s) 1546. Access to memory 1571 and 1572 may be facilitated via a memory controller 1568. The memory controller 1568 may include an internal direct memory access (DMA) controller 1569 or can include logic to perform operations that would otherwise be performed by a DMA controller. In one embodiment, at least one of the one or more CPU(s) 1546 can include one or more accelerators 1545, including but not limited to neural network accelerators.

[0241] The GPGPU 1570 includes multiple global cache memories, including an L2 cache 1553, L1 cache 1554, an instruction cache 1555, and shared memory 1556, at least a portion of which may also be partitioned as a cache memory. The GPGPU 1570 also includes multiple compute units 1560A-1560N. Each compute unit 1560A-1560N includes a set of vector registers 1561, scalar registers 1562, vector logic units 1563, scalar logic units 1564, and a scheduler 1584. The compute units 1560A-1560N can also include local shared memory 1565 and local cache memory 1566. The compute units 1560A-1560N can couple with a constant cache 1567, which can be used to store constant data, which is data that will not change during the run of kernel or shader program that executes on the GPGPU 1570. The constant cache 1567 may be a scalar data cache and cached data can be fetched directly into the scalar registers 1562. In one embodiment, the compute unit 1560A-1560N additionally include at least one matrix unit 1580 to accelerate matrix, tensor, or artificial intelligence operations and at least one ray tracing unit (RT unit 1582) to accelerate ray tracing operations. The at least one matrix unit 1580 and RT unit 1582 can include similar functionality as other matrix/tensor accelerators and ray tracing accelerators described herein.

[0242] During operation, the one or more CPU(s) 1546 can write commands into registers or memory in the GPGPU 1570 that has been mapped into an accessible address space. The command processors 1557 can read the commands from registers or memory and determine how those commands will be processed within the GPGPU 1570. A thread dispatcher 1558 can then be used to dispatch threads to the compute units 1560A-1560N to perform those commands. Each compute unit 1560A-1560N can execute threads independently of the other compute units. Additionally, each compute unit 1560A-1560N can be independently configured for conditional computation and can conditionally output the results of computation to memory. The command processors 1557 can interrupt the one or more CPU(s) 1546 when the submitted commands are complete.

[0243] FIG. 16 is a block diagram of a graphics processor 1600, which may be a discrete graphics processing unit, or may be a graphics processor integrated with a plurality of processing cores, or other semiconductor devices such as, but not limited to, memory devices or network interfaces. The elements of the graphics processor 1600 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise

the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. For example, the graphics processor **1600** may be a variant of the graphics processor **1508** and may be used in place of the graphics processor **1508**. The graphics processor may communicate via a memory mapped I/O interface to registers on the graphics processor and with commands placed into the processor memory. The graphics processor **1600** may include a memory interface **1614** to access local memory, one or more internal caches, one or more shared external caches, and/or system memory.

[0244] The graphics processor **1600** can include a display controller **1602** to drive display output data to a display device **1618**. Display controller **1602** includes hardware for one or more overlay planes for the display and composition of multiple layers of video or user interface elements. The display device **1618** can be an internal or external display device. In one embodiment the display device **1618** is a head mounted display device, such as a VR or AR display device. Graphics processor **1600** may include a video codec engine **1606** to encode, decode, or transcode media to, from, or between one or more media encoding formats, including, but not limited to Moving Picture Experts Group (MPEG) formats such as MPEG-2, Advanced Video Coding (AVC) formats such as H.264/MPEG-4 AVC, H.265/HEVC, Alliance for Open Media (AOMedia) VP8, VP9, as well as the Society of Motion Picture & Television Engineers (SMPTE) 421M/VC-1, and Joint Photographic Experts Group (JPEG) formats such as JPEG, and Motion JPEG (MJPEG) formats.

[0245] The graphics processor **1600** may include a block image transfer (BLIT) engine **1603** to perform two-dimensional (2D) rasterizer operations including, for example, bit-boundary block transfers, or 2D graphics operations may be performed using one or more components of graphics processing engine (GPE **1610**). GPE **1610** may include a 3D pipeline **1612** for performing 3D operations, such as rendering three-dimensional images and scenes using processing functions that act upon 3D primitive shapes (e.g., rectangle, triangle, etc.). The 3D pipeline **1612** includes programmable and fixed function elements that perform various tasks within the element and/or spawn execution threads to a 3D/Media subsystem **1615**. While 3D pipeline **1612** can be used to perform media operations, the GPE **1610** may also include a media pipeline **1616** that is specifically used to perform media operations, such as image or video decode, encode, post-processing, and enhancement. The media pipeline **1616** may include fixed function or programmable logic units to perform one or more specialized media operations, such as video decode acceleration, video de-interlacing, and video encode acceleration in place of, or on behalf of video codec engine **1606**. Media pipeline **1616** may additionally include a thread spawning unit to spawn threads for execution on 3D/Media subsystem **1615**. The spawned threads perform computations for the media operations on one or more graphics execution units included in 3D/Media subsystem **1615**.

[0246] FIG. 17A illustrates a graphics processor **1720**, being a variant of the graphics processor **1600** of FIG. 16 and may be used in place of the graphics processor **1600** and vice versa. Therefore, the disclosure of any features in combination with the graphics processor **1600** herein also discloses a corresponding combination with the graphics processor **1720** but is not limited to such. The graphics processor **1720** has a tiled architecture, according to embodi-

ments described herein. The graphics processor **1720** may include a graphics processing engine cluster **1722** having multiple graphics processing engines within a plurality of graphics engine tiles. Each graphics engine tile **1710A-1710D** can be interconnected via a set of tile interconnects **1723A-1723F**. Each graphics engine tile **1710A-1710D** can also be connected to a memory module or memory devices **1726A-1726D** via memory interconnects **1725A-1725D**. The memory devices **1726A-1726D** can use any graphics memory technology. For example, the memory devices **1726A-1726D** may be graphics double data rate (GDDR) memory. The memory devices **1726A-1726D** may be high-bandwidth memory (HBM) modules that can be on-die with their respective graphics engine tile **1710A-1710D**. The memory devices **1726A-1726D** may be stacked memory devices that can be stacked on top of their respective graphics engine tile **1710A-1710D**. Each graphics engine tile **1710A-1710D** and associated memory devices **1726A-1726D** may reside on separate chiplets, which are bonded to a base die or base substrate, as described in further detail in FIG. 25A-25B.

[0247] The graphics processor **1720** may be configured with a non-uniform memory access (NUMA) system in which memory devices **1726A-1726D** are coupled with associated graphics engine tiles **1710A-1710D**. A given memory device may be accessed by graphics engine tiles other than the tile to which it is directly connected. However, access latency to the memory devices **1726A-1726D** may be lowest when accessing a local tile. In one embodiment, a cache coherent NUMA (ccNUMA) system is enabled that uses the tile interconnects **1723A-1723F** to enable communication between cache controllers within the graphics engine tiles **1710A-1710D** to keep a consistent memory image when more than one cache stores the same memory location.

[0248] The graphics processing engine cluster **1722** can connect with an interconnect fabric **1724**, which may interconnect with an on-chip or on-package fabric. In one embodiment the interconnect fabric **1724** includes a network processor, network on a chip (NoC), or another switching processor to enable the interconnect fabric **1724** to act as a packet switched interconnect fabric that switches data packets between components of the graphics processor **1720**. The interconnect fabric **1724** can enable communication between graphics engine tiles **1710A-1710D** and components such as the video codec engine **1706** and one or more copy engines **1704**. The one or more copy engines **1704** can be used to move data out of, into, and between the memory devices **1726A-1726D** and memory that is external to the graphics processor **1720** (e.g., system memory). The interconnect fabric **1724** can also be used to interconnect the graphics engine tiles **1710A-1710D**. The graphics processor **1720** may optionally include a display controller **1702** to enable a connection with a display device **1718**. The graphics processor may also be configured as a graphics or compute accelerator. In the accelerator configuration, the display controller **1702** and display device **1718** may be omitted.

[0249] The graphics processor **1720** can connect to a host system via a host interface **1728**. The host interface **1728** can enable communication between the graphics processor **1720**, system memory, and/or other system components. The host interface **1728** can be, for example, a PCI express bus or another type of host system interface. For example, the

host interface **1728** may be an NVLink or NVSwitch interface. The host interface **1728** and interconnect fabric **1724** can cooperate to enable multiple instances of the graphics processor **1720** to act as single logical device. Cooperation between the host interface **1728** and interconnect fabric **1724** can also enable the individual graphics engine tiles **1710A-1710D** to be presented to the host system as distinct logical graphics devices.

[0250] FIG. 17B illustrates a compute accelerator **1730**, according to embodiments described herein. The compute accelerator **1730** can include architectural similarities with the graphics processor **1720** of FIG. 17B and is optimized for compute acceleration. A compute engine cluster **1732** can include a set of compute engine tiles **1740A-1740D** that include execution logic that is optimized for parallel or vector-based general-purpose compute operations. In one embodiment, the compute accelerator **1730** may be configured as an AI accelerator or NPU. In such embodiment, the execution logic of the compute engine tiles **1740A-1740D** may be directed primarily towards matrix or tensor operations and include tensor cores or matrix engines described herein. The compute engine tiles **1740A-1740D** may not include fixed function graphics processing logic, although in some embodiments one or more of the compute engine tiles **1740A-1740D** can include logic to perform media acceleration. The compute engine tiles **1740A-1740D** can connect to the memory devices **1726A-1726D** via memory interconnects **1725A-1725D**. The memory devices **1726A-1726D** and memory interconnects **1725A-1725D** may be similar technology as in graphics processor **1720** or can be different. The compute engine tiles **1740A-1740D** can also be interconnected via a set of tile interconnects **1723A-1723F** and may be connected with and/or interconnected by the interconnect fabric **1724**. In one embodiment the compute accelerator **1730** includes a large L3 cache **1736** that can be configured as a device-wide cache. The compute accelerator **1730** can also connect to a host processor and memory via a host interface **1728** in a similar manner as the graphics processor **1720** of FIG. 17B.

[0251] The compute accelerator **1730** can also include an integrated network interface **1742**. In one embodiment the integrated network interface **1742** includes a network processor and controller logic that enables the compute engine cluster **1732** to communicate over a physical layer interconnect **1744** without utilizing data to traverse memory of a host system. In one embodiment, one of the compute engine tiles **1740A-1740D** is replaced by network processor logic and data to be transmitted or received via the physical layer interconnect **1744** may be transmitted directly to or from the memory devices **1726A-1726D**. Multiple instances of the compute accelerator **1730** may be joined via the physical layer interconnect **1744** into a single logical device. Alternatively, the various compute engine tiles **1740A-1740D** may be presented as distinct network accessible compute accelerator devices.

#### Graphics Processing Resources

[0252] FIG. 18A-18C illustrate execution logic including an array of processing elements employed in a graphics processor, according to embodiments described herein. FIG. 18A illustrates graphics core cluster, according to an embodiment. FIG. 18B illustrates a vector engine of a graphics core, according to an embodiment. FIG. 18C illustrates a matrix engine of a graphics core, according to an

embodiment. Elements of FIG. 18A-18C having the same reference numbers as the elements of any other figure herein may operate or function in any manner similar to that described elsewhere herein but are not limited as such. For example, the elements of FIG. 18A-18C can be considered in the context of the graphics processor core block **1519** of FIG. 15B. In one embodiment, the elements of FIG. 18A-18C have similar functionality to equivalent components of the graphics processor **1508** of FIG. 15A or the GPGPU **1570** of FIG. 15C.

[0253] As shown in FIG. 18A, in one embodiment the graphics core cluster **1800** includes a graphics processor core block **1519**, which can include any number of graphics cores (e.g., graphics core **1815A**, graphics core **1815B**, through graphics core **1815N**). Multiple instances of the graphics processor core block **1519** may be included. In one embodiment the elements of the graphics cores **1815A-1815N** have similar or equivalent functionality as the elements of the graphics cores **1521A-1521F** of FIG. 15B. In such embodiment, the graphics cores **1815A-1815N** each include circuitry including but not limited to vector engines **1802A-1802N**, matrix engines **1803A-1803N**, memory load/store units **1804A-1804N**, instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**. The circuitry of the graphics cores **1815A-1815N** can additionally include fixed function logic **1812A-1812N**. The number of vector engines **1802A-1802N** and matrix engines **1803A-1803N** within the graphics cores **1815A-1815N** of a design can vary based on the workload, performance, and power targets for the design.

[0254] With reference to graphics core **1815A**, the vector engine **1802A** and matrix engine **1803A** are configurable to perform parallel compute operations on data in a variety of integer and floating-point data formats based on instructions associated with shader programs. Each vector engine **1802A** and matrix engine **1803A** can act as a programmable general-purpose computational unit that is capable of executing multiple simultaneous hardware threads while processing multiple data elements in parallel for each thread. The vector engine **1802A** and matrix engine **1803A** support the processing of variable width vectors at various SIMD widths, including but not limited to SIMD8, SIMD16, and SIMD32. Input data elements can be stored as a packed data type in a register and the vector engine **1802A** and matrix engine **1803A** can process the various elements based on the data size of the elements. For example, when operating on a 256-bit wide vector, the bits of the vector are stored in a register and the vector is processed as four separate 64-bit packed data elements (Quad-Word (QW) size data elements), eight separate 32-bit packed data elements (Double Word (DW) size data elements), sixteen separate 16-bit packed data elements (Word (W) size data elements), or thirty-two separate 8-bit data elements (byte (B) size data elements). However, different vector widths and register sizes are possible. In one embodiment, the vector engine **1802A** and matrix engine **1803A** are also configurable for SIMT operation on warps or thread groups of various sizes (e.g., 8, 16, or 32 threads).

[0255] Continuing with graphics core **1815A**, the memory load/store unit **1804A** services memory access requests that are issued by the vector engine **1802A**, matrix engine **1803A**, and/or other components of the graphics core **1815A** that have access to memory. The memory access request can

be processed by the memory load/store unit **1804A** to load or store the requested data to or from cache or memory into a register file associated with the vector engine **1802A** and/or matrix engine **1803A**. The memory load/store unit **1804A** can also perform prefetching operations. With additional reference to FIG. 19, in one embodiment, the memory load/store unit **1804A** is configured to provide SIMT scatter/gather prefetching or block prefetching for data stored in memory **1910**, from memory that is local to other tiles via the tile interconnect **1908**, or from system memory. Prefetching can be performed to a specific L1 cache (e.g., data cache/shared local memory **1806A**), the L2 cache **1904** or the L3 cache **1906**. In one embodiment, a prefetch to the L3 cache **1906** automatically results in the data being stored in the L2 cache **1904**.

[0256] The instruction cache **1805A** stores instructions to be executed by the graphics core **1815A**. In one embodiment, the graphics core **1815A** also includes instruction fetch and prefetch circuitry that fetches or prefetches instructions into the instruction cache **1805A**. The graphics core **1815A** also includes instruction decode logic to decode instructions within the instruction cache **1805A**. The data cache/shared local memory **1806A** can be configured as a data cache that is managed by a cache controller that implements a cache replacement policy and/or configured as explicitly managed shared memory. The ray tracing unit **1808A** includes circuitry to accelerate ray tracing operations. The sampler **1810A** provides texture sampling for 3D operations and media sampling for media operations. The fixed function logic **1812A** includes fixed function circuitry that is shared between the various instances of the vector engine **1802A** and matrix engine **1803A**. Graphics cores **1815B-1815N** can operate in a similar manner as graphics core **1815A**.

[0257] Functionality of the instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**, and fixed function logic **1812A-1812N** corresponds with equivalent functionality in the graphics processor architectures described herein. For example, the instruction caches **1805A-1805N** can operate in a similar manner as instruction cache **1555** of FIG. 15C. The data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, and samplers **1810A-1810N** can operate in a similar manner as the cache/SLM **1528A-1528F**, ray tracing units **1527A-1527F**, and samplers **1526A-1526F** of FIG. 15B. The fixed function logic **1812A-1812N** can include elements of the geometry/fixed function pipeline **1531** and/or additional fixed function logic **1538** of FIG. 15B. In one embodiment, the ray tracing units **1808A-1808N** include circuitry to perform ray tracing acceleration operations performed by the ray tracing cores **372** of FIG. 3.

[0258] As shown in FIG. 18B, in one embodiment the vector engine **1802** includes an instruction fetch unit **1837**, a general register file array (GRF **1824**), an architectural register file array (ARF **1826**), a thread arbiter **1822**, a send unit **1830**, a branch unit **1832**, SIMD FPU's **1834**, and in one embodiment, SIMD ALUs **1835**. The GRF **1824** and ARF **1826** includes the set of general register files and architecture register files associated with each hardware thread that may be active in the vector engine **1802**. In one embodiment, per thread architectural state is maintained in the ARF **1826**, while data used during thread execution is stored in the GRF **1824**. The execution state of each thread, including the

instruction pointers for each thread, can be held in thread-specific registers in the ARF **1826**. Register renaming may be used to dynamically allocate registers to hardware threads.

[0259] In one embodiment the vector engine **1802** has an architecture that is a combination of Simultaneous Multi-Threading (SMT) and fine-grained Interleaved Multi-Threading (IMT). The architecture has a modular configuration that can be fine-tuned at design time based on a target number of simultaneous threads and number of registers per graphics core, where graphics core resources are divided across logic used to execute multiple simultaneous threads. The number of logical threads that may be executed by the vector engine **1802** is not limited to the number of hardware threads, and multiple logical threads can be assigned to each hardware thread.

[0260] In one embodiment, the vector engine **1802** can co-issue multiple instructions, which may each be different instructions. The thread arbiter **1822** can dispatch the instructions to one of the send unit **1830**, branch unit **1832**, or SIMD FPU's **1834** for execution. Each execution thread can access **128** general-purpose registers within the GRF **1824**, where each register can store 32 bytes, accessible as a variable width vector of 32-bit data elements. In one embodiment, each thread has access to 4 Kbytes within the GRF **1824**, although embodiments are not so limited, and greater or fewer register resources may be provided in other embodiments. In one embodiment the vector engine **1802** is partitioned into seven hardware threads that can independently perform computational operations, although the number of threads per vector engine **1802** can also vary according to embodiments. For example, in one embodiment up to 16 hardware threads are supported. In an embodiment in which seven threads may access 4 Kbytes, the GRF **1824** can store a total of 28 Kbytes. Where 16 threads may access 4 Kbytes, the GRF **1824** can store a total of 64 Kbytes. Flexible addressing modes can permit registers to be addressed together to build effectively wider registers or to represent strided rectangular block data structures.

[0261] In one embodiment, memory operations, sampler operations, and other longer-latency system communications are dispatched via "send" instructions that are executed by the message passing send unit **1830**. In one embodiment, branch instructions are dispatched to the branch unit **1832** to facilitate SIMD divergence and eventual convergence.

[0262] In one embodiment the SIMD FPU's **1834** of the vector engine **1802** perform floating-point operations. In one embodiment, the SIMD FPU's **1834** also support integer computation. In one embodiment the SIMD FPU's **1834** can execute up to M number of 32-bit floating-point (or integer) operations, or execute up to 2M 16-bit integer or 16-bit floating-point operations. In one embodiment, at least one of the FPU's provides extended math capability to support high-throughput transcendental math functions and double precision 64-bit floating-point. In some embodiments, SIMD ALUs **1835** configured to perform 8-bit integer operations are also present and may be specifically optimized to perform operations associated with machine learning computations. In one embodiment, the SIMD ALUs **1835** are replaced by SIMD FPU's **1834** that are configurable to perform integer and floating-point operations. In one embodiment, the SIMD FPU's **1834** and SIMD ALUs **1835** are configurable to execute SIMT programs. In one embodiment, combined SIMD+SIMT operation is supported.

[0263] In one embodiment, arrays of multiple instances of the vector engine **1802** can be instantiated in a graphics core. For scalability, product architects can choose the exact number of vector engines per graphics core grouping. In one embodiment the vector engine **1802** can execute instructions across a plurality of execution channels. In a further embodiment, each thread executed on the vector engine **1802** is executed on a different channel.

[0264] As shown in FIG. **18C**, in one embodiment the matrix engine **1803** includes an array of processing elements that are configured to perform tensor operations including vector/matrix and matrix/matrix operations, such as but not limited to matrix multiply and/or dot product operations. The matrix engine **1803** is configured with M rows and N columns of processing elements **1852AA-1852MN** that include multiplier and adder circuits organized in a pipelined fashion. In one embodiment, the processing elements **1852AA-1852MN** make up the physical pipeline stages of an N wide and M deep systolic array that can be used to perform vector/matrix or matrix/matrix operations in a data-parallel manner, including matrix multiply, fused multiply-add, dot product or other general matrix-matrix multiplication (GEMM) operations. In one embodiment the matrix engine **1803** supports 16-bit and 8-bit floating-point operations, as well as 8-bit, 4-bit, 2-bit, and binary integer operations. The matrix engine **1803** can also be configured to accelerate specific machine learning operations. In such embodiments, the matrix engine **1803** can be configured with support for the bfloat (brain floating-point) 16-bit floating-point format or a tensor float 32-bit floating-point format (TF32) that have different numbers of mantissa and exponent bits relative to Institute of Electrical and Electronics Engineers (IEEE) 754 formats.

[0265] In one embodiment, during each cycle, each stage can add the result of operations performed at that stage to the output of the previous stage. In other embodiments, the pattern of data movement between the processing elements **1852AA-1852MN** after a set of computational cycles can vary based on the instruction or macro-operation being performed. For example, in one embodiment partial sum loopback is enabled and the processing elements may instead add the output of a current cycle with output generated in the previous cycle. In one embodiment, the final stage of the systolic array can be configured with a loopback to the initial stage of the systolic array. In such embodiment, the number of physical pipeline stages may be decoupled from the number of logical pipeline stages that are supported by the matrix engine **1803**. For example, where the processing elements **1852AA-1852MN** are configured as a systolic array of M physical stages, a loopback from stage M to the initial pipeline stage can enable the processing elements **1852AA-1852MN** to operate as a systolic array of, for example, 2M, 3M, 4M, etc., logical pipeline stages.

[0266] In one embodiment, the matrix engine **1803** includes memory **1841A-1841N**, **1842A-1842M** to store input data in the form of row and column data for input matrices. Memory **1842A-1842M** is configurable to store row elements (A0-Am) of a first input matrix and memory **1841A-1841N** is configurable to store column elements (B0-Bn) of a second input matrix. The row and column elements are provided as input to the processing elements **1852AA-1852MN** for processing. In one embodiment, row and column elements of the input matrices can be stored in a systolic register file **1840** within the matrix engine **1803**

before those elements are provided to the memory **1841A-1841N**, **1842A-1842M**. In one embodiment, the systolic register file **1840** is excluded and the memory **1841A-1841N**, **1842A-1842M** is loaded from registers in an associated vector engine (e.g., GRF **1824** of vector engine **1802** of FIG. **18B**) or other memory of the graphics core that includes the matrix engine **1803** (e.g., data cache/shared local memory **1806A** for matrix engine **1803A** of FIG. **18A**). Results generated by the processing elements **1852AA-1852MN** are then output to an output buffer and/or written to a register file (e.g., systolic register file **1840**, GRF **1824**, data cache/shared local memory **1806A-1806N**) for further processing by other functional units of the graphics processor or for output to memory.

[0267] In some embodiments, the matrix engine **1803** is configured with support for input sparsity, where multiplication operations for sparse regions of input data can be bypassed by skipping multiply operations that have a zero-value operand. In one embodiment, the processing elements **1852AA-1852MN** are configured to skip the performance of certain operations that have zero value input. In one embodiment, sparsity within input matrices can be detected and operations having known zero output values can be bypassed before being submitted to the processing elements **1852AA-1852MN**. The loading of zero value operands into the processing elements can be bypassed and the processing elements **1852AA-1852MN** can be configured to perform multiplications on the non-zero value input elements. The matrix engine **1803** can also be configured with support for output sparsity, such that operations with results that are pre-determined to be zero are bypassed. For input sparsity and/or output sparsity, in one embodiment, metadata is provided to the processing elements **1852AA-1852MN** to indicate, for a processing cycle, which processing elements and/or data channels are to be active during that cycle.

[0268] In one embodiment, the matrix engine **1803** includes hardware to enable operations on sparse data having a compressed representation of a sparse matrix that stores non-zero values and metadata that identifies the positions of the non-zero values within the matrix. Example compressed representations include but are not limited to compressed tensor representations such as compressed sparse row (CSR), compressed sparse column (CSC), compressed sparse fiber (CSF) representations. Support for compressed representations enable operations to be performed on input in a compressed tensor format without the compressed representation having to be decompressed or decoded. In such embodiment, operations can be performed on non-zero input values and the resulting non-zero output values can be mapped into an output matrix. In some embodiments, hardware support is also provided for machine-specific lossless data compression formats that are used when transmitting data within hardware or across system busses. Such data may be retained in a compressed format for sparse input data and the matrix engine **1803** can use the compression metadata for the compressed data to enable operations to be performed on non-zero values, or to enable blocks of zero data input to be bypassed for multiply operations.

[0269] In various embodiments, input data can be provided by a programmer in a compressed tensor representation, or a codec can compress input data into the compressed tensor representation or another sparse data encoding. In addition to support for compressed tensor representations,

streaming compression of sparse input data can be performed before the data is provided to the processing elements **1852AA-1852MN**. In one embodiment, compression is performed on data written to a cache memory associated with the graphics core cluster **1800**, with the compression being performed with an encoding that is supported by the matrix engine **1803**. In one embodiment, the matrix engine **1803** includes support for input having structured sparsity in which a pre-determined level or pattern of sparsity is imposed on input data. This data may be compressed to a known compression ratio, with the compressed data being processed by the processing elements **1852AA-1852MN** according to metadata associated with the compressed data.

[0270] FIG. 19 illustrates a tile **1900** of a multi-tile processor, according to an embodiment. In one embodiment, the tile **1900** is representative of one of the graphics engine tiles **1710A-1710D** of FIG. 17A or compute engine tiles **1740A-1740D** of FIG. 17B. The tile **1900** of the multi-tile graphics processor includes an array of graphics core clusters (e.g., graphics core cluster **1800A**, graphics core cluster **1800B**, through graphics core cluster **1800N**), with each graphics core cluster having an array of graphics cores **515A-515N**. The tile **1900** also includes a global dispatcher **1902** to dispatch threads to processing resources of the tile **1900**.

[0271] The tile **1900** can include or couple with an L3 cache **1906** and memory **1910**. In various embodiments, the L3 cache **1906** may be excluded or the tile **1900** can include additional levels of cache, such as an L4 cache. In one embodiment, each instance of the tile **1900** in the multi-tile graphics processor is associated with memory **1910**, such as in FIG. 17A and FIG. 17B. In one embodiment, a multi-tile processor can be configured as a multi-chip module in which the L3 cache **1906** and/or memory **1910** reside on separate chiplets than the graphics core clusters **1800A-1800N**. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. For example, the L3 cache **1906** can be included in a dedicated cache chiplet or can reside on the same chiplet as the graphics core clusters **1800A-1800N**. In one embodiment, the L3 cache **1906** can be included in an active base die or active interposer.

[0272] A memory fabric **1903** enables communication among the graphics core clusters **1800A-1800N**, L3 cache **1906**, and memory **1910**. An L2 cache **1904** couples with the memory fabric **1903** and is configurable to cache transactions performed via the memory fabric **1903**. A tile interconnect **1908** enables communication with other tiles on the graphics processors and may be one of tile interconnects **1723A-1723F** of FIGS. 17A and 17B. In embodiments in which the L3 cache **1906** is excluded from the tile **1900**, the L2 cache **1904** may be configured as a combined L2/L3 cache. The memory fabric **1903** is configurable to route data to the L3 cache **1906** or memory controllers associated with the memory **1910** based on the presence or absence of the L3 cache **1906** in a specific implementation. The L3 cache **1906** can be configured as a per-tile cache that is dedicated to processing resources of the tile **1900** or may be a partition of a GPU-wide L3 cache.

[0273] FIG. 20 is a block diagram illustrating graphics processor instruction formats **2000**. The graphics processor execution units support an instruction set having instructions in multiple formats. The solid lined boxes illustrate the components that are generally included in an execution unit

instruction, while the dashed lines include components that are optional or that are included in a sub-set of the instructions. In some embodiments the graphics processor instruction formats **2000** described and illustrated are macro-instructions, in that they are instructions supplied to the execution unit, as opposed to micro-operations resulting from instruction decode once the instruction is processed. Thus, a single instruction may cause hardware to perform multiple micro-operations.

[0274] The graphics processor execution units as described herein may natively support instructions in a 128-bit instruction format **2010**. A compacted 64-bit instruction format **2030** is available for some instructions based on the selected instruction, instruction options, and number of operands. The native 128-bit instruction format **2010** provides access to all instruction options, while some options and operations are restricted in the 64-bit instruction format **2030**. The native instructions available in the 64-bit instruction format **2030** vary by embodiment. The instruction is compacted in part using a set of index values in an index field **2013**. The execution unit hardware references a set of compaction tables based on the index values and uses the compaction table outputs to reconstruct a native instruction in the 128-bit instruction format **2010**. Other sizes and formats of instruction can be used.

[0275] For each format, instruction opcode **2012** defines the operation that the execution unit is to perform. The execution units execute each instruction in parallel across the multiple data elements of each operand. For example, in response to an add instruction the execution unit performs a simultaneous add operation across each color channel representing a texture element or picture element. By default, the execution unit performs each instruction across all data channels of the operands. Instruction control field **2014** may enable control over certain execution options, such as channels selection (e.g., predication) and data channel order (e.g., swizzle). For instructions in the 128-bit instruction format **2010** an exec-size field **2016** limits the number of data channels that will be executed in parallel. An exec-size field **2016** may not be available for use in the compacted 64-bit instruction format **2030**.

[0276] Some execution unit instructions have up to three operands including two source operands, src0 **2020**, src1 **2022**, and one destination operand (dest **2018**). Other instructions, such as, for example, data manipulation instructions, dot product instructions, multiply-add instructions, or multiply-accumulate instructions, can have a third source operand (e.g., SRC2 **2024**). The instruction opcode **2012** determines the number of source operands. An instruction's last source operand can be an immediate (e.g., hard-coded) value passed with the instruction. The execution units may also support multiple destination instructions, where one or more of the destinations is implied or implicit based on the instruction and/or the specified destination.

[0277] The 128-bit instruction format **2010** may include an access/address mode field **2026** specifying, for example, whether direct register addressing mode or indirect register addressing mode is used. When direct register addressing mode is used, the register address of one or more operands is directly provided by bits in the instruction.

[0278] The 128-bit instruction format **2010** may also include an access/address mode field **2026**, which specifies an address mode and/or an access mode for the instruction. The access mode may be used to define a data access

alignment for the instruction. Access modes including a 16-byte aligned access mode and a 1-byte aligned access mode may be supported, where the byte alignment of the access mode determines the access alignment of the instruction operands. For example, when in a first mode, the instruction may use byte-aligned addressing for source and destination operands and when in a second mode, the instruction may use 16-byte-aligned addressing for all source and destination operands.

[0279] The address mode portion of the access/address mode field 2026 may determine whether the instruction is to use direct or indirect addressing. When direct register addressing mode is used bits in the instruction directly provide the register address of one or more operands. When indirect register addressing mode is used, the register address of one or more operands may be computed based on an address register value and an address immediate field in the instruction.

[0280] Instructions may be grouped based on instruction opcode 2012 bit-fields to simplify Opcode decode 2040. For an 8-bit opcode, bits 4, 5, and 6 allow the execution unit to determine the type of opcode. The precise opcode grouping shown is merely an example. A move and logic opcode group 2042 may include data movement and logic instructions (e.g., move (mov), compare (cmp)). The move and logic opcode group 2042 may share the five least significant bits (LSB), where move (mov) instructions are in the form of 0000xxxxb and logic instructions are in the form of 0001xxxxb. A flow control instruction group 2044 (e.g., call, jump (jmp)) includes instructions in the form of 0010xxxxb (e.g., 0x20). A miscellaneous instruction group 2046 includes a mix of instructions, including synchronization instructions (e.g., wait, send) in the form of 0011xxxxb (e.g., 0x30). A parallel math instruction group 2048 includes component-wise arithmetic instructions (e.g., add, multiply (mul)) in the form of 0100xxxxb (e.g., 0x40). The parallel math instruction group 2048 performs the arithmetic operations in parallel across data channels. The vector math group 2050 includes arithmetic instructions (e.g., dp4) in the form of 0101xxxxb (e.g., 0x50). The vector math group performs arithmetic such as dot product calculations on vector operands. The opcode decode 2040, in one embodiment, can be used to determine which portion of an execution unit will be used to execute a decoded instruction. For example, some instructions may be designated as systolic instructions that will be performed by a systolic array. Other instructions, such as ray-tracing instructions (not shown) can be routed to a ray-tracing core or ray-tracing logic within a slice or partition of execution logic.

#### Graphics Pipeline

[0281] FIG. 21 is a block diagram of a graphics processor 2100, according to another embodiment. The elements of FIG. 21 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0282] The graphics processor 2100 may include different types of graphics processing pipelines, such as a geometry pipeline 2120, a media pipeline 2130, a display engine 2140, thread execution logic 2150, and a render output pipeline 2170. The graphics processor 2100 may be a graphics

processor within a multi-core processing system that includes one or more general-purpose processing cores. The graphics processor may be controlled by register writes to one or more control registers (not shown) or via commands issued to the graphics processor 2100 via a ring or mesh interconnect 2102. The ring or mesh interconnect 2102 may couple the graphics processor 2100 to other processing components, such as other graphics processors or general-purpose processors. Commands from ring or mesh interconnect 2102 are interpreted by a command streamer 2103, which supplies instructions to individual components of the geometry pipeline 2120 or the media pipeline 2130.

[0283] Command streamer 2103 may direct the operation of a vertex fetcher 2105 that reads vertex data from memory and executes vertex-processing commands provided by command streamer 2103. The vertex fetcher 2105 may provide vertex data to a vertex shader 2107, which performs coordinate space transformation and lighting operations to each vertex. Vertex fetcher 2105 and vertex shader 2107 may execute vertex-processing instructions by dispatching execution threads to graphics cores 2152A-2152B via a thread dispatcher 2131.

[0284] The graphics cores 2152A-2152B may be an array of vector processors having an instruction set for performing graphics and media operations. The graphics cores 2152A-2152B may have an attached L1 cache 2151 that is specific for each array or shared between the arrays. The cache can be configured as a data cache, an instruction cache, or a single cache that is partitioned to contain data and instructions in different partitions.

[0285] A geometry pipeline 2120 may include tessellation components to perform hardware-accelerated tessellation of 3D objects. A programmable hull shader 2111 may configure the tessellation operations. A programmable domain shader 2117 may provide back-end evaluation of tessellation output. A tessellator 2113 may operate at the direction of a programmable hull shader 2111 and contain special purpose logic to generate a set of detailed geometric objects based on a coarse geometric model that is provided as input to geometry pipeline 2120. In addition, if tessellation is not used, tessellation components (e.g., programmable hull shader 2111, tessellator 2113, and programmable domain shader 2117) can be bypassed. The tessellation components can operate based on data received from the vertex shader 2107.

[0286] Complete geometric objects may be processed by a geometry shader 2119 via one or more threads dispatched to graphics cores 2152A-2152B, or can proceed directly to the clipper 2129. The geometry shader may operate on entire geometric objects, rather than vertices or patches of vertices as in previous stages of the graphics pipeline. If the tessellation is disabled, the geometry shader 2119 receives input from the vertex shader 2107. The geometry shader 2119 may be programmable by a geometry shader program to perform geometry tessellation if the tessellation units are disabled.

[0287] Before rasterization, a clipper 2129 processes vertex data. The clipper 2129 may be a fixed function clipper or a programmable clipper having clipping and geometry shader functions. A rasterizer and depth test component 2173 in the render output pipeline 2170 may dispatch pixel shaders to convert the geometric objects into per pixel representations. The pixel shader logic may be included in thread execution logic 2150. Optionally, an application can

bypass the rasterizer and depth test component **2173** and access un-rasterized vertex data via a stream out unit **2123**.

[0288] The graphics processor **2100** has an interconnect bus, interconnect fabric, or some other interconnect mechanism that allows data and message passing amongst the major components of the processor. In some embodiments, graphics cores **2152A-2152B** and associated logic units (e.g., L1 cache **2151**, sampler **2154**, texture cache **2158**, etc.) interconnect via a data port **2156** to perform memory access and communicate with render output pipeline components of the processor. A sampler **2154**, L1 cache **2151**, texture cache **2158** and graphics cores **2152A-2152B** each may have separate memory access paths. Optionally, the texture cache **2158** can also be configured as a sampler cache.

[0289] The render output pipeline **2170** may contain a rasterizer and depth test component **2173** that converts vertex-based objects into an associated pixel-based representation. The rasterizer logic may include a windower/masker unit to perform fixed function triangle and line rasterization. An associated render cache **2178** and depth cache **2179** are also available in some embodiments. A pixel operations component **2177** performs pixel-based operations on the data, though in some instances, pixel operations associated with 2D operations (e.g., bit block image transfers with blending) are performed by the 2D engine **2141** or substituted at display time by the display controller **2143** using overlay display planes. A shared L3 cache **2175** may be available to all graphics components, allowing the sharing of data without the use of main system memory.

[0290] The media pipeline **2130** may include a media engine **2137** and a video front-end **2134**. Video front-end **2134** may receive pipeline commands from the command streamer **2103**. The media pipeline **2130** may include a separate command streamer. Video front-end **2134** may process media commands before sending the command to the media engine **2137**. Media engine **2137** may include thread spawning functionality to spawn threads for dispatch to thread execution logic **2150** via thread dispatcher **2131**.

[0291] The graphics processor **2100** may include a display engine **2140**. This display engine **2140** may be external to the graphics processor **2100** and may couple with the graphics processor via the ring or mesh interconnect **2102**, or some other interconnect bus or fabric. Display engine **2140** may include a 2D engine **2141** and a display controller **2143**. Display engine **2140** may contain special purpose logic capable of operating independently of the 3D pipeline. Display controller **2143** may couple with a display device (not shown), which may be a system integrated display device, as in a laptop computer, or an external display device attached via a display device connector.

[0292] The geometry pipeline **2120** and media pipeline **2130** maybe configurable to perform operations based on multiple graphics and media programming interfaces and are not specific to any one application programming interface (API). A driver software for the graphics processor may translate API calls that are specific to a particular graphics or media library into commands that can be processed by the graphics processor. Support may be provided for the Open Graphics Library (OpenGL), Open Computing Language (OpenCL), and/or Vulkan graphics and compute API, all from the Khronos Group. Support may also be provided for the Direct3D library from the Microsoft Corporation. A combination of these libraries may be supported. Support may also be provided for the Open Source Computer Vision

Library (OpenCV). A future API with a compatible 3D pipeline would also be supported if a mapping can be made from the pipeline of the future API to the pipeline of the graphics processor.

#### Graphics Pipeline Programming

[0293] FIG. 22A is a block diagram illustrating a graphics processor command format **2200** used for programming graphics processing pipelines, such as, for example, the pipelines described herein in conjunction with FIG. 16 and FIG. 21. FIG. 22B is a block diagram illustrating a graphics processor command sequence **2210** according to an embodiment. The solid lined boxes in FIG. 22A illustrate the components that are generally included in a graphics command while the dashed lines include components that are optional or that are included in a sub-set of the graphics commands. The graphics processor command format **2200** of FIG. 22A includes fields to identify a client **2202**, a command operation code (opcode **2204**), and a data field **2206** for the command. A sub-opcode **2205** and a command size **2208** are also included in some commands.

[0294] Client **2202** may specify the client unit of the graphics device that processes the command data. A graphics processor command parser may examine the client field of each command to condition the further processing of the command and route the command data to the appropriate client unit. The graphics processor client units may include a memory interface unit, a render unit, a 2D unit, a 3D unit, and a media unit. Each client unit may have a corresponding processing pipeline that processes the commands. Once the command is received by the client unit, the client unit reads the opcode **2204** and, if present, sub-opcode **2205** to determine the operation to perform. The client unit performs the command using information in data field **2206**. For some commands a command size **2208** is expected to explicitly specify the size of the command. The command parser may automatically determine the size of at least some of the commands based on the command opcode. Commands may be aligned via multiples of a double word. Other command formats can also be used.

[0295] The flow diagram in FIG. 22B illustrates a graphics processor command sequence **2210**. Software or firmware of a data processing system that features an example graphics processor may use a version of the command sequence shown to set up, execute, and terminate a set of graphics operations. A sample command sequence is shown and described for purposes of example and is not limited to these specific commands or to this command sequence. Moreover, the commands may be issued as batch of commands in a command sequence, such that the graphics processor will process the sequence of commands in at least partially concurrence.

[0296] The graphics processor command sequence **2210** may begin with a pipeline flush command **2212** to cause any active graphics pipeline to complete the currently pending commands for the pipeline. Optionally, the 3D pipeline **2222** and the media pipeline **2224** may not operate concurrently. The pipeline flush is performed to cause the active graphics pipeline to complete any pending commands. In response to a pipeline flush, the command parser for the graphics processor will pause command processing until the active drawing engines complete pending operations and the relevant read caches are invalidated. Optionally, any data in the render cache that is marked 'dirty' can be flushed to memory.

Pipeline flush command **2212** can be used for pipeline synchronization or before placing the graphics processor into a low power state.

[0297] A pipeline select command **2213** may be used when a command sequence utilizes the graphics processor to explicitly switch between pipelines. A pipeline select command **2213** may be utilized once within an execution context before issuing pipeline commands unless the context is to issue commands for both pipelines. A pipeline flush command **2212** may have to be utilized immediately before a pipeline switch via the pipeline select command **2213**.

[0298] A pipeline control command **2214** may configure a graphics pipeline for operation and may be used to program the 3D pipeline **2222** and the media pipeline **2224**. The pipeline control command **2214** may configure the pipeline state for the active pipeline. The pipeline control command **2214** may be used for pipeline synchronization and to clear data from one or more cache memories within the active pipeline before processing a batch of commands.

[0299] Commands related to the return buffer state **2216** may be used to configure a set of return buffers for the respective pipelines to write data. Some pipeline operations utilize the allocation, selection, or configuration of one or more return buffers into which the operations write intermediate data during processing. The graphics processor may also use one or more return buffers to store output data and to perform cross thread communication. The return buffer state **2216** may include selecting the size and number of return buffers to use for a set of pipeline operations.

[0300] The remaining commands in the command sequence differ based on the active pipeline for operations. Based on a pipeline determination **2220**, the command sequence is tailored to the 3D pipeline **2222** beginning with the 3D pipeline state **2230** or the media pipeline **2224** beginning at the media pipeline state **2240**.

[0301] The commands to configure the 3D pipeline state **2230** include 3D state setting commands for vertex buffer state, vertex element state, constant color state, depth buffer state, and other state variables that are to be configured before 3D primitive commands are processed. The values of these commands are determined at least in part based on the particular 3D API in use. The 3D pipeline state **2230** commands may also be able to selectively disable or bypass certain pipeline elements if those elements will not be used.

[0302] A 3D primitive **2232** command may be used to submit 3D primitives to be processed by the 3D pipeline. Commands and associated parameters that are passed to the graphics processor via the 3D primitive **2232** command are forwarded to the vertex fetch function in the graphics pipeline. The vertex fetch function uses the 3D primitive **2232** command data to generate vertex data structures. The vertex data structures are stored in one or more return buffers. The 3D primitive **2232** command may be used to perform vertex operations on 3D primitives via vertex shaders. To process vertex shaders, 3D pipeline **2222** dispatches shader execution threads to graphics processor execution units.

[0303] The 3D pipeline **2222** may be triggered via an execute **2234** command or event. A register may write trigger command executions. An execution may be triggered via a 'go' or 'kick' command in the command sequence. Command execution may be triggered using a pipeline synchronization command to flush the command sequence through the graphics pipeline. The 3D pipeline will perform

geometry processing for the 3D primitives. Once operations are complete, the resulting geometric objects are rasterized and the pixel engine colors the resulting pixels. Additional commands to control pixel shading and pixel back-end operations may also be included for those operations.

[0304] The graphics processor command sequence **2210** may follow the media pipeline **2224** path when performing media operations. In general, the specific use and manner of programming for the media pipeline **2224** depends on the media or compute operations to be performed. Specific media decode operations may be offloaded to the media pipeline during media decode. The media pipeline can also be bypassed and media decode can be performed in whole or in part using resources provided by one or more general-purpose processing cores. The media pipeline may also include elements for general-purpose graphics processor unit (GPGPU) operations, where the graphics processor is used to perform SIMD vector operations using computational shader programs that are not explicitly related to the rendering of graphics primitives.

[0305] Media pipeline **2224** may be configured in a similar manner as the 3D pipeline **2222**. A set of commands to configure the media pipeline state **2240** are dispatched or placed into a command queue before the media object commands **2242**. Commands for the media pipeline state **2240** may include data to configure the media pipeline elements that will be used to process the media objects. This includes data to configure the video decode and video encode logic within the media pipeline, such as encode or decode format. Commands for the media pipeline state **2240** may also support the use of one or more pointers to "indirect" state elements that contain a batch of state settings.

[0306] Media object commands **2242** may supply pointers to media objects for processing by the media pipeline. The media objects include memory buffers containing video data to be processed. Optionally, all media pipeline states should be valid before issuing a media object command **2242**. Once the pipeline state is configured and media object commands **2242** are queued, the media pipeline **2224** is triggered via an execute command **2244** or an equivalent execute event (e.g., register write). Output from media pipeline **2224** may then be post processed by operations provided by the 3D pipeline **2222** or the media pipeline **2224**. GPGPU operations may be configured and executed in a similar manner as media operations.

#### Graphics Software Architecture

[0307] FIG. 23 illustrates an example graphics software architecture for a data processing system **2300**. Such a software architecture may include a 3D graphics application **2310**, an operating system **2320**, and a processor **2330**. The processor **2330** may include a graphics processor **2332** and one or more general-purpose processor core(s) **2334**. The processor **2330** may be a variant of one of the processor(s) **1402** or any other of the processors described herein, and may be used in place thereof. Therefore, the disclosure of any features in combination with the processor(s) **1402** or any other of the processors described herein also discloses a corresponding combination with the graphics processor **2332** but is not limited to such. Moreover, the elements of FIG. 23 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to

that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. The 3D graphics application **2310** and operating system **2320** are each executed in the system memory **2350** of the data processing system.

**[0308]** 3D graphics application **2310** may contain one or more shader programs including shader instructions **2312**. The shader language instructions may be in a high-level shader language, such as the High-Level Shader Language (HLSL) of Direct3D, the OpenGL Shader Language (GLSL), and so forth. The application may also include executable instructions **2314** in a machine language suitable for execution by the general-purpose processor core(s) **2334**. The application may also include graphics objects **2316** defined by vertex data.

**[0309]** The operating system **2320** may be a Microsoft® Windows® operating system from the Microsoft Corporation, a proprietary UNIX-like operating system, or an open source UNIX-like operating system using a variant of the Linux kernel. The operating system **2320** can support a graphics API **2322** such as the Direct3D API, the OpenGL API, or the Vulkan API. When the Direct3D API is in use, the operating system **2320** uses a front-end shader compiler **2324** to compile any shader instructions **2312** in HLSL into a lower-level shader language. The compilation may be a just-in-time (JIT) compilation or the application can perform shader pre-compilation. High-level shaders may be compiled into low-level shaders during the compilation of the 3D graphics application **2310**. The shader instructions **2312** may be provided in an intermediate form, such as a version of the Standard Portable Intermediate Representation (SPIR) used by the Vulkan API.

**[0310]** User mode graphics driver **2326** may contain a back-end shader compiler **2327** to convert the shader instructions **2312** into a hardware specific representation. When the OpenGL API is in use, shader instructions **2312** in the GLSL high-level language are passed to a user mode graphics driver **2326** for compilation. The user mode graphics driver **2326** may use operating system kernel mode functions **2328** to communicate with a kernel mode graphics driver **2329**. The kernel mode graphics driver **2329** may communicate with graphics processor **2332** to dispatch commands and instructions.

#### IP Core Implementations

**[0311]** One or more aspects may be implemented by representative code stored on a machine-readable medium which represents and/or defines logic within an integrated circuit such as a processor. For example, the machine-readable medium may include instructions which represent various logic within the processor. When read by a machine, the instructions may cause the machine to fabricate the logic to perform the techniques described herein. Such representations, known as “IP cores,” are reusable units of logic for an integrated circuit that may be stored on a tangible, machine-readable medium as a hardware model that describes the structure of the integrated circuit. The hardware model may be supplied to various customers or manufacturing facilities, which load the hardware model on fabrication machines that manufacture the integrated circuit. The integrated circuit may be fabricated such that the circuit performs operations described in association with any of the embodiments described herein.

**[0312]** FIG. **24** is a block diagram illustrating an IP core development system **2400** that may be used to manufacture an integrated circuit to perform operations according to an embodiment. The IP core development system **2400** may be used to generate modular, re-usable designs that can be incorporated into a larger design or used to construct an entire integrated circuit (e.g., an SOC integrated circuit). A design facility **2430** can generate a software simulation **2410** of an IP core design in a high-level programming language (e.g., C/C++). The software simulation **2410** can be used to design, test, and verify the behavior of the IP core using a simulation model **2412**. The simulation model **2412** may include functional, behavioral, and/or timing simulations. A register transfer level design (RTL design **2415**) can then be created or synthesized from the simulation model **2412**. The RTL design **2415** is an abstraction of the behavior of the integrated circuit that models the flow of digital signals between hardware registers, including the associated logic performed using the modeled digital signals. In addition to an RTL design **2415**, lower-level designs at the logic level or transistor level may also be created, designed, or synthesized. Thus, the particular details of the initial design and simulation may vary.

**[0313]** The RTL design **2415** or equivalent may be further synthesized by the design facility into a hardware model **2420**, which may be in a hardware description language (HDL), or some other representation of physical design data. The HDL may be further simulated or tested to verify the IP core design. The IP core design can be stored for delivery to a fabrication facility **2465** using non-volatile memory **2440** (e.g., hard disk, flash memory, or any non-volatile storage medium). The fabrication facility **2465** may be a 3<sup>rd</sup> party fabrication facility. Alternatively, the IP core design may be transmitted (e.g., via the Internet) over a wired connection **2450** or wireless connection **2460**. The fabrication facility **2465** may then fabricate an integrated circuit that is based at least in part on the IP core design. The fabricated integrated circuit can be configured to perform operations in accordance with at least one embodiment described herein.

**[0314]** FIG. **25A** illustrates cross-section side view of a package assembly **2590** for an integrated circuit that includes multiple units of hardware logic chiplets connected to a substrate **2580** (e.g., base die). A graphics processing unit, parallel processor, and/or compute accelerator as described herein can be composed from diverse silicon chiplets that are separately manufactured. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. A diverse set of chiplets with different IP core logic can be assembled into a single device. Additionally, the chiplets can be integrated into a base die or base chiplet using active interposer technology. The concepts described herein enable the interconnection and communication between the different forms of IP within the GPU. IP cores can be manufactured using different process technologies and composed during manufacturing, which avoids the complexity of converging multiple IPs, especially on a large SoC with several flavors of IPs, to the same manufacturing process. Enabling the use of multiple process technologies improves the time to market and provides a cost-effective way to create multiple product SKUs. Additionally, the disaggregated IPs are more amenable to

being power gated independently, components that are not in use on a given workload can be powered off, reducing overall power consumption.

[0315] In various embodiments a package assembly 2590 can include fewer or greater number of components and chiplets that are interconnected by an interconnect fabric 2585 or a bridge structure 2587. The bridge structure 2587 may be used to facilitate a point-to-point interconnect between, for example, a logic or I/O chiplet 2574 and memory chiplets 2575. In some implementations, the bridge structure 2587 may also be embedded within the substrate 2580. The chiplets within the package assembly 2590 may have a 2.5D arrangement using Chip-on-Wafer-on-Substrate (CoWoS) stacking in which multiple dies are stacked side-by-side on a silicon interposer that includes through-silicon vias (TSVs) to couple the chiplets with the substrate 2580, which includes electrical connections to the package interconnect 2583.

[0316] In one embodiment, silicon interposer is an active interposer 2589 that includes embedded logic in addition to TSVs. In such embodiment, the chiplets within the package assembly 2590 are arranged using 3D face to face die stacking on top of the active interposer 2589. The active interposer 2589 can include I/O hardware logic 2591, cache memory 2592, and other hardware logic 2593, in addition to the interconnect fabric 2585 and the bridge structure 2587. The interconnect fabric 2585 enables communication between the various logic chiplets within the active interposer 2589. The interconnect fabric 2585 may be an NoC interconnect or another form of packet switched fabric that switches data packets between components of the package assembly. For complex assemblies, the interconnect fabric 2585 may be a dedicated chiplet enables communication between the various hardware logic of the package assembly 2590.

[0317] The hardware logic chiplets can include special purpose hardware logic chiplets 2572, a logic or I/O chiplet 2574, and/or memory chiplets 2575. The special purpose hardware logic chiplets 2572 and logic or I/O chiplet 2574 may be implemented at least partly in configurable logic or fixed-functionality logic hardware and can include one or more portions of any of the processor core(s), graphics processor(s), parallel processors, or other accelerator devices described herein. The memory chiplets 2575 can be DRAM (e.g., GDDR, HBM) memory or cache (SRAM) memory. Cache memory 2592 within the active interposer 2589 (or substrate 2580) can act as a global cache for the package assembly 2590, part of a distributed global cache, or as a dedicated cache for the interconnect fabric 2585.

[0318] Each chiplet can be fabricated as separate semiconductor die and coupled with a base die that is embedded within or coupled with the substrate 2580. The coupling with the substrate 2580 can be performed via an interconnect structure 2573. The interconnect structure 2573 may be configured to route electrical signals between the various chiplets and logic within the substrate 2580. The interconnect structure 2573 can include interconnects such as, but not limited to bumps or pillars. In some embodiments, the interconnect structure 2573 may be configured to route electrical signals such as, for example, I/O signals and/or power or ground signals associated with the operation of the logic, I/O and memory chiplets. In one embodiment, an additional interconnect structure couples the active interposer 2589 with the substrate 2580.

[0319] The substrate 2580 may be an epoxy-based laminate substrate and/or may also include other suitable types of substrates. The package assembly 2590 can be connected to other electrical devices via a package interconnect 2583. The package interconnect 2583 may be coupled to a surface of the substrate 2580 to route electrical signals to other electrical devices, such as a motherboard, other chipset, or multi-chip module.

[0320] A logic or I/O chiplet 2574 and a memory chiplet 2575 may be electrically coupled via a bridge structure 2587 that is configured to route electrical signals between the logic or I/O chiplet 2574 and a memory chiplet 2575. The bridge structure 2587 may be a dense interconnect structure that provides a route for electrical signals. The bridge structure 2587 may include a bridge substrate composed of glass or a suitable semiconductor material. Electrical routing features can be formed on the bridge substrate to provide a chip-to-chip connection between the logic or I/O chiplet 2574 and a memory chiplet 2575. The bridge structure 2587 may also be referred to as a silicon bridge or an interconnect bridge. For example, the bridge structure 2587 is an Embedded Multi-die Interconnect Bridge (EMIB). Alternatively, the bridge structure 2587 may simply be a direct connection from one chiplet to another chiplet.

[0321] FIG. 25B illustrates a package assembly 2594 including interchangeable chiplets 2595, according to an embodiment. The interchangeable chiplets 2595 can be assembled into standardized chiplet slots or chiplet sockets on base chiplets 2596, 2598. The base chiplets 2596, 2598 can be coupled via a bridge interconnect 2597, which can be similar to the other bridge interconnects described herein and may be, for example, an EMIB. Memory chiplets can also be connected to logic or I/O chiplets via a bridge interconnect. I/O and logic chiplets can communicate via an interconnect fabric. The base chiplets can each support one or more slots in a standardized format for one of logic or I/O or memory/cache.

[0322] SRAM and power delivery circuits may be fabricated into one or more of the base chiplets 2596, 2598, which can be fabricated using a different process technology relative to the interchangeable chiplets 2595 that are stacked on top of the base chiplets. For example, the base chiplets 2596, 2598 can be fabricated using a larger process technology, while the interchangeable chiplets can be manufactured using a smaller process technology. One or more of the interchangeable chiplets 2595 may be memory (e.g., DRAM) chiplets. Different memory densities can be selected for the package assembly 2594 based on the power, and/or performance targeted for the product that uses the package assembly 2594. Additionally, logic chiplets with a different number of type of functional units can be selected at time of assembly based on the power, and/or performance targeted for the product. Additionally, chiplets containing IP logic cores of differing types can be inserted into the interchangeable chiplet slots, enabling hybrid processor designs that can mix and match different technology IP blocks.

#### Example SoC/SIP Processor

[0323] FIG. 26 illustrates an example processor that may be fabricated using one or more IP cores. In addition to what is illustrated, other logic and circuits may be included, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores.

The elements of FIG. 26 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0324] The processor 2600 may be an SoC or SIP that includes one or more application processor(s) 2605 (e.g., CPUs), a graphics processor 2610, which may be a variant of the graphics processor(s) 1408, or of any graphics processor described herein and may be used in place of any graphics processor described. Therefore, the disclosure of any features in combination with a graphics processor herein also discloses a corresponding combination with the graphics processor 2610 but is not limited to such. The processor 2600 may additionally include a vision processor 2615 and/or a media processor 2620, any of which may be a modular IP core from the same or multiple different design facilities. The processor 2600 may include peripheral or bus logic including a USB controller 2625, UART controller 2630, an SPI/SDIO controller 2635, and an I<sup>2</sup>S/I<sup>2</sup>C controller 2640. Additionally, the integrated circuit can include a display device 2645 coupled to one or more of a high-definition multimedia interface (HDMI) controller 2650 and a reliability, availability, and serviceability engine (RAS engine 2655). The RAS engine 2655 is used to identify potential faults that may occur during device runtime to minimize the downtime that would result if those potential faults were to occur. Storage may be provided by a flash memory subsystem 2660 including flash memory and a flash memory controller. The processor 2600 can include a memory 2665 that is an on-chip or on-package memory. The processor 2600 may also include memory controllers to enable access to an off-package memory device. Some integrated circuits additionally include an embedded security engine 2670.

[0325] In one embodiment, the processing components of the processor 2600 can be optimized for low power operation to enable deployment to a variety of machine learning platforms, including autonomous vehicles and autonomous robots. For example, one implementation of the processor 2600 can be used as a portion of the main control system for an autonomous vehicle. When configured for use in autonomous vehicles, the processor 2600 is designed and configured for compliance with the relevant functional safety standards of the deployment jurisdiction.

[0326] During operation, the media processor 2620 and vision processor 2615 can work in concert to accelerate computer vision operations. The media processor 2620 can enable low latency decode of multiple high-resolution (e.g., 4K, 8K) video streams. The decoded video streams can be written to a buffer in the memory 2665. The vision processor 2615 can then parse the decoded video and perform preliminary processing operations on the frames of the decoded video in preparation of processing the frames using a trained image recognition model. For example, the vision processor 2615 can accelerate convolution operations for a CNN that is used to perform image recognition on the high-resolution video data, while back-end model computations are performed by the GPGPU 1306.

[0327] The application processor(s) 2605 can include control logic to assist with sequencing and synchronization of data transfers and shared memory operations performed by the media processor 2620 and the vision processor 2615.

The application processor(s) 2605 can also function as an application processor to execute software applications that can make use of the inferencing compute capability of the GPGPU 1306. For example, at least a portion of the navigation and driving logic can be implemented in software executing on the application processor(s) 2605. Such software can directly issue computational workloads to the graphics processor 2610 or the computational workloads can be issued to the application processor(s) 2605, which can offload at least a portion of those operations to the graphics processor 2610.

[0328] The graphics processor 2610 can include compute clusters including a low power configuration of the processing clusters 706A-706H of FIG. 7 or the graphics core clusters 1800A-1800N of FIG. 19. The compute clusters within the graphics processor 2610 can support instruction that are specifically optimized to perform inferencing computations on a trained neural network. For example, the graphics processor 2610 can support instructions to perform low precision computations such as 8-bit and 4-bit integer and floating-point operations, including operations on integer and floating-point input in MX formats or other block scaled formats.

Secure Virtualized Telemetry with Save and Restore Over Global Counters

[0329] Parallel rendering graphics architectures are increasingly expected to deliver near zero-wait responsiveness and superb performance while taking on large workloads and increased performance expectations. Examples of large workloads include neural networks, artificial intelligence (AI), machine learning, etc. Such workloads have become more prevalent as they have been implemented in a number of computing devices, such as personal computing devices, business-related computing devices, etc. Furthermore, with the growing use of large machine learning and neural network workloads, new silicon has been produced that is targeted at running large workloads. Such new silicon includes dedicated hardware accelerators (e.g., graphics processing unit (GPU), field-programmable gate array (FPGA), vision processing unit (VPU), etc.) customized for processing data using data parallelism.

[0330] Artificial intelligence (AI), including machine learning (ML), deep learning (DL), neural networks, and/or other artificial machine-driven logic, enables machines (e.g., computers, logic circuits, etc.) to use a model to process input data to generate an output based on patterns and/or associations previously learned by the model via a training process. For instance, the model may be trained with data to recognize patterns and/or associations and follow such patterns and/or associations when processing input data such that other input(s) result in output(s) consistent with the recognized patterns and/or associations.

[0331] With respect to AI applications, parallel rendering graphics architectures and/or other hardware accelerator architectures that execute such AI applications tend to gather a significant amount of telemetry data (e.g., for AI-based pattern analysis, etc.). Such telemetry data can include free running counters, bandwidth measurements, and other types of measurement and metrics that the GPU/hardware accelerator collects. In the case of a virtualized environment, such as the GPU hosting separate virtualized resources for one or more tenant (e.g., virtual machine (VM), containers, etc.), this telemetry data may be collected in the context of both physical functions (PFs) (e.g., hypervisor level and/or

kernel level) and/or virtual functions (VFs) (e.g., associated with user workload and profiles at the VM level).

**[0332]** In the case of a confidential compute environment that is virtualized, confidential VMs may be provided which maintains confidentiality of user workloads and/or data running on the VMs. However, in the case of virtualized confidential compute, the hypervisor may not have to guarantee any level of confidentiality for the virtualized environments. For example, some conventional solutions for intelligent telemetry procurement and relies on a centralized telemetry services stack to aggregate telemetry over a cloud tenant network. Such conventional solutions rely on hypervisor/OS/bare-metal managed telemetry collection and dissemination. A large number of this telemetry applies to VF-specific registers, workload and finger-printing state from the targeted GPU device within the telemetry manageability pool. However, in these conventional approaches, the Hypervisor, Host OS and Platform (base management controller (BMC)) are outside a Trusted Computing Base (TCB) of the confidential VM. As such, the telemetry collection architecture of these conventional approaches is therefore outside the TCB of the confidential VM.

**[0333]** A malicious hypervisor can use data collected to conduct side-channel attacks on the confidential VM, such as:

- [0334]** (1) Analysis of workload patterns
- [0335]** (2) Injecting side-channel workloads inline with confidential VM's workload and then assessing how that perturbs the confidential VM's workloads as evidenced by variations in their telemetry.
- [0336]** (3) Detect specific timing sequences and use those to exploit Time-Of-Check-Time-Of-Use (TOC-TOU) type attack surfaces.
- [0337]** (4) Confidentiality breach—VF telemetry data may reveal specific user profile information, especially if collected off Graphics, Display, Media or AI engines within the SOC that are responsible for servicing VF render, compute or AI workloads.

**[0338]** Implementations herein address the above noted technical problems by providing secure virtualized telemetry with save and restore (S&R) schemes over global counters. Implementations herein provide for deployment of a SOC trusted telemetry endpoint (also referred to herein as a secure telemetry endpoint) that is independently-mapped within each confidential VM's VF base address register (BAR) as a "logical" endpoint. The SOC trusted telemetry endpoint supports Single Root IO Virtualization (SR-IOV) over PCIe, as well as trusted session establishment schemes such as Security Protocols and Data Models (SPDM) over PCIe Data Object Exchange (DOE).

**[0339]** The SOC trusted telemetry endpoint of implementations herein can provide hardware (HW) isolation between PF (hypervisor) level telemetry and VF (confidential VM) level telemetry and provides a HW-assisted path for PCIe to Fabric to IP memory (e.g., DRAM) lookup for telemetry procurement.

**[0340]** Furthermore, implementations provide for a save and restore over global counters technique. For example, for hypervisor scheduling schemes, such as time-slicing (also referred to herein as "temporal partitioning"), where multiple confidential VMs share the same SOC resources, the SOC trusted telemetry endpoint provides a save and restore (SR) component (e.g., engine, hardware circuitry, firmware,

etc.) over trusted scheduling frame boundaries to allow for free-running global counters used for telemetry collection to be updated on VF switches.

**[0341]** Implementations herein provide for the following technical advantages. SOC trusted telemetry acceleration where VFs are able to directly access their SOC telemetry data without relying on Host OS, Hypervisor or Platform Tenant Networks/Servers from procuring this information or being privy to it. HW isolation across PF and VF Telemetry counters both for telemetry aggregation by SOC, as well as procurement by confidential VM via its VF BAR. HW-assisted fast path for telemetry aggregation and dissemination with minimal SOC firmware involvement. Using VF temporal partition/world-switch boundaries to invoke SR finite state machines (FSMs) for resetting free-running global counters mapped as per VF telemetry.

**[0342]** FIGS. 27-33 provide further details on the approach to implement secure virtualized telemetry with save and restore over global counters.

**[0343]** FIG. 27 illustrates an example of a graphics multiprocessor 2700 providing a secure telemetry endpoint for secure virtualized telemetry collection with save and restore over global counters, in accordance with implementations herein. The elements of FIG. 27 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. Therefore, the discussion of any features in combination with a graphics processor herein also discloses a corresponding combination with the example graphics multiprocessor 2700, but is not limited to such.

**[0344]** In one implementation, graphics multiprocessor 2700 may be the same as GPGPUs 806A-806D described with respect to FIG. 8, and/or GPGPU 1570 described with respect to FIG. 15C, for example. In one implementation, graphics multiprocessor 2700 may be part of an SOC/SOP implementing secure virtualized telemetry with SR schemes over global counters, in accordance with implementations herein.

**[0345]** The graphics multiprocessor 2700 may include at least one PCIe endpoint 2720 (also may be referred to as a PCIe endpoint controller) connected to PCIe lanes 2710 of the graphics multiprocessor 2700. The PCIe endpoint 2720 can also have a standard advance extensible interface (AXI) towards SOC fabric/interconnect 2730, which connects the PCIe endpoint 2720 to telemetry endpoint 2740 (also referred to herein as secure telemetry endpoint 2740). Graphics multiprocessor 2700 further includes one or more processing engines, such as a graphics processing 2750, media processing 2760, and/or AI processing 2770. The processing engines are communicably coupled to the SOC fabric/interconnect 2730 and telemetry endpoint 2740.

**[0346]** In implementations herein, telemetry endpoint 2740 can include transmit (Tx)/receive (Rx) processing circuitry 2742 that is communicably coupled to the SoC fabric/interconnect 2730. Tx/Rx processing circuitry 2742 can host one or more translation tables 2743. Telemetry endpoint 2740 also include a micro controller 2744 hosting mailboxes 2745, multiplexor (mux) circuitry 2749, save and restore (SR) circuitry 2746, SR SRAM 2747, and telemetry SRAM 2748. Although SRAM is illustrated as the memory utilized in telemetry endpoint 2740, implementations herein

may utilize other types of memory and/or storage for telemetry endpoint **2740**. Further details of how the illustrated components of FIG. 27 operate to provide for secure virtualized telemetry with save and restore over global counters of implementations herein is provided below with respect to FIG. 28.

[0347] FIG. 28 is a block diagram illustrating an example SOC/SOP **2800** implementing secure virtualized telemetry with SR schemes over global counters, in accordance with implementations herein. The elements of FIG. 28 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. Therefore, the discussion of any features in combination with a graphics processor herein also discloses a corresponding combination with the example graphics multiprocessor **2700**, but is not limited to such.

[0348] In one implementation, SOC/SOP **2800** may host the graphics multiprocessor **2700** described with respect to FIG. 27. In one implementation, SOC/SOP **2800** includes telemetry endpoint **2810**. Telemetry endpoint **2810** may be the same as telemetry endpoint **2740** described with respect to FIG. 27. Telemetry endpoint **2810** may be provided as IP of SOC/SOP **2800**.

[0349] In implementations herein, the telemetry endpoint may include at least three ingress/egress ports:

[0350] (1) Trusted ROT fabric endpoint **2820** may be a privileged fabric component accessible by the SoC Silicon root-of-trust (ROT) (e.g., a TEE data security manager (DSM)) **2702**.

[0351] (2) Trusted PCIe endpoint **2840** may be a PCIe downstream port to SOC fabric by which a VF (e.g., confidential VM **2804**) communicates with the telemetry endpoint **2810** (functioning as a SR-IOV compliant PCIe EP). This can include PCIe DOE Mailboxes (see, e.g., mailboxes **2745** of FIG. 27) for a confidential VM to establish SPDM-over-DOE session with the telemetry endpoint **2810**.

[0352] (3) Trusted telemetry collection endpoint **2815** may be fabric unicast and/or multicast ports which enables (i) a manageability microcontroller **2830** performs telemetry aggregation by reading various IP-level telemetry counters (e.g., from graphics processing **2750**, media processing **2760**, AI processing **2770** of FIG. 27), (ii) SR circuitry **2845** saves and restores certain free-running global counters over VF temporal partitions (e.g., world-switches), and (iii) allowing for import wires or ingress messages from various scheduling IPs (e.g., graphics scheduler **2870**, media scheduler **2880**, AI scheduler **2890**) of VF world switch events for their various scheduling domains.

[0353] The SoC/SOP **2800** can implement the telemetry endpoint **2810** to provide the secure virtualized telemetry with SR schemes over global counters of implementations herein, detailed as follows with references to the circled numbers illustrated in FIG. 28.

[0354] At a first stage shown as circle **1** in FIG. 28, the Silicon RoT **2702** functioning as a TEE DSM can measure and attest a patching engine **2825** of the telemetry endpoint **2810** as part of including it in the TEE's TCB. The patching engine **2825** may be a fixed function block (e.g., FSM) with no mutable code or data. Measurement of this block (patch-

ing engine **2825**) may include the silicon RoT **2702** examining the patching engine **2825** FSM's SOC Straps, register state, and/or the validity of any high volume manufacturing (HVM) fuses it consumes.

[0355] At a second stage shown as circle **2** in FIG. 28, after the silicon RoT **2702** has expanded its TCB to include the patching engine **2825**, it interfaces (via the trusted ROT fabric endpoint **2820**) with the patching engine **2825** to load the mutable FW code or data into the telemetry SRAM **2860** (or register files in some implementations). In implementations herein, the patching engine **2825** may not have to proxy the loads on behalf of the Silicon RoT **2802**, but rather the patching engine **2825** can provide the silicon RoT **2802** a trusted, privileged path to the telemetry SRAM **2860** (and other memories).

[0356] At a third stage shown as circle **3** in FIG. 28, the mutable FW and data are loaded into the SR circuitry **2845** (which may be referred to herein as a SR-IOV Save and Restore Engine), the manageability microcontroller **2830** (which is responsible for telemetry aggregation), and the translation table **2850** (which handles the incoming VF telemetry read requests).

[0357] At a fourth stage shown as circle **4** in FIG. 28, the silicon RoT **2702** functioning as TEE DSM informs a confidential VM **2804** (also referred to herein as VM) of the expansion of the TEE DSM's TCB to now include the telemetry endpoint **2810**. The TEE DSM can then provide attestation measurements (e.g., cryptographic hashes of loaded FW, code) of the telemetry endpoint **2810** to the confidential VM **2804**. If the confidential VM **2804** successfully attests these measurements, then it can bind to the telemetry endpoint **2810** (e.g., as it is already exposed as a PCIe SR-IOV device by the hypervisor as part of standard services) and establish an SPDM-over-DOE session to negotiate a session key.

[0358] At a fifth stage shown as circle **5** in FIG. 28, both the telemetry endpoint **2810** and the confidential VM **2804** may use the session key as a "session protection asset" and either wrap all subsequent telemetry exchanges via this key (or establish a requester-authenticated Diffie-Helman session). Registration of this session allows for the telemetry endpoint's **2810** manageability microcontroller **2830** and SR circuitry **2845** to initiate their telemetry aggregation and global counter SR cycles over the trusted telemetry collection endpoint **2815**. Save and restore (SR) SRAM **2855** may be utilized to store data corresponding to the SR cycles, described in further detail below with respect to FIG. 30. Independently, the various schedulers **2870**, **2880**, **2890** (e.g., for the Graphics Compute, AI and Media resources) managing their respective VF temporal pools can feed VF SR and active VF events into the telemetry endpoint **2810** that can be used by the SR circuitry **2845** and the manageability microcontroller **2830** for their operations. In one implementation, PF/VF IDs corresponding to the access for telemetry fetch can be transmitted on the Ax user bits. Further details of the SR cycles are described below with respect to FIG. 30.

[0359] At a sixth stage shown as circle **6** in FIG. 28, the confidential VM **2804** then issues a telemetry read request via the telemetry endpoint's **2810** MMIO mailbox address mapped within each VF's PCIe IO BAR by the hypervisor. The confidential VM **2804** can then issue a telemetry read for a given telemetry counter's logical address. To ensure that information on various per-VF telemetry counters

remains obfuscated across confidential VMs **2804**, each telemetry counter can be represented by its logical address. The telemetry endpoint's **2810** translation table **2850** can consume from the incoming VF MMIO read request to identify the Function ID, any Security Policy (SAI) that gets tagged with the MMIO, and the logical address. The translation table **2850** can then translate that information to the physical address within the telemetry SRAM **2860** where the manageability microcontroller **2830** has stored the aggregated telemetry data for this VF. In implementations herein, no firmware is involved within the procurement of this telemetry data as the host device engine is able to directly loop into the telemetry SRAM **2860** to procure the aggregated data.

[0360] From the perspective of the confidential VM **2804**, the VF BAR-based telemetry reads and the subsequent logical-to-physical translation are illustrated as shown in FIG. 29. FIG. 29 is a block diagram depicting an example VF-BAR based telemetry read and translation **2900**, in accordance with implementations herein. The VF-BAR based telemetry read and translation **2900** depicts, a VM BAR view **2910**, a secure telemetry translation table **2920**, and a secure telemetry SRAM **2930**.

[0361] The secure telemetry endpoint, such as telemetry endpoint **2810** of FIG. 28, can support a number of deployment configurations including:

[0362] PF Only: All telemetry Mapped to PF BAR;

[0363] 1 VF: Bifurcation of all telemetry across PF and a single confidential VM. No temporal partitioning/world-switching, therefore SR circuitry (e.g., SR engine) in secure telemetry endpoint is disabled and not enabled by TEE DSM (SilROT);

[0364] Multi-VF: Same as above except multiple confidential VMs now sharing resources via temporal partitioning.

[0365] In one example, for multi-VF, the VM BAR view **2910** may include VM0 (VF0) BAR space **2912** for a first VM VM0 (VF0), VM1 (VF1) BAR space **2914** for a second VM VM1 (VF1), and hypervisor (PF) BAR space **2916** for a hypervisor of the VMs. This BAR space **2912**, **2914**, **2916** may host telemetry space that includes pointers that map to the secure telemetry translation table **2920**. In one implementation, the secure telemetry translation table **2920** may be the same as translation table **2850** described with respect to FIG. 28. The secure telemetry translation table **2920** can translate the incoming pointer from the VM BAR space **2912**, **2914**, **2916** to a physical address within the secure telemetry SRAM **2930**. The SRAM **2930** may include separate telemetry space for each VM and the hypervisor, including telemetry space VM0 (VF0) **2932**, telemetry space VM1 (VF1) **2934**, and telemetry space hypervisor (PF) **2936**. In implementations herein, the manageability microcontroller of the secure telemetry endpoint can store aggregated telemetry data for each VM (VF) into its designated telemetry space **2932**, **2934**, **2936**.

[0366] Referring back to FIG. 28, in one implementation, the secure telemetry translation table **2850** may be a local translation module (LTM) that is an in-memory database that provides a mapping of all endpoints in the SoC and how their registers can be accessed. The secure telemetry translation table **2850** may specify the path to access different endpoints in the SoC/SoP **2800**. The secure telemetry translation table **2850** may specify security rules for accessing registers. Furthermore, the secure telemetry translation table

**2850** may map MMIO addresses for telemetry/debug capabilities to corresponding endpoint registers, and map logical addresses to the corresponding physical address for the host transactions. Table 1 below is one example of contents of a secure telemetry translation table in accordance with implementations herein.

TABLE 1

| Secure Telemetry Translation Table |        |                       |                        |                |                |                          |
|------------------------------------|--------|-----------------------|------------------------|----------------|----------------|--------------------------|
| Function ID                        | BAR ID | Logical Start Address | Physical Start Address | End-point Name | End-point Path | Security Policy          |
| 0                                  | 0      | 0x0000_0000           | 0x69470                | EP1            | Path to EP1    | EP1 BAR0 Security Policy |
| 0                                  | 1      | 0x0000_1000           | 0x80000                | EP1            | Path to EP1    | EP1 BAR1 Security Policy |
| 0                                  | 2      | 0x0000_2000           | 0x90000                | EP1            | Path to EP1    | EP1 BAR2 Security Policy |
| 1                                  | 0      | 0x0000_3000           | 0x92000                | EP2            | Path to EP2    | EP2 BAR0 Security Policy |
| 1                                  | 1      | 0x0000_3400           | 0x9A000                | EP2            | Path to EP2    | EP2 BAR1 Security Policy |
| 2                                  | 0      | 0x0000_4000           | 0xA0000                | EP3            | Path to EP3    | EP3 BAR0 Security Policy |
| 3                                  | 0      | 0x0000_5000           | 0xB0000                | EP4            | Path to EP4    | EP4 BAR0 Security Policy |

[0367] As previously noted, the telemetry endpoint's **2810** manageability microcontroller **2830** and SR circuitry **2845** can initiate telemetry aggregation and global counter SR cycles over the trusted telemetry collection endpoint **2815**. With respect to the SR cycles over global counters provided by implementations herein, FIG. 30 is an example sequence diagram illustrating a sequence of events for an example SR cycle **3000**, in accordance with implementations herein. The example SR cycle **3000** includes both firmware **3010** and hardware **3015** components. The firmware components **3010** may include a VF scheduler **3001** and a VF quanta **3002**. The hardware components **3015** may include a manageability microcontroller (mgmt uC) **3003**, per-VM telemetry (telem) SRAM **3004**, SR circuitry **3005**, SR SRAM **3006**, processor engines **3007**, and telemetry SRAM **3008**.

[0368] The example SR cycle **3000** sequence of events includes a VF quant start **3020** where free running counters in graphics processing components (such as graphics, media, AI IP, and so on) are initialized to zero on a VF quanta switch. The VF quanta start **3020** is notified to the firmware **3010** by graphics/media/AI processing. FW **3010** starts an iterative loop **3022** of telemetry gathering from SoC components (e.g., processor engines **3007**) including reading telemetry **3025** from processor engines **3007** and performing this read for each processor engine **3030**. For each counter fetched (receive telemetry **3035**) from the target SoC endpoints from via telemetry SRAM **3008**, the FW **3010** notifies **3040** the SR circuitry **3005** to restore **3045** a corresponding saved count value from SR SRAM **3006**.

[0369] The FW **3010** adds **3050** the new count value fetched from the telemetry SRAM **3008** with the restored count value from SR SRAM **3006**. The sum value is stored in the telemetry SRAM **3008** by populating the telemetry **3055** to the mgmt uC **3003**.

[0370] The loop **3022** continues until the occurrence of a VF quanta exit **3060**. In some implementations, the VF quanta exit **3060** is notified to the FW **3010** by graphics/media/AI processing. The FW **3010** then notifies **3065** the

SR circuitry **3005** to save **3070** all counters corresponding to the last active VF. While these iterations are happening, the host can issue telemetry read commands in parallel, which can directly access the latest value stored in telemetry SRAM **3008**.

[**0371**] FIG. **31** is a flow diagram illustrating an embodiment of a method **3100** for providing implementation of secure virtualized telemetry with save and restore over global counters. Method **3100** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, etc.), software (such as instructions run on a processing device), or a combination thereof. The process of method **3100** is illustrated in linear sequences for brevity and clarity in presentation; however, it is contemplated that any number of them can be performed in parallel, asynchronously, or in different orders. Further, for brevity, clarity, and ease of understanding, many of the components and processes described with respect to FIGS. **1-30** may not be repeated or discussed hereafter. In one implementation, a processor, such as graphics multiprocessor **2700** of FIG. **27**, may perform method **3100**.

[**0372**] Method **3100** begins at processing block **3110** where the processor may map, responsive to attesting secure telemetry endpoint IP as part of trusted computing base (TCB) of a trusted execution environment (TEE), within virtual function (VF) BAR space of a confidential VM a secure telemetry endpoint IP as a logical endpoint for the confidential VM, where the confidential VM is hosted by the TEE. Then, at block **3120**, the processor may establish a trusted session between the VM and the secure telemetry endpoint IP.

[**0373**] Subsequently, at block **3130**, the processor may procure, by the secure telemetry endpoint IP, telemetry data corresponding to operation of the VM on the graphics processor. In one implementation, the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint IP that is accessible via the VF BAR space of the VM. Lastly, at block **3140**, the processor may utilize, by the VM, a hardware assisted path established by the secure telemetry endpoint IP to access telemetry data procured for the VM in a secure manner.

[**0374**] FIG. **32** is a flow diagram illustrating an embodiment of a method **3200** for providing establishing a secure connection for secure virtualized telemetry with save and restore over global counters. Method **3200** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, etc.), software (such as instructions run on a processing device), or a combination thereof. The process of method **3200** is illustrated in linear sequences for brevity and clarity in presentation; however, it is contemplated that any number of them can be performed in parallel, asynchronously, or in different orders. Further, for brevity, clarity, and ease of understanding, many of the components and processes described with respect to FIGS. **1-31** may not be repeated or discussed hereafter. In one implementation, a processor, such as graphics multiprocessor **2700** of FIG. **27**, may perform method **3200**.

[**0375**] Method **3200** begins at processing block **3210** where the processor may measure and attest secure telemetry endpoint as part of including it in TCB of TEE. At block **3220**, the processor may load mutable code and data into secure telemetry endpoint SRAM. Then, at block **3230**, the

processor may load mutable code and data into secure telemetry endpoint microcontroller, translation table, and save and restore component.

[**0376**] Subsequently, at block **3240**, the processor may inform a confidential VM of expansion of TEE's TCB to include the secure telemetry endpoint and provide secure telemetry endpoint measurements to the confidential VM for attestation and binding. At block **3250**, the processor may utilize session key as a session protection asset between the secure telemetry endpoint and the confidential VM. Lastly, at block **3260**, the processor may issue, using the session key, telemetry requests from the confidential VM to the secure telemetry endpoint via a secure telemetry MMIO address mapped within the VM's I/O BAR.

[**0377**] FIG. **33** is a flow diagram illustrating an embodiment of a method **3300** for providing save and restore cycles for secure virtualized telemetry with save and restore over global counters. Method **3300** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, etc.), software (such as instructions run on a processing device), or a combination thereof. The process of method **3300** is illustrated in linear sequences for brevity and clarity in presentation; however, it is contemplated that any number of them can be performed in parallel, asynchronously, or in different orders. Further, for brevity, clarity, and ease of understanding, many of the components and processes described with respect to FIGS. **1-32** may not be repeated or discussed hereafter. In one implementation, a processor, such as graphics multiprocessor **2700** of FIG. **27**, may perform method **3300**.

[**0378**] Method **3300** begins at processing block **3310** where the processor may initialize telemetry counters to zero on a VF quanta switch for a VM. At block **3320**, the processor may notify VF quanta start for the VM to secure telemetry endpoint microcontroller. Then, at block **3330**, the processor may begin an iterative loop of telemetry gather from SoC components

[**0379**] Subsequently, at block **3340**, the processor may, for each counter fetched from target SoC component, notify secure telemetry endpoint SR engine to restore corresponding saved count value from SR SRAM. At block **3350**, the processor may add new count value with the restored count value as a sum value that is stored in secure telemetry endpoint SRAM. Then, at block **3360**, the processor may end the telemetry collection iterative loop at VF quanta exit. Lastly, at block **3370**, the processor may notify SR engine to save all counters corresponding to the last active VM.

[**0380**] The following examples pertain to further embodiments. Example 1 is an apparatus to facilitate secure virtualized telemetry with save and restore over global counters. The apparatus of Example 1 includes processor circuitry comprising: execution circuitry to provide a trusted execution environment (TEE) to host a virtual machine (VM), the TEE to: responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of the TEE, map, within virtual function (VF) BAR space of the VM, the secure telemetry endpoint circuitry as a logical endpoint for the VM; and establish a trusted session between the VM and the secure telemetry endpoint circuitry; and the secure telemetry endpoint circuitry communicably coupled to the execution circuitry, the secure telemetry endpoint circuitry to: procure telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry

endpoint circuitry that is accessible via the VF BAR space of the VM; and provide a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

**[0381]** In Example 2, the subject matter of Example 1 can optionally include wherein the telemetry data comprises counter values incremented by one or more of a graphics engine of the processor circuitry, a media engine of the processor circuitry, or an artificial intelligence (AI) engine of the processor circuitry. In Example 3, the subject matter of any one of Examples 1-2 can optionally include wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises: measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry.

**[0382]** In Example 4, the subject matter of any one of Examples 1-3 can optionally include wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry. In Example 5, the subject matter of any one of Examples 1-4 can optionally include wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry comprises: notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry; providing secure telemetry endpoint measurements to the VM for attestation and binding; and generating a session key for the trusted session; wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

**[0383]** In Example 6, the subject matter of any one of Examples 1-5 can optionally include wherein procuring the telemetry data further comprise the secure telemetry endpoint circuitry to: receive notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of the processor circuitry are initialized to zero upon occurrence of the VF quanta start; initiate a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry; receive notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminate the telemetry collection iterative loop.

**[0384]** In Example 7, the subject matter of any one of Examples 1-6 can optionally include wherein the telemetry collection iterative loop of telemetry gathering comprises: for each telemetry counter fetched, notify a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and add a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory. In Example 8, the subject matter of any one of Examples 1-7 can optionally include wherein the execution circuitry comprises a tensor core of a graphics processing unit (GPU). In Example 9, the subject matter of any one of Examples 1-8 can optionally include wherein the processor circuitry is at least one of a single instruction multiple data (SIMD) machine or a single instruction multiple thread (SIMT) machine.

**[0385]** Example 10 is a method for facilitating secure virtualized telemetry with save and restore over global counters. The method of Example 10 can include responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of a trusted execution environment (TEE), mapping, by execution circuitry providing the TEE to host a virtual machine (VM), the secure telemetry endpoint circuitry as a logical endpoint for the VM within virtual function (VF) BAR space of the VM; establishing, by the TEE, a trusted session between the VM and the secure telemetry endpoint circuitry; procuring, by the secure telemetry endpoint circuitry, telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and providing, by the secure telemetry endpoint circuitry, a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

**[0386]** In Example 11, the subject matter of Example 10 can optionally include wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises: measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry. In Example 12, the subject matter of Examples 10-11 can optionally include wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry.

**[0387]** In Example 13, the subject matter of Examples 10-12 can optionally include wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry further comprises: notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry; providing secure telemetry endpoint measurements to the VM for attestation and binding; and generating a session key for the trusted session; wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM. In Example 14, the subject matter of Examples 10-13 can optionally include wherein procuring the telemetry data further comprises: receiving notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of processor circuitry comprising the execution circuitry are initialized to zero upon occurrence of the VF quanta start; initiating a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry; receiving notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminating the telemetry collection iterative loop.

**[0388]** In Example 15, the subject matter of Examples 10-14 can optionally include wherein the telemetry collection iterative loop of telemetry gathering comprises: for each telemetry counter fetched, notifying a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and adding a new count value of the telemetry counter to the

corresponding saved count value as a sum value that is stored in the save and restore memory.

**[0389]** Example 16 is a non-transitory computer-readable storage medium for facilitating secure virtualized telemetry with save and restore over global counters. The non-transitory computer-readable storage medium of Example 16 having instructions stored thereon, which when executed by one or more processors, cause the one or more processors to perform operations comprising: responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of a trusted execution environment (TEE), mapping, by execution circuitry providing the TEE to host a virtual machine (VM), the secure telemetry endpoint circuitry as a logical endpoint for the VM within virtual function (VF) BAR space of the VM; establishing, by the TEE, a trusted session between the VM and the secure telemetry endpoint circuitry; procuring, by the secure telemetry endpoint circuitry, telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and providing, by the secure telemetry endpoint circuitry, a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

**[0390]** In Example 17, the subject matter of Example 16 can optionally include wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises: measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry; wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry.

**[0391]** In Example 18, the subject matter of Examples 16-17 can optionally include wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry further comprises: notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry; providing secure telemetry endpoint measurements to the VM for attestation and binding; and generating a session key for the trusted session; wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

**[0392]** In Example 19, the subject matter of Examples 16-18 can optionally include wherein procuring the telemetry data further comprises: receiving notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of processor circuitry comprising the execution circuitry are initialized to zero upon occurrence of the VF quanta start; initiating a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry; receiving notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminating the telemetry collection iterative loop.

**[0393]** In Example 20, the subject matter of Examples 16-19 can optionally include wherein the telemetry collection iterative loop of telemetry gathering comprises: for each

telemetry counter fetched, notifying a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and adding a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory.

**[0394]** Example 21 is a system for facilitating secure virtualized telemetry with save and restore over global counters. The system of Example 21 can optionally include a memory; and processor circuitry communicably coupled to the memory, the processor circuitry comprising: execution circuitry to provide a trusted execution environment (TEE) to host a virtual machine (VM), the TEE to: responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of the TEE, map, within virtual function (VF) BAR space of the VM, the secure telemetry endpoint circuitry as a logical endpoint for the VM; and establish a trusted session between the VM and the secure telemetry endpoint circuitry; and the secure telemetry endpoint circuitry communicably coupled to the execution circuitry, the secure telemetry endpoint circuitry to: procure telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and provide a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

**[0395]** In Example 22, the subject matter of Example 21 can optionally include wherein the telemetry data comprises counter values incremented by one or more of a graphics engine of the processor circuitry, a media engine of the processor circuitry, or an artificial intelligence (AI) engine of the processor circuitry. In Example 23, the subject matter of any one of Examples 21-22 can optionally include wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises: measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry.

**[0396]** In Example 24, the subject matter of any one of Examples 21-23 can optionally include wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry. In Example 25, the subject matter of any one of Examples 21-24 can optionally include wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry comprises: notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry; providing secure telemetry endpoint measurements to the VM for attestation and binding; and generating a session key for the trusted session; wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

**[0397]** In Example 26, the subject matter of any one of Examples 21-25 can optionally include wherein procuring the telemetry data further comprise the secure telemetry endpoint circuitry to: receive notification of a virtual function (VF) quanta start for the VM, wherein telemetry coun-

ters of the processor circuitry are initialized to zero upon occurrence of the VF quanta start; initiate a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry; receive notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminate the telemetry collection iterative loop.

**[0398]** In Example 27, the subject matter of any one of Examples 21-26 can optionally include wherein the telemetry collection iterative loop of telemetry gathering comprises: for each telemetry counter fetched, notify a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and add a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory. In Example 28, the subject matter of any one of Examples 21-27 can optionally include wherein the execution circuitry comprises a tensor core of a graphics processing unit (GPU). In Example 29, the subject matter of any one of Examples 21-28 can optionally include wherein the processor circuitry is at least one of a single instruction multiple data (SIMD) machine or a single instruction multiple thread (SMT) machine.

**[0399]** Example 30 is an apparatus for facilitating secure virtualized telemetry with save and restore over global counters, comprising means for responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of a trusted execution environment (TEE), means for mapping, using execution circuitry providing the TEE to host a virtual machine (VM), the secure telemetry endpoint circuitry as a logical endpoint for the VM within virtual function (VF) BAR space of the VM; means for establishing, via the TEE, a trusted session between the VM and the secure telemetry endpoint circuitry; means for procuring, via the secure telemetry endpoint circuitry, telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and means for providing, via the secure telemetry endpoint circuitry, a hardware-assisted path for the VM to request access to the telemetry data procured for the VM. In Example 31, the subject matter of Example 30 can optionally include the apparatus further configured to perform the method of any one of the Examples 11-15.

**[0400]** Example 32 is at least one machine readable medium comprising a plurality of instructions that in response to being executed on a computing device, cause the computing device to carry out a method according to any one of Examples 10-15. Example 33 is an apparatus for facilitating secure virtualized telemetry with save and restore over global counters, configured to perform the method of any one of Examples 10-15. Example 34 is an apparatus for facilitating secure virtualized telemetry with save and restore over global counters, comprising means for performing the method of any one of Examples 10-15. Specifics in the Examples may be used anywhere in one or more embodiments.

**[0401]** The foregoing description and drawings are to be regarded in an illustrative rather than a restrictive sense. Persons skilled in the art will understand that various modifications and changes may be made to the embodiments

described herein without departing from the broader spirit and scope of the features set forth in the appended claims.

What is claimed is:

1. An apparatus comprising:

processor circuitry comprising:

execution circuitry to provide a trusted execution environment (TEE) to host a virtual machine (VM), the TEE to:

responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of the TEE, map, within virtual function (VF) BAR space of the VM, the secure telemetry endpoint circuitry as a logical endpoint for the VM; and

establish a trusted session between the VM and the secure telemetry endpoint circuitry; and

the secure telemetry endpoint circuitry communicably coupled to the execution circuitry, the secure telemetry endpoint circuitry to:

procure telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and

provide a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

2. The apparatus of claim 1, wherein the telemetry data comprises counter values incremented by one or more of a graphics engine of the processor circuitry, a media engine of the processor circuitry, or an artificial intelligence (AI) engine of the processor circuitry.

3. The apparatus of claim 1, wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises:

measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry.

4. The apparatus of claim 3, wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry.

5. The apparatus of claim 1, wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry comprises:

notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry;

providing secure telemetry endpoint measurements to the VM for attestation and binding; and

generating a session key for the trusted session;

wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

6. The apparatus of claim 1, wherein procuring the telemetry data further comprise the secure telemetry endpoint circuitry to:

receive notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of the processor circuitry are initialized to zero upon occurrence of the VF quanta start;

initiate a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry;

receive notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminate the telemetry collection iterative loop.

7. The apparatus of claim 6, wherein the telemetry collection iterative loop of telemetry gathering comprises:

- for each telemetry counter fetched, notify a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and
- add a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory.

8. The apparatus of claim 1, wherein the execution circuitry comprises a tensor core of a graphics processing unit (GPU).

9. The apparatus of claim 1, wherein the processor circuitry is at least one of a single instruction multiple data (SIMD) machine or a single instruction multiple thread (SIMT) machine.

10. A method comprising:

- responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of a trusted execution environment (TEE), mapping, by execution circuitry providing the TEE to host a virtual machine (VM), the secure telemetry endpoint circuitry as a logical endpoint for the VM within virtual function (VF) BAR space of the VM;
- establishing, by the TEE, a trusted session between the VM and the secure telemetry endpoint circuitry;
- procuring, by the secure telemetry endpoint circuitry, telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and
- providing, by the secure telemetry endpoint circuitry, a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

11. The method of claim 10, wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises:

- measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and
- loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry.

12. The method of claim 11, wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry.

13. The method of claim 10, wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry further comprises:

- notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry;
- providing secure telemetry endpoint measurements to the VM for attestation and binding; and

- generating a session key for the trusted session; wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

14. The method of claim 10, wherein procuring the telemetry data further comprises:

- receiving notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of processor circuitry comprising the execution circuitry are initialized to zero upon occurrence of the VF quanta start;
- initiating a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry;
- receiving notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminating the telemetry collection iterative loop.

15. The method of claim 14, wherein the telemetry collection iterative loop of telemetry gathering comprises:

- for each telemetry counter fetched, notifying a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and
- adding a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory.

16. At least one non-transitory computer readable medium having instructions stored thereon, which when executed by one or more processors, cause the one or more processors to perform operations comprising:

- responsive to attesting a secure telemetry endpoint circuitry as part of trusted computing base (TCB) of a trusted execution environment (TEE), mapping, by execution circuitry providing the TEE to host a virtual machine (VM), the secure telemetry endpoint circuitry as a logical endpoint for the VM within virtual function (VF) BAR space of the VM;
- establishing, by the TEE, a trusted session between the VM and the secure telemetry endpoint circuitry;
- procuring, by the secure telemetry endpoint circuitry, telemetry data corresponding to operation of the VM, wherein the telemetry data is stored in a portion of telemetry data storage of the secure telemetry endpoint circuitry that is accessible via the VF BAR space of the VM; and
- providing, by the secure telemetry endpoint circuitry, a hardware-assisted path for the VM to request access to the telemetry data procured for the VM.

17. The at least one non-transitory computer readable medium of claim 16, wherein attesting the secure telemetry endpoint circuitry as part of the TCB of the TEE comprises:

- measuring and attesting a fixed function block of the secure telemetry endpoint circuitry, wherein the fixed function block comprises immutable code and data; and
- loading mutable code and mutable data of the TEE into memory of the secure telemetry endpoint circuitry;

wherein the mutable code and the mutable data are loaded into a microcontroller of the secure telemetry endpoint circuitry, a translation table of the secure telemetry endpoint circuitry, and a save and restore component of the secure telemetry endpoint circuitry.

**18.** The at least one non-transitory computer readable medium of claim **16**, wherein establishing the trusted session between the VM and the secure telemetry endpoint circuitry further comprises:

notifying the VM of expansion of the TCB of the TEE to include the secure telemetry endpoint circuitry;  
providing secure telemetry endpoint measurements to the VM for attestation and binding; and  
generating a session key for the trusted session;  
wherein the session key is utilized as part of the hardware-assisted path for the VM to request the access to the telemetry data procured for the VM.

**19.** The at least one non-transitory computer readable medium of claim **16**, wherein procuring the telemetry data further comprises:

receiving notification of a virtual function (VF) quanta start for the VM, wherein telemetry counters of processor circuitry comprising the execution circuitry are initialized to zero upon occurrence of the VF quanta start;

initiating a telemetry collection iterative loop of telemetry gathering from the telemetry counters comprising fetching values from the telemetry counters to store in telemetry storage of the secure telemetry endpoint circuitry;

receiving notification of a VF quanta end for the VM; and responsive to the VF quanta end, terminating the telemetry collection iterative loop.

**20.** The at least one non-transitory computer readable medium of claim **19**, wherein the telemetry collection iterative loop of telemetry gathering comprises:

for each telemetry counter fetched, notifying a save and restore engine of the secure telemetry endpoint circuitry to restore a corresponding saved count value from save and restore memory of the secure telemetry endpoint circuitry; and

adding a new count value of the telemetry counter to the corresponding saved count value as a sum value that is stored in the save and restore memory.

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